

accumulatedMutations_MEF

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1 Env

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure5/")
```

2 Libraries

```
library(FSA)
## ## FSA v0.10.0. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.
library(ineq)
library(vegan)
## Loading required package: permute
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(tidyr)
library(ggpubr)
## Loading required package: ggplot2
## Registered S3 methods overwritten by 'car':
##   method      from
##   hist.boot    FSA
##   confint.boot FSA
library(Seurat)
## Loading required package: SeuratObject
## Loading required package: sp
## 'SeuratObject' was built under R 4.5.0 but the current version is
## 4.5.1; it is recommended that you reinstall 'SeuratObject' as the ABI
## for R may have changed
##
## Attaching package: 'SeuratObject'
## The following objects are masked from 'package:base':
##
```

```

##      intersect, t
library(igraph)
##
## Attaching package: 'igraph'
## The following object is masked from 'package:Seurat':
##
##      components
## The following object is masked from 'package:tidyr':
##
##      crossing
## The following objects are masked from 'package:dplyr':
##
##      as_data_frame, groups, union
## The following object is masked from 'package:vegan':
##
##      diversity
## The following object is masked from 'package:permute':
##
##      permute
## The following objects are masked from 'package:stats':
##
##      decompose, spectrum
## The following object is masked from 'package:base':
##
##      union
library(ggraph)
##
## Attaching package: 'ggraph'
## The following object is masked from 'package:sp':
##
##      geometry
library(webshot)
library(ggplot2)
library(ggpmisc)
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##      method                  from
##      heightDetails.titleGrob ggplot2
##      widthDetails.titleGrob  ggplot2
##
## Attaching package: 'ggpp'
## The following objects are masked from 'package:ggpubr':
##
##      as_npc, as_npcx, as_npcy
## The following object is masked from 'package:ggplot2':
##
##      annotate
library(stringr)
library(corrplot)
## corrplot 0.95 loaded
library(pheatmap)
library(reshape2)
##

```

```

## Attaching package: 'reshape2'
## The following object is masked from 'package:tidyr':
##
##      smiths
library(tidygraph)
##
## Attaching package: 'tidygraph'
## The following object is masked from 'package:igraph':
##
##      groups
## The following object is masked from 'package:stats':
##
##      filter
library(networkD3)
##
## Attaching package: 'networkD3'
## The following object is masked from 'package:Seurat':
##
##      JS
## The following object is masked from 'package:SeuratObject':
##
##      JS
library(tidyverse)
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr 2.1.5
## v lubridate 1.9.4    v tibble 3.3.0
## v purrr 1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%--%()      masks igraph::%--%()
## x ggpp::annotate()      masks ggplot2::annotate()
## x tibble::as_data_frame() masks igraph::as_data_frame(), dplyr::as_data_frame()
## x purrr::compose()      masks igraph::compose()
## x igraph::crossing()     masks tidyr::crossing()
## x tidygraph::filter()   masks dplyr::filter(), stats::filter()
## x dplyr::lag()          masks stats::lag()
## x purrr::simplify()     masks igraph::simplify()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(patchwork)

```

3 Def function

```

retriveCellsByLocation <- function(CSM, locList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[, c("CellTag",locList)]
  data_MutPerCell_CSM_tmp <- data.frame(
    CellTag = rownames(CSM_tmp),
    MutNum = rowSums(CSM_tmp == "Mut"),
    WTNNum = rowSums(CSM_tmp == "WT")
  )
  cells_Total <- data_MutPerCell_CSM_tmp[rowSums(data_MutPerCell_CSM_tmp[,c("MutNum", "WTNum")]) > 0, ]
  cells_WT <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "WTNum"] > 0,]$CellTag
}

```

```

cells_Mut <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "MutNum"] > 0,]$CellTag
cells <- list(Total = cells_Total, Mut = cells_Mut, WT = cells_WT)
return(cells)
}

retriveLocsByCells <- function(CSM, cellList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[CSM$CellTag %in% cellList,]
  data_CellPerMut_CSM_tmp <- data.frame(
    Location = colnames(CSM_tmp),
    CellMutNum = colSums(CSM_tmp == "Mut"),
    CellWTNum = colSums(CSM_tmp == "WT")
  )
  pos_Total <- data_CellPerMut_CSM_tmp[rowSums(data_CellPerMut_CSM_tmp[, c("CellMutNum", "CellWTNum")]) > 0,]$Location
  pos_Mut <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellMutNum"] > 0,]$Location
  pos_WT <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellWTNum"] > 0,]$Location
  pos <- list(Total = pos_Total, Mut = pos_Mut, WT = pos_WT)
  return(pos)
}

cal_MultiSpe <- function(matrixData) {
  res <- data.frame(
    Shannon = vegan::diversity(matrixData, index = "shannon"),
    Simpson = vegan::diversity(matrixData, index = "simpson"),
    Gini = apply(matrixData, 1, function(x) ineq(x + 0.00001, type = "Gini")),
    sumMut = rowSums(matrixData),
    Tau = NA
  )
  # Tau = sum(1 - (x / max(x))) / (n - 1)
  res$Tau <- apply(matrixData, 1, function(x) {
    if (all(x == 0)) return(NA)
    x <- x / max(x)
    sum(1 - x) / (length(x) - 1)
  })
  rownames(res) <- rownames(matrixData)
  return(res)
}

cal_scMutScore <- function(data_cell_mut, data_mut_time, data_meta){
  # data_cell_mut [cell * muts]
  # data_mut_time [muts * tissues/celltype/days]
  # data_meta [Cellbarcode, CellMutTag(tissues/celltype/days)]
  # data_cell_time_matrix
  replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

  data_cell_mut_num <- as.data.frame(lapply(data_cell_mut, function(x) {
    ifelse(x %in% names(replacement_map), replacement_map[x], x)
  })))

  rownames(data_cell_mut_num) <- rownames(data_cell_mut)

```

```

print(dim(data_cell_mut_num))
print(dim(data_mut_time))

data_cell_time_matrix <- as.matrix(data_cell_mut_num) %*% as.matrix(data_mut_time)
rownames(data_cell_time_matrix) <- rownames(data_cell_mut)
colnames(data_cell_time_matrix) <- colnames(data_mut_time)

mut_s_total <- unique(rownames(data_mut_time))

# get different tissues/time/celltype names
col_names_data <- colnames(data_cell_time_matrix)

col_length <- length(col_names_data)

col_coefficient_fix <- setNames(
  rep(1/col_length, col_length),
  col_names_data
)

data_cell_time_matrix <- data_cell_time_matrix[,col_names_data]

# Gini
res_index <- cal_MultiSpe(data_cell_time_matrix)

res_diversity <- as.data.frame((data_cell_time_matrix > 1) * rep(col_coefficient_fix[colnames(data_cell_time_matrix)],
res_diversity$Diversity <- rowSums(res_diversity)

res_diversity$Diversity[is.na(res_diversity$Diversity)] <- 0.00001

res_index$Gini[is.na(res_index$Gini)] <- 0.00001
res_index$Tau[is.na(res_index$Tau)] <- 0.00001

res_index$GiniTau <- res_index$Gini * res_index$Tau
res_index$scMutScore <- 2 / (
  (1 / (0.1 * res_index$GiniTau + 0.00001)) +
  (1 / (0.9 * res_diversity$Diversity + 0.00001))
)

res_index$Diversity <- res_diversity$Diversity

res_index$sumMut_norm <- (res_index$sumMut - min(res_index$sumMut)) /
  (max(res_index$sumMut) - min(res_index$sumMut))

res_index$GiniTau_norm <- (res_index$GiniTau - min(res_index$GiniTau)) /
  (max(res_index$GiniTau) - min(res_index$GiniTau))

res_index$scMutScoreQun <- res_index$sumMut_norm

res_index$scMutScore <- scales::rescale(res_index$scMutScore, to = c(0, 1))
res_index$scMutScoreQun <- scales::rescale(res_index$scMutScoreQun, to = c(0, 1))

res <- list(diversity = res_diversity, score = res_index, matrixD = data_cell_time_matrix)
return(res)

```

```

}

propagate_mutations <- function(data, time_points) {
  for (i in 1:nrow(data)) {
    mut_time <- which(data[i, time_points] == "Mut")
    if (length(mut_time) > 0) {
      first_mut <- min(mut_time)
      data[i, time_points[first_mut:length(time_points)]] <- "Mut"
    }
  }
  return(data)
}

```

4 Load Data

```

MEF_CSM <- readRDS("../Data/MEF/MEF_CSM.rds")
MEF_CSM_hf1 <- MEF_CSM$hf1
MEF_CSM_hf2 <- MEF_CSM$hf2

MEF_SSM <- readRDS("../Data/MEF/MEF_SSM.rds")
MEF_SSM_hf1 <- MEF_SSM$hf1
MEF_SSM_hf2 <- MEF_SSM$hf2

MEF_Meta <- readRDS("../Data/MEF/MEF_Meta.rds")
MEF_Meta_hf1 <- MEF_Meta$hf1
MEF_Meta_hf2 <- MEF_Meta$hf2

# retrieved from MPG database, see https://github.com/QunATCG/scMutrace-tutorial/tree/main/AdvancedFunc
MPG_data <- read.csv("../Data/MEF/MPG_combined.csv")

```

5 Observed Accumulated Mutation Num

```

Reps_sequencing <- c("Rep1_all", "Rep2_all")
Days_sequencing <- c("d6", "d9", "d12", "d15", "d21", "d28")

col_hf1_names <- c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28")
col_hf2_names <- c("hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")

data_accMutNum_res <- as.data.frame(matrix(nrow = length(Days_sequencing), ncol = length(Reps_sequencing),
colnames(data_accMutNum_res) <- Reps_sequencing
rownames(data_accMutNum_res) <- Days_sequencing

for (i in rownames(data_accMutNum_res)){
  data_accMutNum_res[i, "Rep1_all"] <- length(MEF_SSM_hf1[MEF_SSM_hf1[, paste0("hf1.", i)] == "Mut",]$mutNum)
  data_accMutNum_res[i, "Rep2_all"] <- length(MEF_SSM_hf2[MEF_SSM_hf2[, paste0("hf2.", i)] == "Mut",]$mutNum)
}

data_accMutNum_res$Days <- rownames(data_accMutNum_res)

```

```

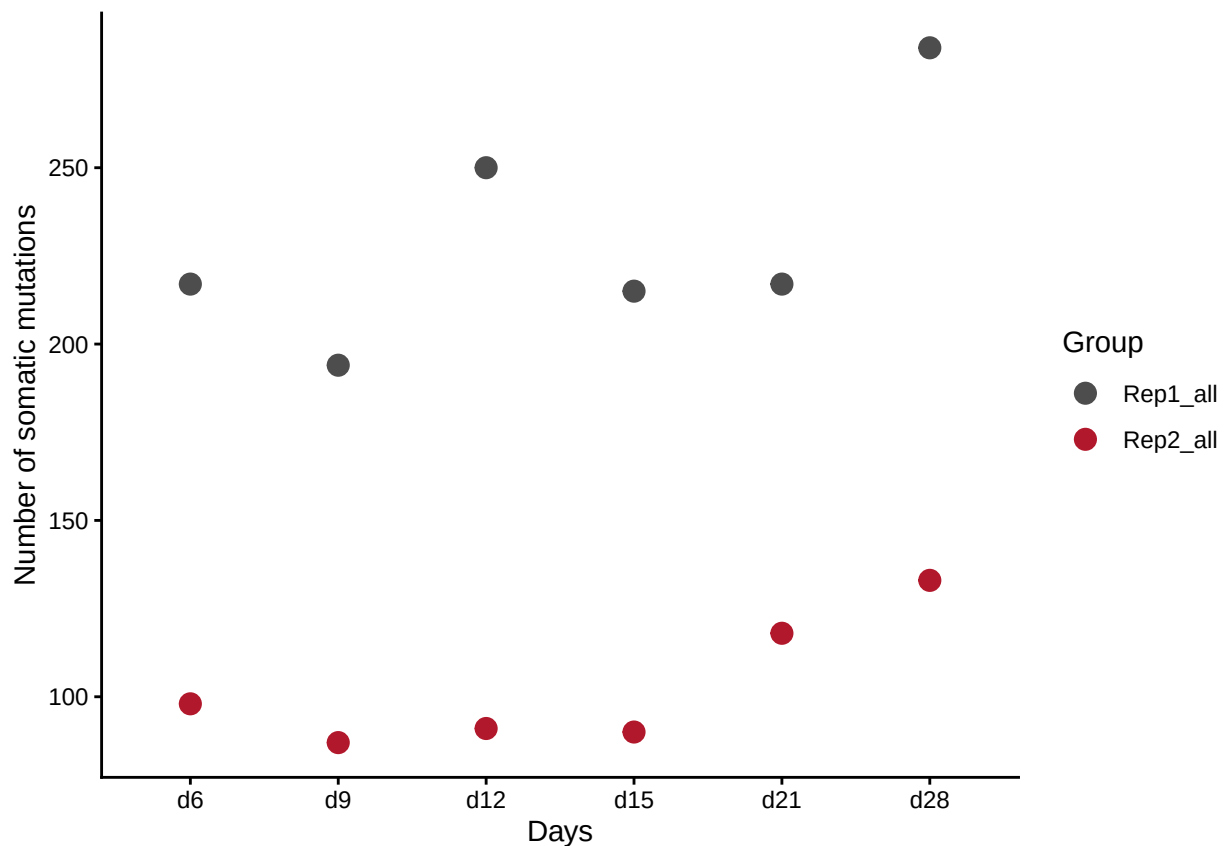
data_accMutNum_res_plot <- melt(data_accMutNum_res)
## Using Days as id variables
colnames(data_accMutNum_res_plot) <- c("Days", "Group", "MutNum")

data_accMutNum_res_plot$Days <- factor(data_accMutNum_res_plot$Days, levels = c("d6", "d9", "d12", "d15", "d21", "d28"))

p_reps_somaticNum <- ggplot(data_accMutNum_res_plot, aes(x = Days, y = MutNum, colour = Group, fill = Group)) +
  geom_point(shape = 21, size = 3, stroke = 0.8) +
  ylab("Number of somatic mutations") +
  scale_colour_manual(values = c("#4d4d4d", "#b2182b")) +
  scale_fill_manual(values = c("#4d4d4d", "#b2182b")) +
  theme_classic()

p_reps_somaticNum

```



```

replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

MEF_SSM_hf1_hf2 <- merge(MEF_SSM_hf1[,c("mutPos", col_hf1_names)], MEF_SSM_hf2[,c("mutPos", col_hf2_names)], by = "mutPos")
MEF_SSM_hf1_hf2[is.na(MEF_SSM_hf1_hf2)] <- "NoCall"

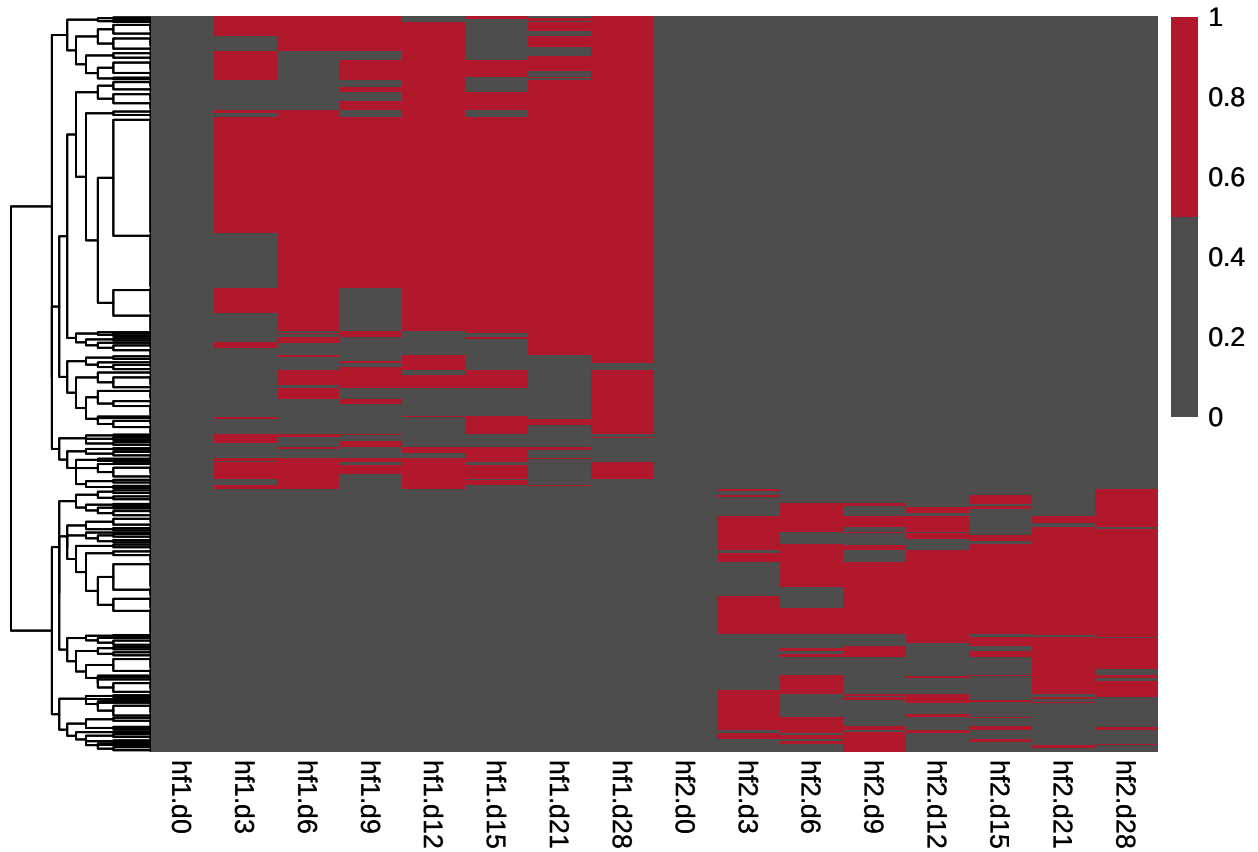
MEF_SSM_hf1_hf2_num <- as.data.frame(lapply(MEF_SSM_hf1_hf2, function(x) {
  ifelse(x %in% names(replacement_map), replacement_map[x], x)
}))

rownames(MEF_SSM_hf1_hf2_num) <- MEF_SSM_hf1_hf2$mutPos

```



```
# propagate_mutations
p_accSomaticMut_heatmap_observed <- pheatmap(MEF_SSM_hf1_hf2_num[,c(2:17)], show_rownames = F, color = c(
p_accSomaticMut_heatmap_observed
```



6 Imputed Accumulated Mutation Num

```
color_settings <- c(d0 = "#FFDC91FF", d3 = "#7876B1FF", d6 = "#4575b4",
d9 = "#91bfdb", d12 = "#e0f3f8", d15 = "#fee090",
d21 = "#fc8d59", d28 = "#BC3C29FF")

color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
Day_9 = "#91bfdb", Day_12 = "#e0f3f8", Day_15 = "#fee090",
Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

Reps_sequencing <- c("Rep1_all", "Rep2_all")
Days_sequencing <- c("d0", "d3", "d6", "d9", "d12", "d15", "d21", "d28")

col_hf1_names <- c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28")
col_hf2_names <- c("hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")

MEF_SSM_hf1_impute <- propagate_mutations(MEF_SSM_hf1, col_hf1_names)
MEF_SSM_hf2_impute <- propagate_mutations(MEF_SSM_hf2, col_hf2_names)
```

```

data_accMutNum_res <- as.data.frame(matrix(nrow = length(Days_sequencing), ncol = length(Reps_sequencing),
colnames(data_accMutNum_res) <- Reps_sequencing
rownames(data_accMutNum_res) <- Days_sequencing

for (i in rownames(data_accMutNum_res)){
  data_accMutNum_res[i, "Rep1_all"] <- length(MEF_SSM_hf1_impute[MEF_SSM_hf1_impute[, paste0("hf1.", i)],
  data_accMutNum_res[i, "Rep2_all"] <- length(MEF_SSM_hf2_impute[MEF_SSM_hf2_impute[, paste0("hf2.", i)],
}

data_accMutNum_res$Days <- rownames(data_accMutNum_res)
data_accMutNum_res_plot <- melt(data_accMutNum_res)
## Using Days as id variables
colnames(data_accMutNum_res_plot) <- c("Days", "Group", "MutNum")

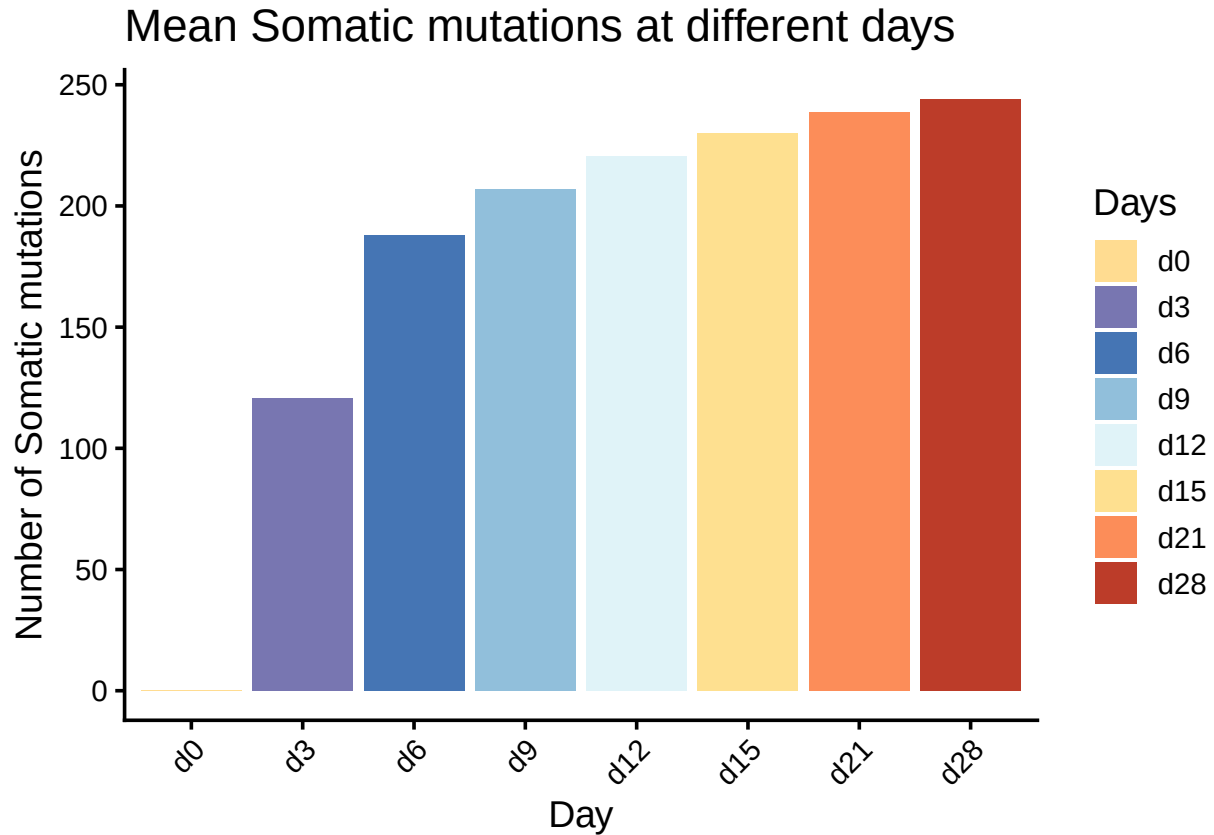
data_accMutNum_res_mean_plot <- aggregate(MutNum ~ Days, data = data_accMutNum_res_plot, FUN = mean)

data_accMutNum_res_mean_plot$Days <- factor(data_accMutNum_res_mean_plot$Days, levels = c("d0", "d3", "d7"))

p_reps_somaticNum <- ggplot(data_accMutNum_res_mean_plot, aes(x = Days, y = MutNum, fill = Days)) +
  geom_col(position = "dodge") +
  theme_minimal(base_size = 14) +
  labs(title = "Mean Somatic mutations at different days",
       x = "Day",
       y = "Number of Somatic mutations") +
  scale_fill_manual(values = color_settings) +
  theme_classic(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p_reps_somaticNum

```



```

replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

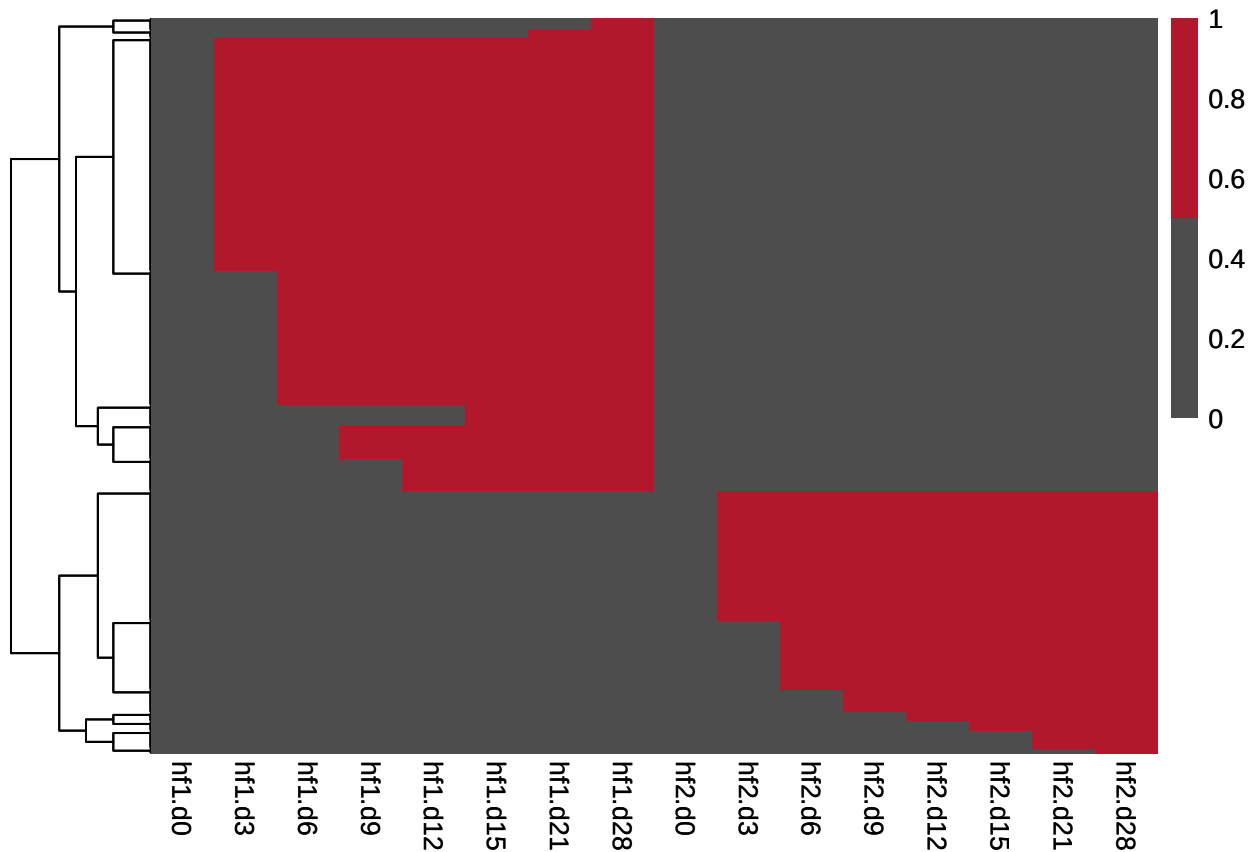
MEF_SSM_hf1_hf2_impute <- merge(MEF_SSM_hf1_impute[,c("mutPos", col_hf1_names)], MEF_SSM_hf2_impute[,c(
MEF_SSM_hf1_hf2_impute[is.na(MEF_SSM_hf1_hf2_impute)] <- "NoCall"

MEF_SSM_hf1_hf2_impute_num <- as.data.frame(lapply(MEF_SSM_hf1_hf2_impute, function(x) {
  ifelse(x %in% names(replacement_map), replacement_map[x], x)
})))

rownames(MEF_SSM_hf1_hf2_impute_num) <- MEF_SSM_hf1_hf2_impute$mutPos

# propagate_mutations
p_accSomaticMut_heatmap_impute <- pheatmap(MEF_SSM_hf1_hf2_impute_num[,c(2:17)], show_rownames = F, col
p_accSomaticMut_heatmap_impute

```



```
# cell number
hf1_cellNum <- as.data.frame(table(MEF_Meta_hf1$Day)) %>%
  mutate(Sample = "hf1")
hf2_cellNum <- as.data.frame(table(MEF_Meta_hf2$Day)) %>%
  mutate(Sample = "hf2")

hf_cellNum <- bind_rows(hf1_cellNum, hf2_cellNum)
colnames(hf_cellNum) <- c("Day", "Count", "Sample")

hf_cellNum_dayMean <- aggregate(Count ~ Day, data = hf_cellNum, FUN = mean)

days_order_row <- c("Day_0", "Day_3", "Day_6", "Day_9", "Day_12",
  "Day_15", "Day_21", "Day_28")

hf_cellNum_dayMean <- hf_cellNum_dayMean %>% filter(Day %in% days_order_row)
hf_cellNum_dayMean$Day <- factor(hf_cellNum_dayMean$Day, levels = days_order_row)

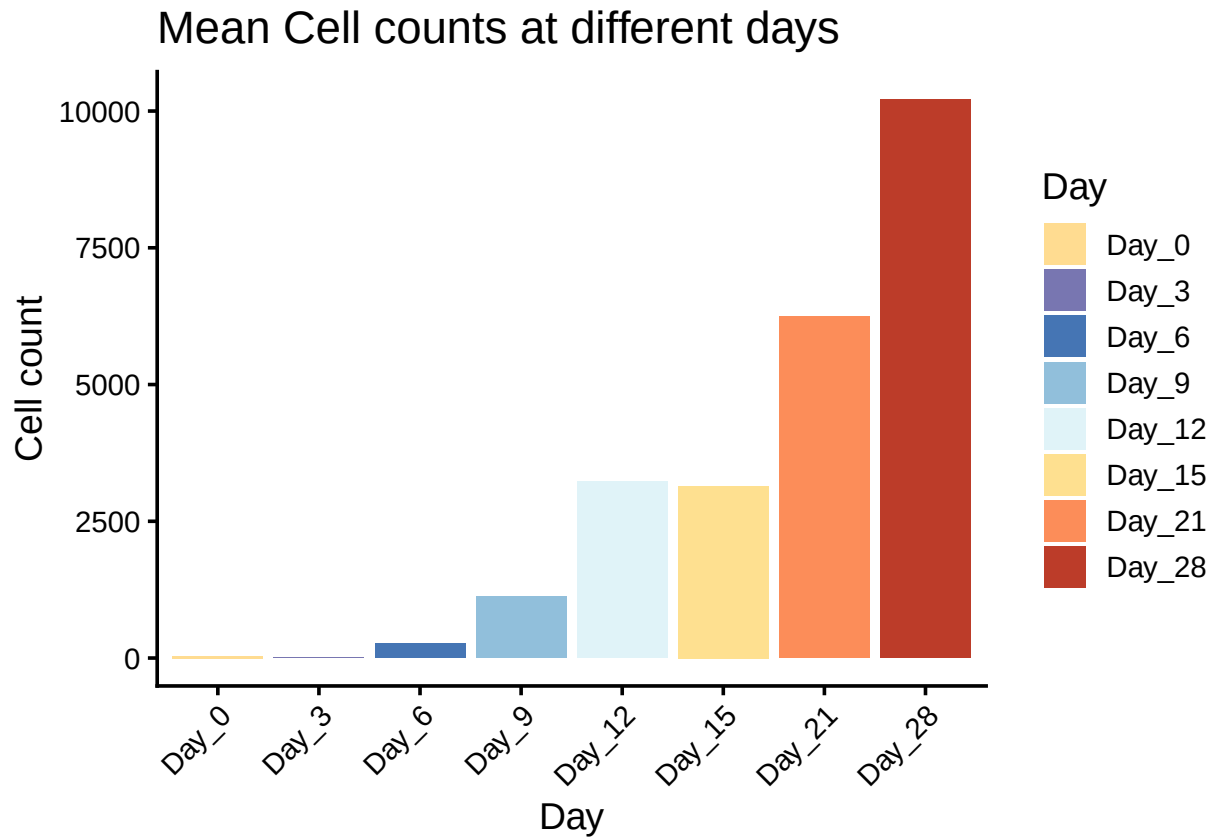
color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
  Day_9 = "#91bffd", Day_12 = "#e0f3f8", Day_15 = "#fee090",
  Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

p_hf_cellNum <- ggplot(hf_cellNum_dayMean, aes(x = Day, y = Count, fill = Day)) +
  geom_col(position = "dodge") +
  theme_minimal(base_size = 14) +
```

```

labs(title = "Mean Cell counts at different days",
     x = "Day",
     y = "Cell count") +
scale_fill_manual(values = color_settings_2) +
theme_classic(base_size = 14) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
p_hf_cellNum

```



7 Accumulated Mutation Num normalized by cell number

```

color_settings_randomCells <- c(hf1.d3 = "#7876B1FF", hf1.d6 = "#4575b4", hf1.d9 = "#91bfdb", hf1.d12 = "#e0f3f8",
                                hf2.d3 = "#7876B1FF", hf2.d6 = "#4575b4", hf2.d9 = "#91bfdb", hf2.d12 = "#e0f3f8")

color_settings_randomCells_day <- c(d6 = "#4575b4", d9 = "#91bfdb", d12 = "#e0f3f8", d15 = "#fee090", d21 = "#ff7f0e", d28 = "#d62728")

totalCells <- c(rownames(MEF_CSM_hf1), rownames(MEF_CSM_hf2))

selectRandomNum <- c(1000, 2000, 3000, 4000, 5000)

dataSelect <- c("hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28",
                "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")

```

```

days <- c("d6", "d9", "d12", "d15", "d21", "d28")
reps <- 20
data_random_counts <- as.data.frame(matrix(nrow = (length(selectRandomNum) * length(dataSelect) * reps)
colnames(data_random_counts) <- c("ID", "reps", "selectNum", "MutNum")

set.seed(2025)
t = 0
for (i in dataSelect){
  dataCells <- totalCells[grep(i, totalCells)]
  selectNum <- selectRandomNum
  for (j in 1:reps){
    for (k in selectNum){
      dataCells_random <- sample(dataCells, k, replace = TRUE)
      hf1_selected <- dataCells_random[dataCells_random %in% rownames(MEF_CSM_hf1)]
      hf2_selected <- dataCells_random[dataCells_random %in% rownames(MEF_CSM_hf2)]

      data_random_hf1 <- MEF_CSM_hf1[hf1_selected,]
      data_random_hf2 <- MEF_CSM_hf2[hf2_selected,]

      num_hf1 <- sum(colSums(data_random_hf1 == "Mut") > 0)
      num_hf2 <- sum(colSums(data_random_hf2 == "Mut") > 0)
      t = t + 1
      if(t %% 400 == 0){
        print(paste0("processed ", t, " iterations"))
      }
      data_random_counts[t,]$ID <- i
      data_random_counts[t,]$reps <- j
      data_random_counts[t,]$selectNum <- k
      data_random_counts[t,]$MutNum <- num_hf1 + num_hf2
    }
  }
}

## [1] "processed 400 iterations"
## [1] "processed 800 iterations"
## [1] "processed 1200 iterations"

data_random_counts <- separate(data_random_counts,col = "ID",into = c("Timecourse", "Day"), sep = "\\.")
data_random_counts$ID <- paste(data_random_counts$Timecourse, data_random_counts$Day, sep = ".")
data_random_counts$Day <- factor(data_random_counts$Day, levels = days)
data_random_counts$ID <- factor(data_random_counts$ID, levels = dataSelect)

p_random_somaticNum <- ggplot(data_random_counts,
  aes(x = Day, y = MutNum, color = factor(selectNum))) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  facet_wrap(~ Timecourse, ncol = 1, scales = "free_y") +
  ylab("Number of Somatic Mutations") +
  xlab("Day") +
  labs(color = "SelectNum") +
  scale_color_manual(values = c(
    "1000" = "#4d4d4d",
    "2000" = "#bababa",
    "3000" = "#fddbc7",

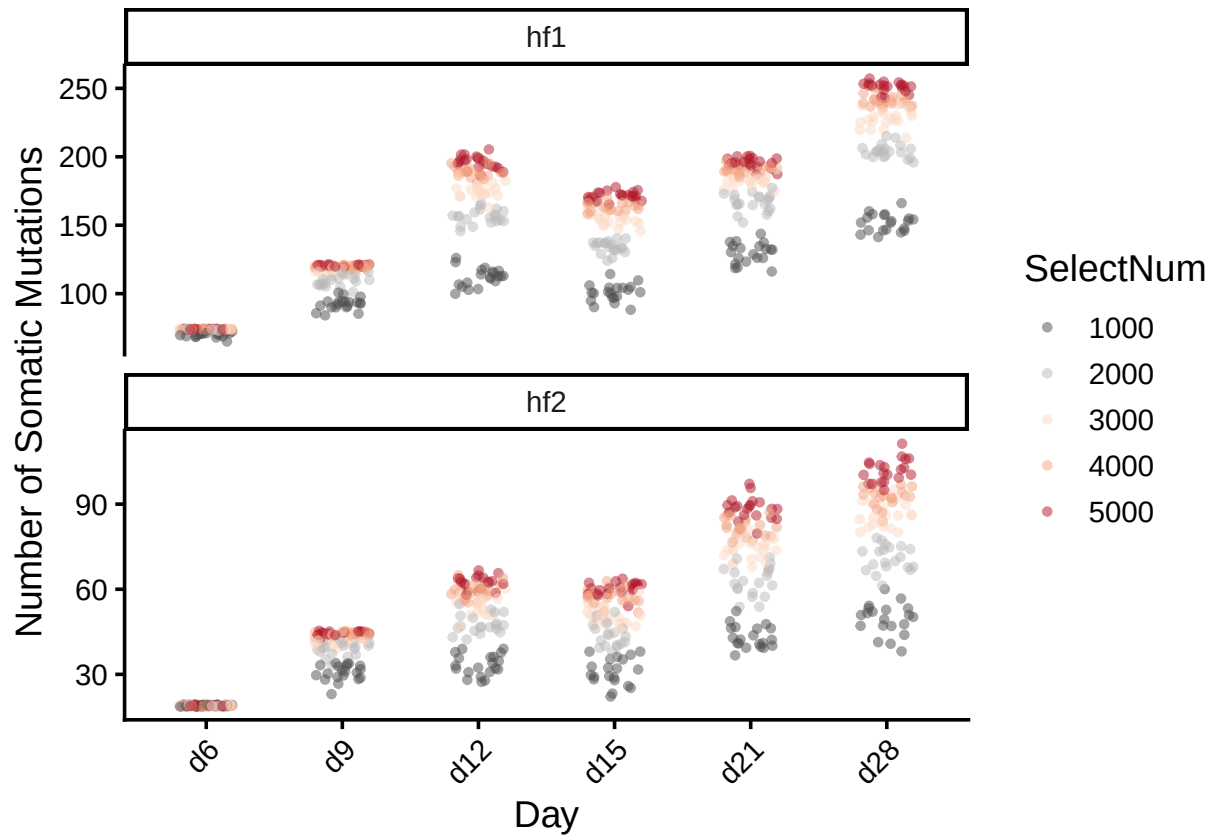
```

```

"4000" = "#f4a582",
"5000" = "#b2182b"
)) +
theme_classic(base_size = 14) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

p_random_somaticNum

```



8 MPGgenotypecheck

```

# check position
MPG_data <- read.csv("../Data/MEF/MPG_combined.csv")
MEF_detected_pos <- unique(union(MEF_SSM_hf1$position_cor, MEF_SSM_hf2$position_cor))
MEF_detected_pos[MEF_detected_pos %in% MPG_data$position_cor]
## [1] "10_28334861" "10_28384990" "11_103070309" "11_59698972" "11_80397502"
## [6] "12_59094027" "12_81947565" "13_44240902" "13_59527224" "15_102984869"
## [11] "15_31472794" "15_96305641" "15_96697854" "16_20124088" "16_37897824"
## [16] "16_38799521" "17_28695383" "1_162857222" "1_178800830" "1_61723999"
## [21] "2_169768737" "2_75700504" "3_19993055" "3_22093358" "3_55288508"
## [26] "4_103363073" "4_6980700" "5_36660914" "5_66596487" "6_128290883"
## [31] "6_90475221" "7_137044144" "7_141638786" "7_16356122" "8_45733398"
## [36] "8_45807946" "8_71690586" "8_94675603" "9_124330134" "9_21429385"

```

```

## [41] "9_60385333" "10_32940591" "10_78338121" "13_119610824" "13_49547531"
## [46] "14_103312866" "17_62865029" "18_60642899" "1_137967589" "1_57915729"
## [51] "2_121793360" "5_98880003" "6_120931717" "7_13047148" "7_133890940"
## [56] "8_45786901" "8_45802144" "9_100773522" "9_102813915"

MPG_data <- MPG_data[,c(3,8:44)]
MPG_data[MPG_data == ""] <- NA

MEF_SSM_refAlt <- rbind(MEF_SSM_hf1[,c("position_cor", "ref", "alt")], MEF_SSM_hf2[,c("position_cor", "
MPG_data_refAlt <- merge(MEF_SSM_refAlt, MPG_data, by = "position_cor", all.x = T)
MPG_data_refAlt[is.na(MPG_data_refAlt)] <- 0

MPG_data_refAlt[, 4:ncol(MPG_data_refAlt)] <- apply(MPG_data_refAlt[, 4:ncol(MPG_data_refAlt)], 2, fun
  refCheck <- MPG_data_refAlt[, "ref"]
  ifelse(col != 0 & col != refCheck, 1, 0)
})

MPG_data_refAlt <- MPG_data_refAlt[!duplicated(MPG_data_refAlt$position_cor),]
rownames(MPG_data_refAlt) <- MPG_data_refAlt$position_cor

MPG_data_refAlt$MutNum <- rowSums(MPG_data_refAlt[,4:40])
MPG_data_refAlt_sorted <- MPG_data_refAlt[order(-MPG_data_refAlt$MutNum), ]

colnames_order <- c("C57BL_6J", "C57BL_10J", "C57BL_6NJ", "C57BR_cdJ", "C57L_J",
  "AKR_J", "A_J", "BALB_cJ", "BTBRT_Itpr3_J", "BUB_BnJ",
  "C3H_HeH", "C3H_HeJ", "C58_J", "CAST_EiJ", "CBA_J",
  "DBA_1J", "DBA_2J", "FVB_NJ", "I_LnJ", "KK_HlJ",
  "LEWES_EiJ", "LP_J", "MOLF_EiJ",
  "NOD_ShilTJ", "NZB_B1NJ", "NZO_H1LtJ", "NZW_LacJ",
  "PWK_PhJ", "RF_J", "SEA_GnJ", "SPRET_EiJ", "ST_bJ", "WSB_EiJ",
  "ZALENDE_EiJ", "X129P2_01aHsd", "X129S1_SvImJ", "X129S5SvEv_Brd")

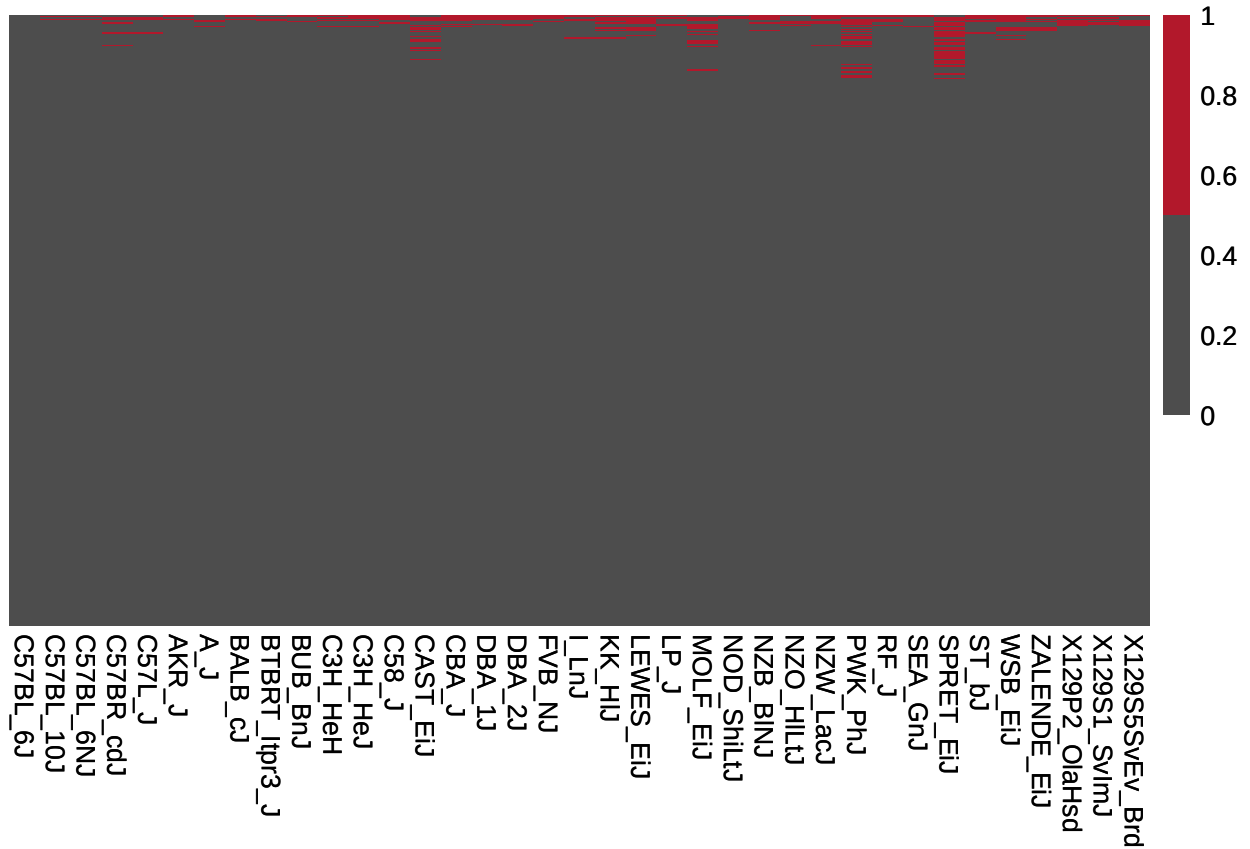
colnames_order[!colnames_order %in% colnames(MPG_data_refAlt_sorted)]
## character(0)

heatmap_data <- MPG_data_refAlt_sorted[, colnames_order]

custom_colors <- c("#4d4d4d", "#b2182b")

p_heatmapMPG <- pheatmap(heatmap_data, show_rownames = FALSE, cluster_rows = FALSE, cluster_cols = FALSE)
p_heatmapMPG

```

9 Calculationg mutations of each cell and cells of each mutation

```
data_MutPerCell_hf1 <- data.frame(
  CellTag = rownames(MEF_CSM_hf1),
  MutNum_somatic = rowSums(MEF_CSM_hf1 == "Mut"),
  WTNum_somatic = rowSums(MEF_CSM_hf1 == "WT")
)

data_CellPerMut_hf1 <- data.frame(
  Location = colnames(MEF_CSM_hf1),
  CellMutNum = colSums(MEF_CSM_hf1 == "Mut"),
  CellWTNum = colSums(MEF_CSM_hf1 == "WT")
)

data_MutPerCell_hf2 <- data.frame(
  CellTag = rownames(MEF_CSM_hf2),
  MutNum_somatic = rowSums(MEF_CSM_hf2 == "Mut"),
  WTNum_somatic = rowSums(MEF_CSM_hf2 == "WT")
)

data_CellPerMut_hf2 <- data.frame(
  Location = colnames(MEF_CSM_hf2),
  CellMutNum = colSums(MEF_CSM_hf2 == "Mut"),
```

```

CellWTNum = colSums(MEF_CSM_hf2 == "WT")
)

```

10 calMulti

```

# [Cell * Time] = [Cell * Mut] * [Mut * Time]
time_types <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")
cell_num = 1
mut_num = 1

# _hf1
cells_candidate_hf1 <- data_MutPerCell_hf1[data_MutPerCell_hf1$MutNum_somatic >= cell_num,]$CellTag
cells_candidate_hf1 <- cells_candidate_hf1[cells_candidate_hf1 %in% MEF_Meta_hf1$Cellbarcode]
mut_candidate_hf1 <- data_CellPerMut_hf1[data_CellPerMut_hf1$CellMutNum >= mut_num,]$Location

cell_time_num_hf1 <- sapply(col_hf1_names, function(i) sum(str_count(cells_candidate_hf1, i)))
total_cells_hf1 <- sum(cell_time_num_hf1)
cell_time_num_ratio_hf1 <- cell_time_num_hf1/total_cells_hf1 + 2

# data_Cell_Mut
data_cell_mut_raw_hf1 <- MEF_CSM_hf1[cells_candidate_hf1, mut_candidate_hf1]

# data_Mut_Time
data_mut_time_hf1 <- as.data.frame(matrix(nrow = length(mut_candidate_hf1), ncol = length(time_types)),
rownames(data_mut_time_hf1) <- mut_candidate_hf1
colnames(data_mut_time_hf1) <- time_types

for (i in 1:nrow(data_mut_time_hf1)){
  mut_tmp <- rownames(data_mut_time_hf1)[i]
  cell_tmp <- retrieveCellsByLocation(data_cell_mut_raw_hf1, mut_tmp)
  selected_meta_tags <- MEF_Meta_hf1[MEF_Meta_hf1$Cellbarcode %in% cell_tmp$Mut,]$Day
  data_mut_time_hf1[i,]$Day_6 <- sum(str_count(selected_meta_tags, "Day_6"))
  data_mut_time_hf1[i,]$Day_9 <- sum(str_count(selected_meta_tags, "Day_9"))
  data_mut_time_hf1[i,]$Day_12 <- sum(str_count(selected_meta_tags, "Day_12"))
  data_mut_time_hf1[i,]$Day_15 <- sum(str_count(selected_meta_tags, "Day_15"))
  data_mut_time_hf1[i,]$Day_21 <- sum(str_count(selected_meta_tags, "Day_21"))
  data_mut_time_hf1[i,]$Day_28 <- sum(str_count(selected_meta_tags, "Day_28"))
}

data_mut_time_hf1_nor <- sweep(data_mut_time_hf1, 2, log2(cell_time_num_ratio_hf1[3:8]), "/")

# _hf2
cells_candidate_hf2 <- data_MutPerCell_hf2[data_MutPerCell_hf2$MutNum_somatic >= cell_num,]$CellTag
cells_candidate_hf2 <- cells_candidate_hf2[cells_candidate_hf2 %in% MEF_Meta_hf2$Cellbarcode]
mut_candidate_hf2 <- data_CellPerMut_hf2[data_CellPerMut_hf2$CellMutNum >= mut_num,]$Location

cell_time_num_hf2 <- sapply(col_hf2_names, function(i) sum(str_count(cells_candidate_hf2, i)))
total_cells_hf2 <- sum(cell_time_num_hf2)
cell_time_num_ratio_hf2 <- cell_time_num_hf2/total_cells_hf2 + 2

# data_Cell_Mut

```

```

data_cell_mut_raw_hf2 <- MEF_CSM_hf2[cells_candidate_hf2, muts_candidate_hf2]

# data_Mut_Time
data_mut_time_hf2 <- as.data.frame(matrix(nrow = length(muts_candidate_hf2), ncol = length(time_types)))
rownames(data_mut_time_hf2) <- muts_candidate_hf2
colnames(data_mut_time_hf2) <- time_types

for (i in 1:nrow(data_mut_time_hf2)){
  mut_tmp <- rownames(data_mut_time_hf2)[i]
  cell_tmp <- retrieveCellsByLocation(data_cell_mut_raw_hf2, mut_tmp)
  selected_meta_tags <- MEF_Meta_hf2[MEF_Meta_hf2$Cellbarcode %in% cell_tmp$Mut,]$Day
  data_mut_time_hf2[i,]$Day_6 <- sum(str_count(selected_meta_tags, "Day_6"))
  data_mut_time_hf2[i,]$Day_9 <- sum(str_count(selected_meta_tags, "Day_9"))
  data_mut_time_hf2[i,]$Day_12 <- sum(str_count(selected_meta_tags, "Day_12"))
  data_mut_time_hf2[i,]$Day_15 <- sum(str_count(selected_meta_tags, "Day_15"))
  data_mut_time_hf2[i,]$Day_21 <- sum(str_count(selected_meta_tags, "Day_21"))
  data_mut_time_hf2[i,]$Day_28 <- sum(str_count(selected_meta_tags, "Day_28"))
}

data_mut_time_hf2_nor <- sweep(data_mut_time_hf2, 2, log2(cell_time_num_ratio_hf2[3:8]), "/")

res_index_hf1 <- cal_scMutScore(data_cell_mut_raw_hf1, data_mut_time_hf1_nor, MEF_Meta_hf1)
## [1] 4371 297
## [1] 297 6
res_index_hf2 <- cal_scMutScore(data_cell_mut_raw_hf2, data_mut_time_hf2_nor, MEF_Meta_hf2)
## [1] 2627 156
## [1] 156 6
res_index <- rbind(res_index_hf1$score, res_index_hf2$score)

res_index$cb <- rownames(res_index)
res_index <- separate(res_index, col = "cb", into = c("celltype", "cell"), sep = "_")
res_index$Tag <- res_index$celltype
res_index <- separate(res_index, col = "celltype", into = c("rep", "Day"), sep = "\\.")

res_index$Tag <- factor(res_index$Tag, levels = c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28",
"hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28"))

res_index$Day <- factor(res_index$Day, levels = c("d0", "d3", "d6", "d9", "d12", "d15", "d21", "d28"))

res_index <- res_index[res_index$Diversity > 0, ]

color_settings <- c(d0 = "#FFDC91FF", d3 = "#7876B1FF", d6 = "#4575b4",
d9 = "#91b9db", d12 = "#e0f3f8", d15 = "#fee090",
d21 = "#fc8d59", d28 = "#BC3C29FF")

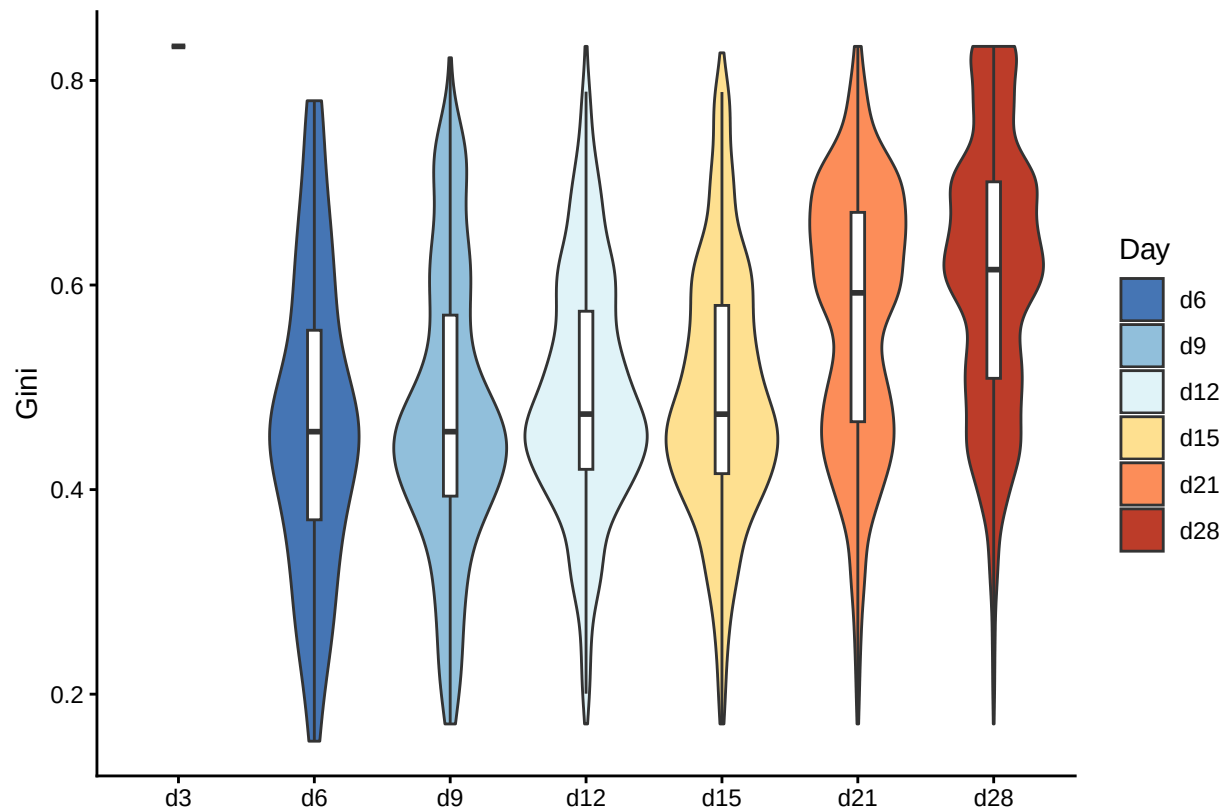
color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
Day_9 = "#91b9db", Day_12 = "#e0f3f8", Day_15 = "#fee090",
Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

res_index <- res_index[!is.na(res_index$scMutScore),]

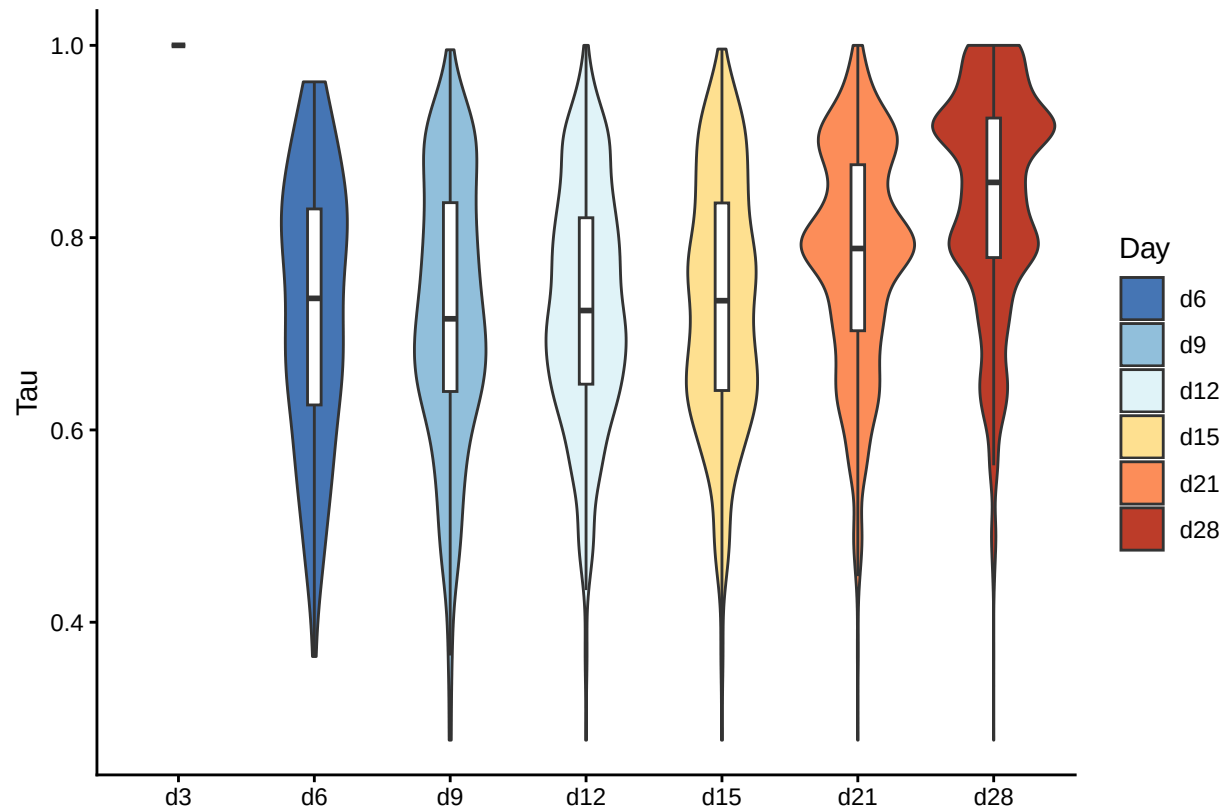
ggplot(data = res_index, mapping = aes(x = Day, y = Gini, fill = Day)) +
  geom_violin(trim = TRUE) +

```

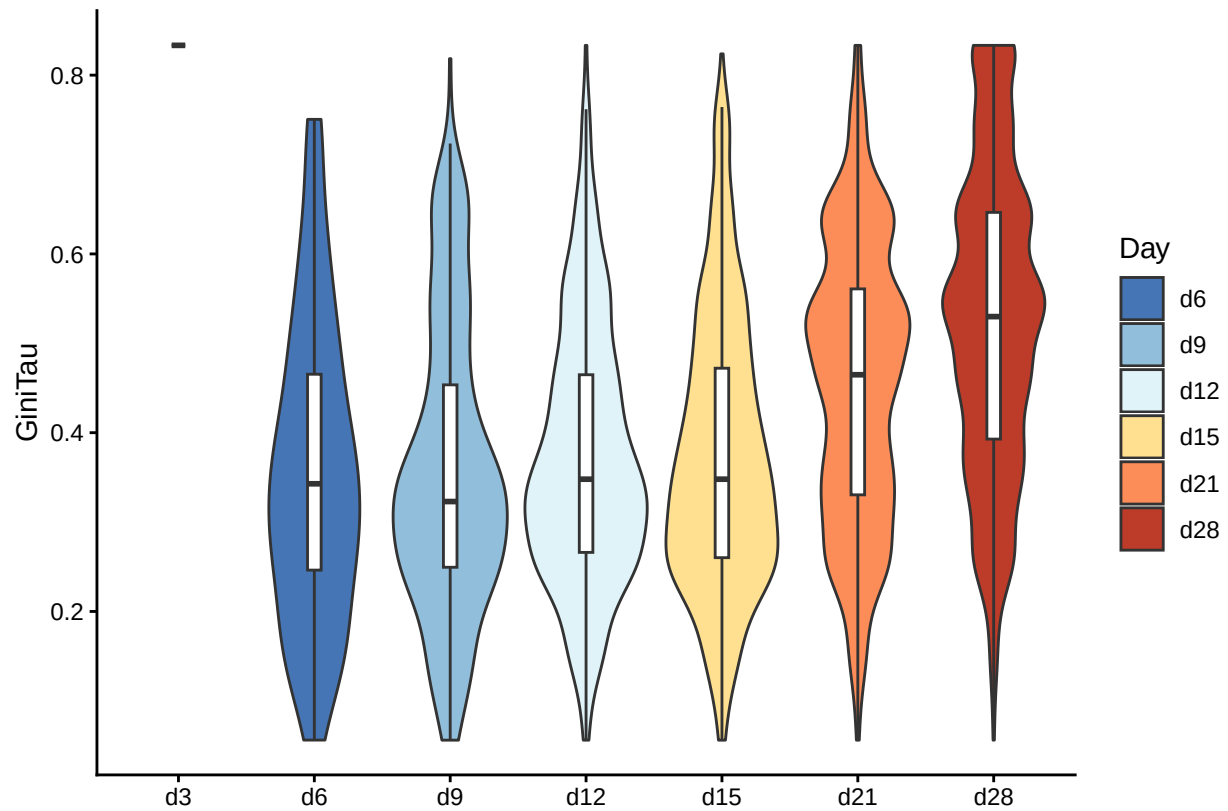
```
geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
labs(x = "", y = "Gini") +
scale_fill_manual(values = color_settings) +
theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



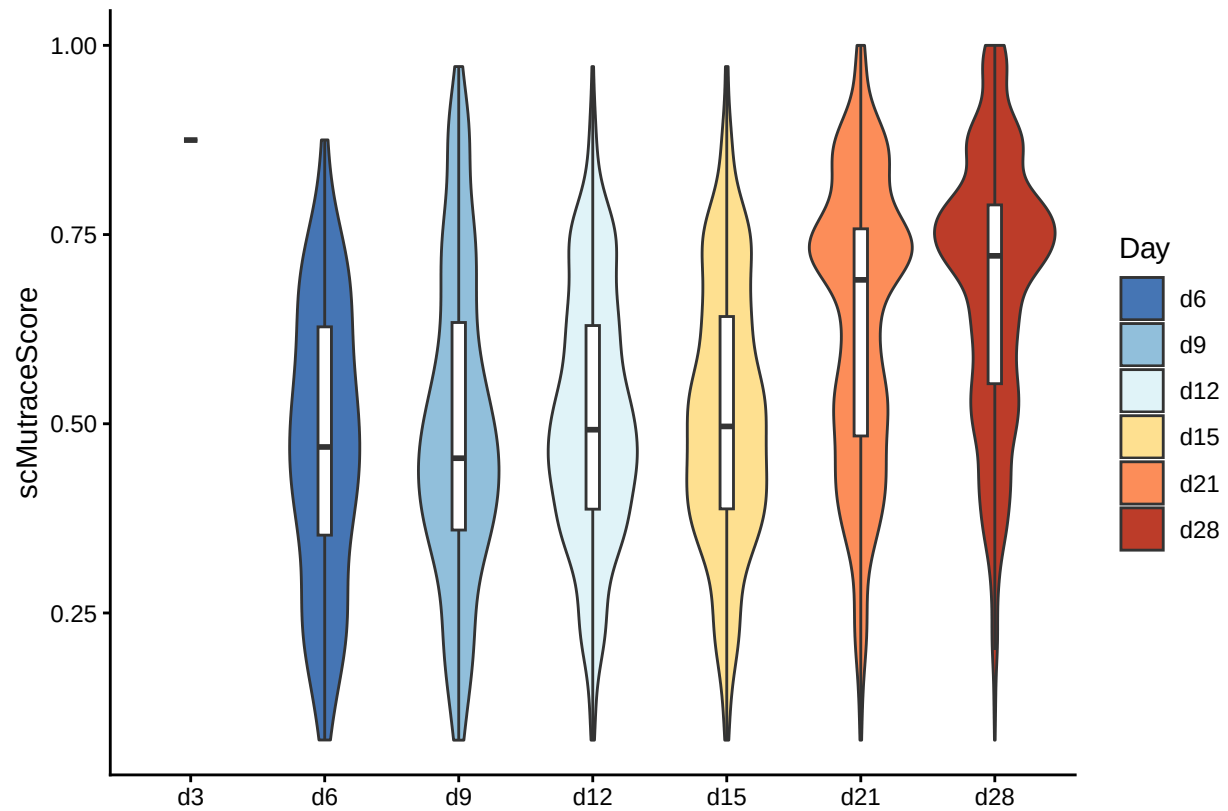
```
ggplot(data = res_index, mapping = aes(x = Day, y = Tau, fill = Day)) +
geom_violin(trim = TRUE) +
geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
labs(x = "", y = "Tau") +
scale_fill_manual(values = color_settings) +
theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



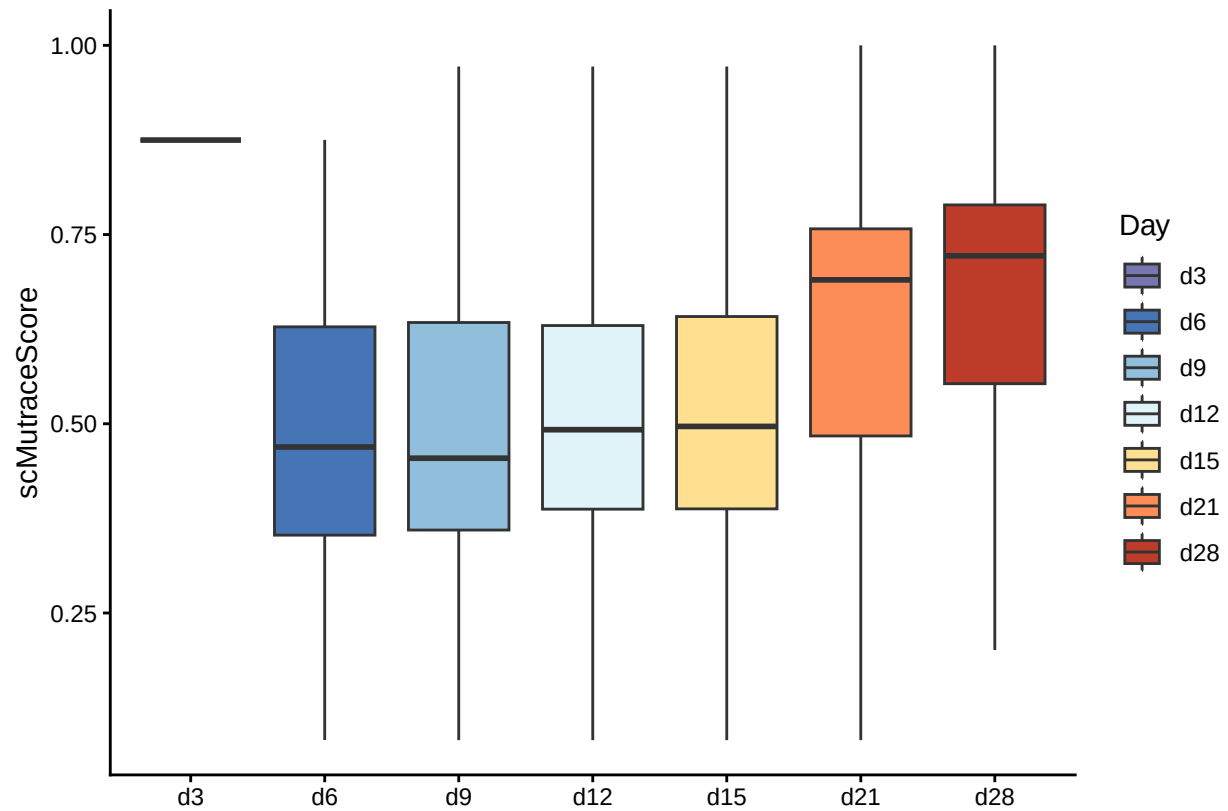
```
ggplot(data = res_index, mapping = aes(x = Day, y = GiniTau, fill = Day)) +
  geom_violin(trim = TRUE) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  labs(x = "", y = "GiniTau") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



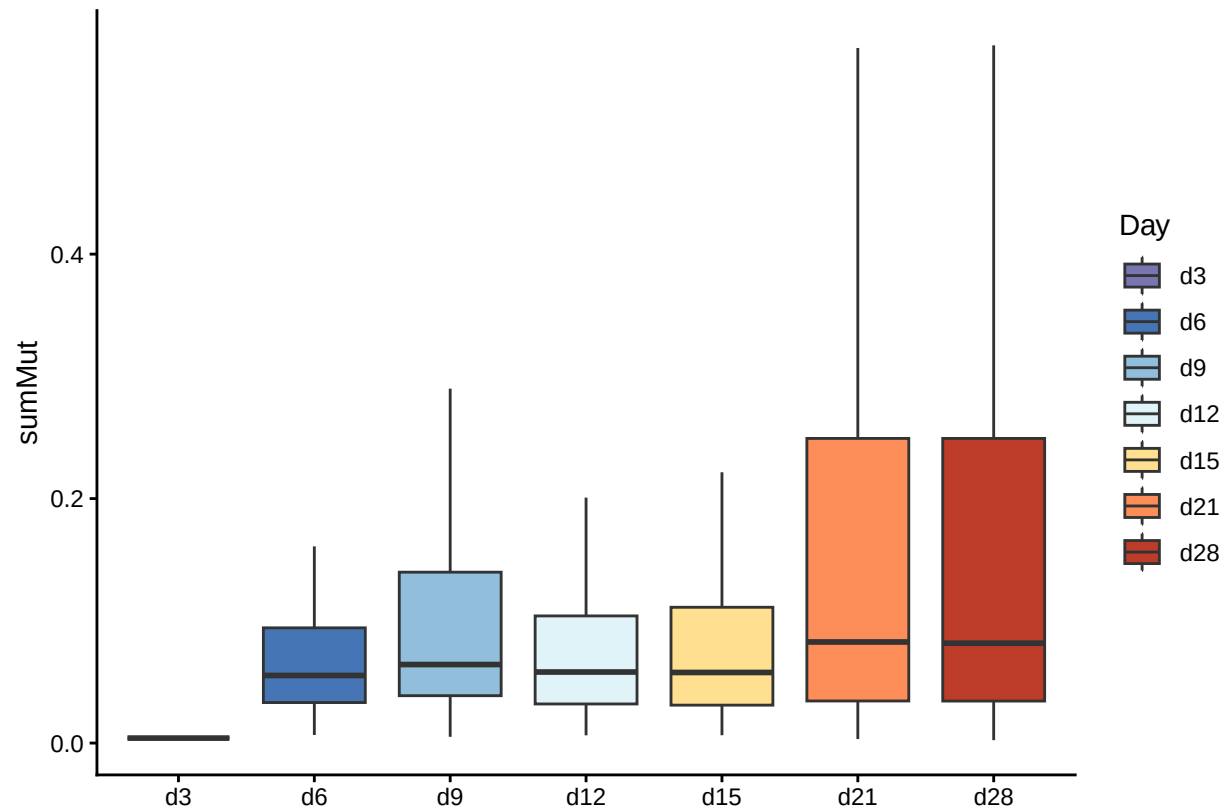
```
ggplot(data = res_index, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



```
ggplot(data = res_index, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
```

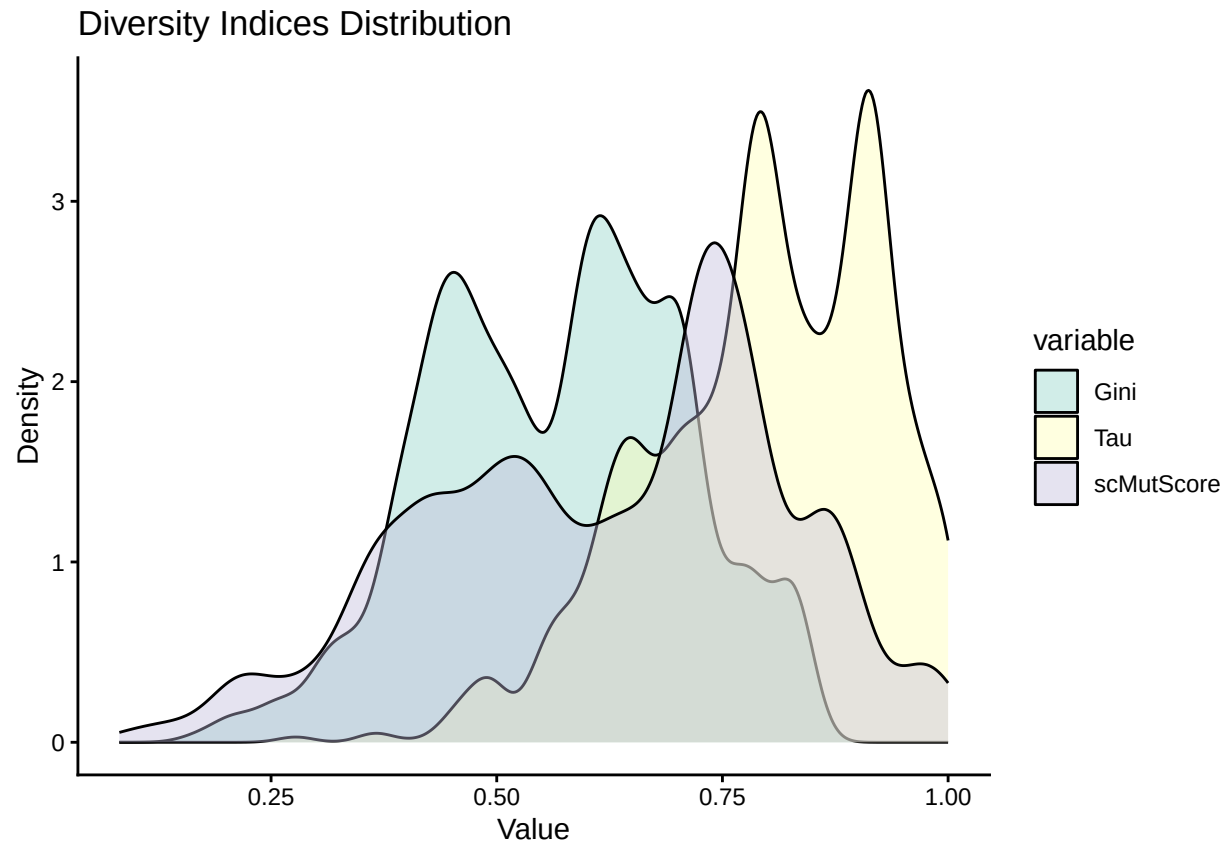


```
ggplot(data = res_index, mapping = aes(x = Day, y = scMutScoreQun, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "sumMut") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
```

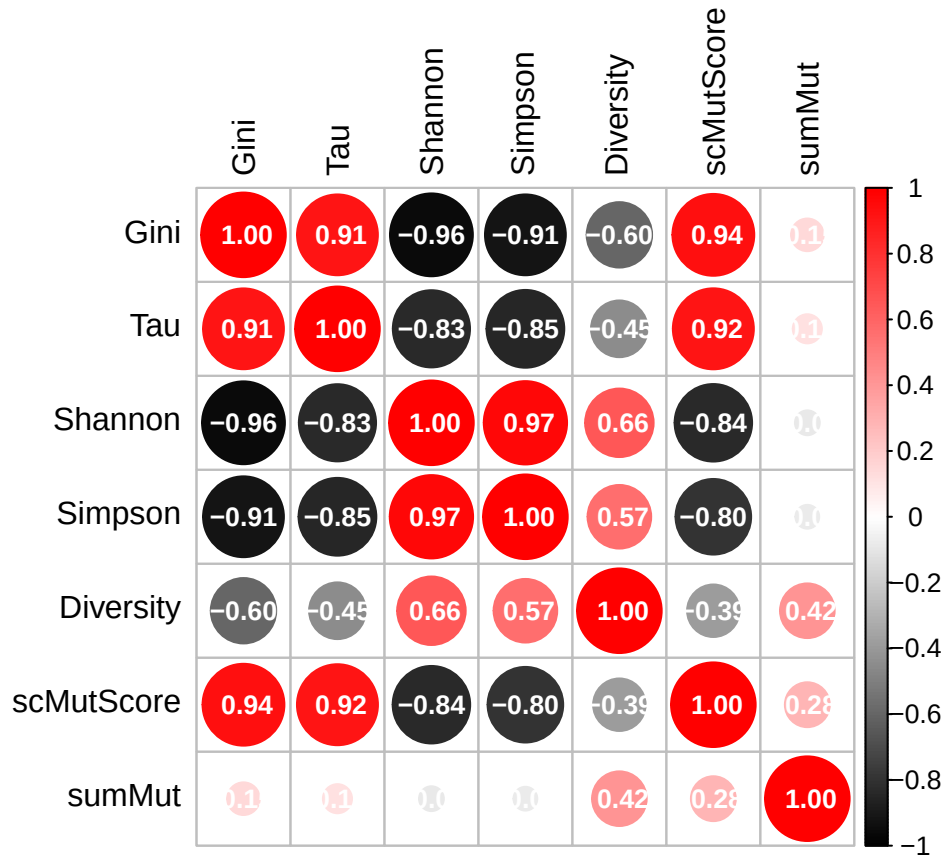



```
res_index_long <- melt(res_index[,c("Gini", "Tau", "scMutScore")])
## No id variables; using all as measure variables

ggplot(res_index_long, aes(x=value, fill=variable)) +
  geom_density(alpha=0.4) +
  theme_classic() +
  labs(x="Value", y="Density", title="Diversity Indices Distribution") +
  scale_fill_brewer(palette="Set3")
```



```
corrplot(cor(res_index[,c("Gini", "Tau", "Shannon", "Simpson", "Diversity", "scMutScore", "sumMut")], us
method="circle", col = colorRampPalette(c("black", "white", "red"))(200),
addCoef.col = "white", tl.col = "black", number.cex = 0.8)
```



11 scMutScore Model Validation

```
set.seed(2025)

# 1. init
data_check_CM <- data_cell_mut_raw_hf1
data_check_Meta <- MEF_Meta_hf1[MEF_Meta_hf1$Cellbarcode %in% rownames(data_check_CM),]
col_check_names <- c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28")
time_types <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")

cell_num <- 1
mut_num <- 1
mut_loss_ratio <- 0.5
cell_loss_ratio <- 0.5
iteration_times <- 100

# list for save data
data_scMutModel_list <- vector("list", iteration_times)

num_cells <- nrow(data_check_CM)
num_mutations <- ncol(data_check_CM)
cells_all <- rownames(data_check_CM)
mut_all <- colnames(data_check_CM)
```

```

# 2. loop
for (iter in seq_len(iteration_times)) {

  # 2.1 randomly loss info
  loss_cell_indices <- sample(seq_len(num_cells), ceiling(cell_loss_ratio * num_cells))
  loss_mut_indices <- sample(seq_len(num_mutations), ceiling(mut_loss_ratio * num_mutations))

  C_partial_missing <- data_check_CM
  # new data
  if(length(loss_cell_indices) > 0 && length(loss_mut_indices) > 0){
    C_partial_missing[loss_cell_indices, loss_mut_indices] <- "NoCall"
  }

  # 2.2 count cell/mut number
  MutPerCell <- rowSums(C_partial_missing == "Mut")
  CellPerMut <- colSums(C_partial_missing == "Mut")

  cells_candidate <- cells_all[MutPerCell >= cell_num & cells_all %in% data_check_Meta$Cellbarcode]
  muts_candidate <- muts_all[CellPerMut >= mut_num]

  message(paste0("Iteration: ", iter, " loss_cell_indices: ", length(loss_cell_indices), " loss_mut_ind",
  message(paste0("Iteration: ", iter, " cells_candidate: ", length(cells_candidate), " muts_candidate: ")

  data_cell_mut_raw_check <- C_partial_missing[cells_candidate, muts_candidate, drop = TRUE]

  # 2.3 Mut × Time
  cell_time_num <- sapply(col_check_names, function(i) sum(str_count(cells_candidate, i)))
  total_cells <- sum(cell_time_num)
  cell_time_ratio <- cell_time_num / total_cells + 2 # +2 Avoidance 0

  # init Mut × Time
  M_partial_missing <- matrix(0, nrow = length(muts_candidate), ncol = length(time_types),
                             dimnames = list(muts_candidate, time_types))

  # Calculate the number of each mutation at each time
  for (t in seq_along(time_types)) {
    time_label <- time_types[t]
    # Find out in which cells each mutation occurs
    mut_cell_list <- apply(data_cell_mut_raw_check, 2, function(x) which(x == "Mut"))
    # Count the number of times each mutation occurs at that point in time
    for (mut in muts_candidate) {
      cells_with_mut <- rownames(data_cell_mut_raw_check)[data_cell_mut_raw_check[, mut] == "Mut"]
      selected_days <- data_check_Meta$Day[data_check_Meta$Cellbarcode %in% cells_with_mut]
      M_partial_missing[mut, t] <- sum(str_count(selected_days, time_label))
    }
  }

  # 2.4 normalized Mut × Time
  M_partial_missing_norm <- sweep(M_partial_missing, 2, log2(cell_time_ratio[3:8]), "/")

  # 2.5 scMutScore
  res_index_tmp <- cal_scMutScore(data_cell_mut_raw_check, M_partial_missing_norm, data_check_Meta)$score
  res_index_tmp$iteration <- paste0("iter", iter)
}

```

```

res_index_tmp <- res_index_tmp[res_index_tmp$Diversity > 0, ]

data_scMutModel_list[[iter]] <- res_index_tmp
}
## Iteration: 1 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 1 cells_candidate: 3452 muts_candidate: 287
## [1] 3452 287
## [1] 287 6
## Iteration: 2 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 2 cells_candidate: 3307 muts_candidate: 293
## [1] 3307 293
## [1] 293 6
## Iteration: 3 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 3 cells_candidate: 3402 muts_candidate: 287
## [1] 3402 287
## [1] 287 6
## Iteration: 4 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 4 cells_candidate: 3330 muts_candidate: 290
## [1] 3330 290
## [1] 290 6
## Iteration: 5 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 5 cells_candidate: 3451 muts_candidate: 292
## [1] 3451 292
## [1] 292 6
## Iteration: 6 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 6 cells_candidate: 3324 muts_candidate: 296
## [1] 3324 296
## [1] 296 6
## Iteration: 7 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 7 cells_candidate: 3345 muts_candidate: 291
## [1] 3345 291
## [1] 291 6
## Iteration: 8 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 8 cells_candidate: 3516 muts_candidate: 290
## [1] 3516 290
## [1] 290 6
## Iteration: 9 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 9 cells_candidate: 3324 muts_candidate: 292
## [1] 3324 292
## [1] 292 6
## Iteration: 10 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 10 cells_candidate: 3449 muts_candidate: 291
## [1] 3449 291
## [1] 291 6
## Iteration: 11 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 11 cells_candidate: 3451 muts_candidate: 288
## [1] 3451 288
## [1] 288 6
## Iteration: 12 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 12 cells_candidate: 3446 muts_candidate: 291
## [1] 3446 291
## [1] 291 6
## Iteration: 13 loss_cell_indices: 2186 loss_mut_indices: 149

```

```

## Iteration: 13 cells_candidate: 3444 muts_candidate: 293
## [1] 3444 293
## [1] 293 6
## Iteration: 14 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 14 cells_candidate: 3402 muts_candidate: 293
## [1] 3402 293
## [1] 293 6
## Iteration: 15 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 15 cells_candidate: 3291 muts_candidate: 288
## [1] 3291 288
## [1] 288 6
## Iteration: 16 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 16 cells_candidate: 3393 muts_candidate: 293
## [1] 3393 293
## [1] 293 6
## Iteration: 17 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 17 cells_candidate: 3225 muts_candidate: 294
## [1] 3225 294
## [1] 294 6
## Iteration: 18 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 18 cells_candidate: 3418 muts_candidate: 290
## [1] 3418 290
## [1] 290 6
## Iteration: 19 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 19 cells_candidate: 3449 muts_candidate: 296
## [1] 3449 296
## [1] 296 6
## Iteration: 20 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 20 cells_candidate: 3221 muts_candidate: 290
## [1] 3221 290
## [1] 290 6
## Iteration: 21 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 21 cells_candidate: 3246 muts_candidate: 293
## [1] 3246 293
## [1] 293 6
## Iteration: 22 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 22 cells_candidate: 3334 muts_candidate: 292
## [1] 3334 292
## [1] 292 6
## Iteration: 23 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 23 cells_candidate: 3167 muts_candidate: 295
## [1] 3167 295
## [1] 295 6
## Iteration: 24 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 24 cells_candidate: 3211 muts_candidate: 292
## [1] 3211 292
## [1] 292 6
## Iteration: 25 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 25 cells_candidate: 3502 muts_candidate: 291
## [1] 3502 291
## [1] 291 6
## Iteration: 26 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 26 cells_candidate: 3241 muts_candidate: 292

```

```

## [1] 3241 292
## [1] 292 6
## Iteration: 27 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 27 cells_candidate: 3384 muts_candidate: 292
## [1] 3384 292
## [1] 292 6
## Iteration: 28 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 28 cells_candidate: 3368 muts_candidate: 291
## [1] 3368 291
## [1] 291 6
## Iteration: 29 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 29 cells_candidate: 3437 muts_candidate: 293
## [1] 3437 293
## [1] 293 6
## Iteration: 30 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 30 cells_candidate: 3338 muts_candidate: 294
## [1] 3338 294
## [1] 294 6
## Iteration: 31 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 31 cells_candidate: 3394 muts_candidate: 293
## [1] 3394 293
## [1] 293 6
## Iteration: 32 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 32 cells_candidate: 3410 muts_candidate: 294
## [1] 3410 294
## [1] 294 6
## Iteration: 33 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 33 cells_candidate: 3371 muts_candidate: 290
## [1] 3371 290
## [1] 290 6
## Iteration: 34 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 34 cells_candidate: 3504 muts_candidate: 294
## [1] 3504 294
## [1] 294 6
## Iteration: 35 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 35 cells_candidate: 3368 muts_candidate: 290
## [1] 3368 290
## [1] 290 6
## Iteration: 36 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 36 cells_candidate: 3378 muts_candidate: 295
## [1] 3378 295
## [1] 295 6
## Iteration: 37 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 37 cells_candidate: 3323 muts_candidate: 294
## [1] 3323 294
## [1] 294 6
## Iteration: 38 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 38 cells_candidate: 3321 muts_candidate: 291
## [1] 3321 291
## [1] 291 6
## Iteration: 39 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 39 cells_candidate: 3420 muts_candidate: 290
## [1] 3420 290

```

```

## [1] 290 6
## Iteration: 40 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 40 cells_candidate: 3215 muts_candidate: 294
## [1] 3215 294
## [1] 294 6
## Iteration: 41 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 41 cells_candidate: 3474 muts_candidate: 291
## [1] 3474 291
## [1] 291 6
## Iteration: 42 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 42 cells_candidate: 3525 muts_candidate: 285
## [1] 3525 285
## [1] 285 6
## Iteration: 43 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 43 cells_candidate: 3364 muts_candidate: 288
## [1] 3364 288
## [1] 288 6
## Iteration: 44 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 44 cells_candidate: 3305 muts_candidate: 294
## [1] 3305 294
## [1] 294 6
## Iteration: 45 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 45 cells_candidate: 3362 muts_candidate: 292
## [1] 3362 292
## [1] 292 6
## Iteration: 46 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 46 cells_candidate: 3323 muts_candidate: 294
## [1] 3323 294
## [1] 294 6
## Iteration: 47 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 47 cells_candidate: 3417 muts_candidate: 291
## [1] 3417 291
## [1] 291 6
## Iteration: 48 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 48 cells_candidate: 3456 muts_candidate: 291
## [1] 3456 291
## [1] 291 6
## Iteration: 49 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 49 cells_candidate: 3506 muts_candidate: 292
## [1] 3506 292
## [1] 292 6
## Iteration: 50 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 50 cells_candidate: 3379 muts_candidate: 295
## [1] 3379 295
## [1] 295 6
## Iteration: 51 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 51 cells_candidate: 3239 muts_candidate: 294
## [1] 3239 294
## [1] 294 6
## Iteration: 52 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 52 cells_candidate: 3373 muts_candidate: 291
## [1] 3373 291
## [1] 291 6

```



```

## Iteration: 53 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 53 cells_candidate: 3338 muts_candidate: 287
## [1] 3338 287
## [1] 287 6
## Iteration: 54 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 54 cells_candidate: 3232 muts_candidate: 292
## [1] 3232 292
## [1] 292 6
## Iteration: 55 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 55 cells_candidate: 3425 muts_candidate: 289
## [1] 3425 289
## [1] 289 6
## Iteration: 56 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 56 cells_candidate: 3218 muts_candidate: 289
## [1] 3218 289
## [1] 289 6
## Iteration: 57 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 57 cells_candidate: 3285 muts_candidate: 294
## [1] 3285 294
## [1] 294 6
## Iteration: 58 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 58 cells_candidate: 3260 muts_candidate: 292
## [1] 3260 292
## [1] 292 6
## Iteration: 59 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 59 cells_candidate: 3532 muts_candidate: 290
## [1] 3532 290
## [1] 290 6
## Iteration: 60 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 60 cells_candidate: 3368 muts_candidate: 292
## [1] 3368 292
## [1] 292 6
## Iteration: 61 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 61 cells_candidate: 3414 muts_candidate: 295
## [1] 3414 295
## [1] 295 6
## Iteration: 62 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 62 cells_candidate: 3468 muts_candidate: 289
## [1] 3468 289
## [1] 289 6
## Iteration: 63 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 63 cells_candidate: 3327 muts_candidate: 289
## [1] 3327 289
## [1] 289 6
## Iteration: 64 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 64 cells_candidate: 3384 muts_candidate: 288
## [1] 3384 288
## [1] 288 6
## Iteration: 65 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 65 cells_candidate: 3232 muts_candidate: 290
## [1] 3232 290
## [1] 290 6
## Iteration: 66 loss_cell_indices: 2186 loss_mut_indices: 149

```

```

## Iteration: 66 cells_candidate: 3428 muts_candidate: 293
## [1] 3428 293
## [1] 293 6
## Iteration: 67 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 67 cells_candidate: 3204 muts_candidate: 290
## [1] 3204 290
## [1] 290 6
## Iteration: 68 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 68 cells_candidate: 3336 muts_candidate: 290
## [1] 3336 290
## [1] 290 6
## Iteration: 69 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 69 cells_candidate: 3297 muts_candidate: 288
## [1] 3297 288
## [1] 288 6
## Iteration: 70 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 70 cells_candidate: 3453 muts_candidate: 294
## [1] 3453 294
## [1] 294 6
## Iteration: 71 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 71 cells_candidate: 3295 muts_candidate: 293
## [1] 3295 293
## [1] 293 6
## Iteration: 72 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 72 cells_candidate: 3283 muts_candidate: 291
## [1] 3283 291
## [1] 291 6
## Iteration: 73 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 73 cells_candidate: 3265 muts_candidate: 293
## [1] 3265 293
## [1] 293 6
## Iteration: 74 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 74 cells_candidate: 3445 muts_candidate: 290
## [1] 3445 290
## [1] 290 6
## Iteration: 75 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 75 cells_candidate: 3306 muts_candidate: 293
## [1] 3306 293
## [1] 293 6
## Iteration: 76 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 76 cells_candidate: 3277 muts_candidate: 287
## [1] 3277 287
## [1] 287 6
## Iteration: 77 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 77 cells_candidate: 3423 muts_candidate: 288
## [1] 3423 288
## [1] 288 6
## Iteration: 78 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 78 cells_candidate: 3331 muts_candidate: 292
## [1] 3331 292
## [1] 292 6
## Iteration: 79 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 79 cells_candidate: 3371 muts_candidate: 290

```

```

## [1] 3371 290
## [1] 290 6
## Iteration: 80 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 80 cells_candidate: 3364 muts_candidate: 289
## [1] 3364 289
## [1] 289 6
## Iteration: 81 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 81 cells_candidate: 3534 muts_candidate: 292
## [1] 3534 292
## [1] 292 6
## Iteration: 82 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 82 cells_candidate: 3522 muts_candidate: 290
## [1] 3522 290
## [1] 290 6
## Iteration: 83 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 83 cells_candidate: 3356 muts_candidate: 288
## [1] 3356 288
## [1] 288 6
## Iteration: 84 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 84 cells_candidate: 3363 muts_candidate: 292
## [1] 3363 292
## [1] 292 6
## Iteration: 85 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 85 cells_candidate: 3351 muts_candidate: 289
## [1] 3351 289
## [1] 289 6
## Iteration: 86 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 86 cells_candidate: 3360 muts_candidate: 290
## [1] 3360 290
## [1] 290 6
## Iteration: 87 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 87 cells_candidate: 3358 muts_candidate: 291
## [1] 3358 291
## [1] 291 6
## Iteration: 88 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 88 cells_candidate: 3482 muts_candidate: 291
## [1] 3482 291
## [1] 291 6
## Iteration: 89 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 89 cells_candidate: 3390 muts_candidate: 291
## [1] 3390 291
## [1] 291 6
## Iteration: 90 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 90 cells_candidate: 3309 muts_candidate: 289
## [1] 3309 289
## [1] 289 6
## Iteration: 91 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 91 cells_candidate: 3510 muts_candidate: 290
## [1] 3510 290
## [1] 290 6
## Iteration: 92 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 92 cells_candidate: 3434 muts_candidate: 293
## [1] 3434 293

```

```

## [1] 293    6
## Iteration: 93 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 93 cells_candidate: 3429 muts_candidate: 292
## [1] 3429  292
## [1] 292    6
## Iteration: 94 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 94 cells_candidate: 3228 muts_candidate: 289
## [1] 3228  289
## [1] 289    6
## Iteration: 95 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 95 cells_candidate: 3352 muts_candidate: 292
## [1] 3352  292
## [1] 292    6
## Iteration: 96 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 96 cells_candidate: 3267 muts_candidate: 290
## [1] 3267  290
## [1] 290    6
## Iteration: 97 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 97 cells_candidate: 3366 muts_candidate: 290
## [1] 3366  290
## [1] 290    6
## Iteration: 98 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 98 cells_candidate: 3296 muts_candidate: 289
## [1] 3296  289
## [1] 289    6
## Iteration: 99 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 99 cells_candidate: 3295 muts_candidate: 293
## [1] 3295  293
## [1] 293    6
## Iteration: 100 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 100 cells_candidate: 3405 muts_candidate: 292
## [1] 3405  292
## [1] 292    6

# 3. merge res
data_scMutModel_res <- do.call(rbind, data_scMutModel_list)

# 4. column
data_scMutModel_res$cb <- rownames(data_scMutModel_res)
data_scMutModel_res <- separate(data_scMutModel_res, col = "cb", into = c("celltype", "cell"), sep = "_")
data_scMutModel_res$Tag <- data_scMutModel_res$celltype
data_scMutModel_res <- separate(data_scMutModel_res, col = "celltype", into = c("rep", "Day"), sep = "\\.")

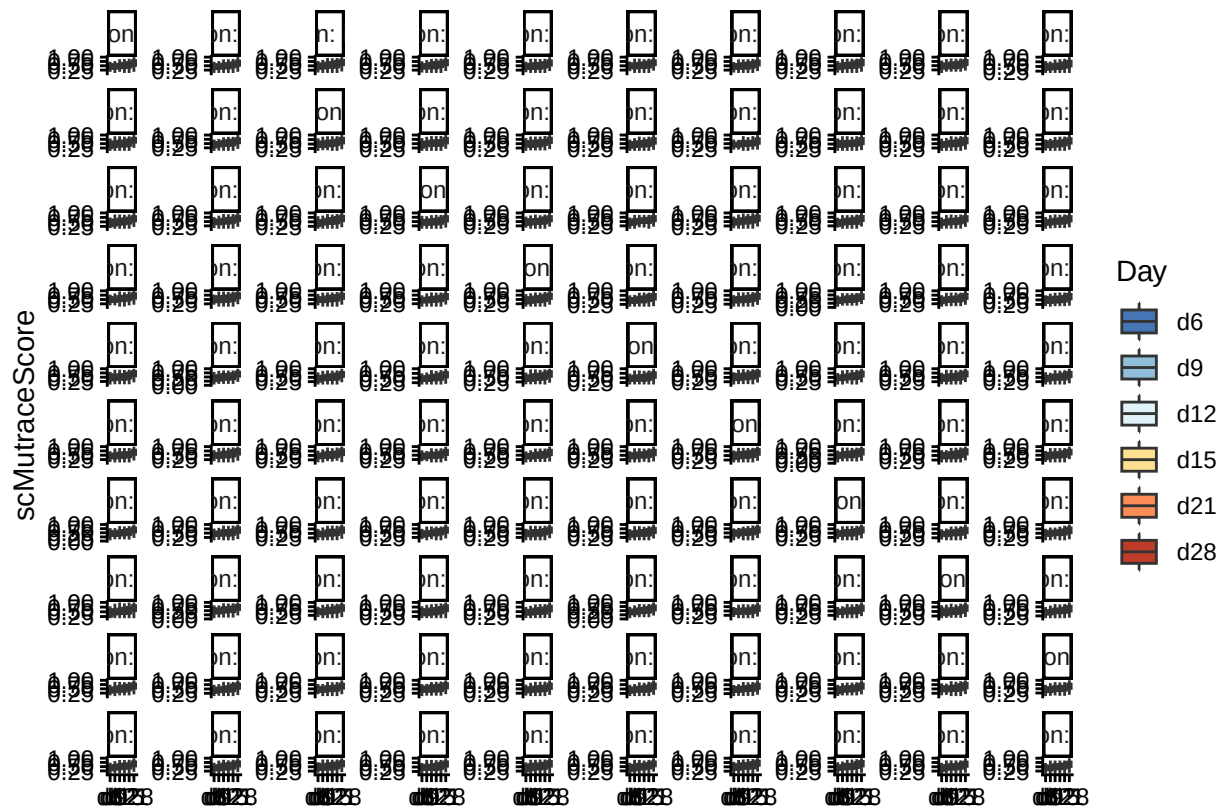
data_scMutModel_res$Tag <- factor(data_scMutModel_res$Tag, levels = col_check_names)
data_scMutModel_res$Day <- factor(data_scMutModel_res$Day, levels = c("d0", "d3", "d6", "d9", "d12", "d15"))

# 5. plot
p_scMutModel_randomly <- ggplot(data = data_scMutModel_res, mapping = aes(x = Day, y = scMutScore, fill = rep)) +
  #geom_violin(trim = TRUE, alpha = 0.6) +
  geom_boxplot(outliers = FALSE) +
  facet_wrap(~ iteration, nrow = 10, ncol = iteration_times / 10, scales = "free_y", labeller = label_both) +
  labs(x = "", y = "scMuttraceScore") +
  scale_fill_manual(values = color_settings) +

```

```
theme_classic()

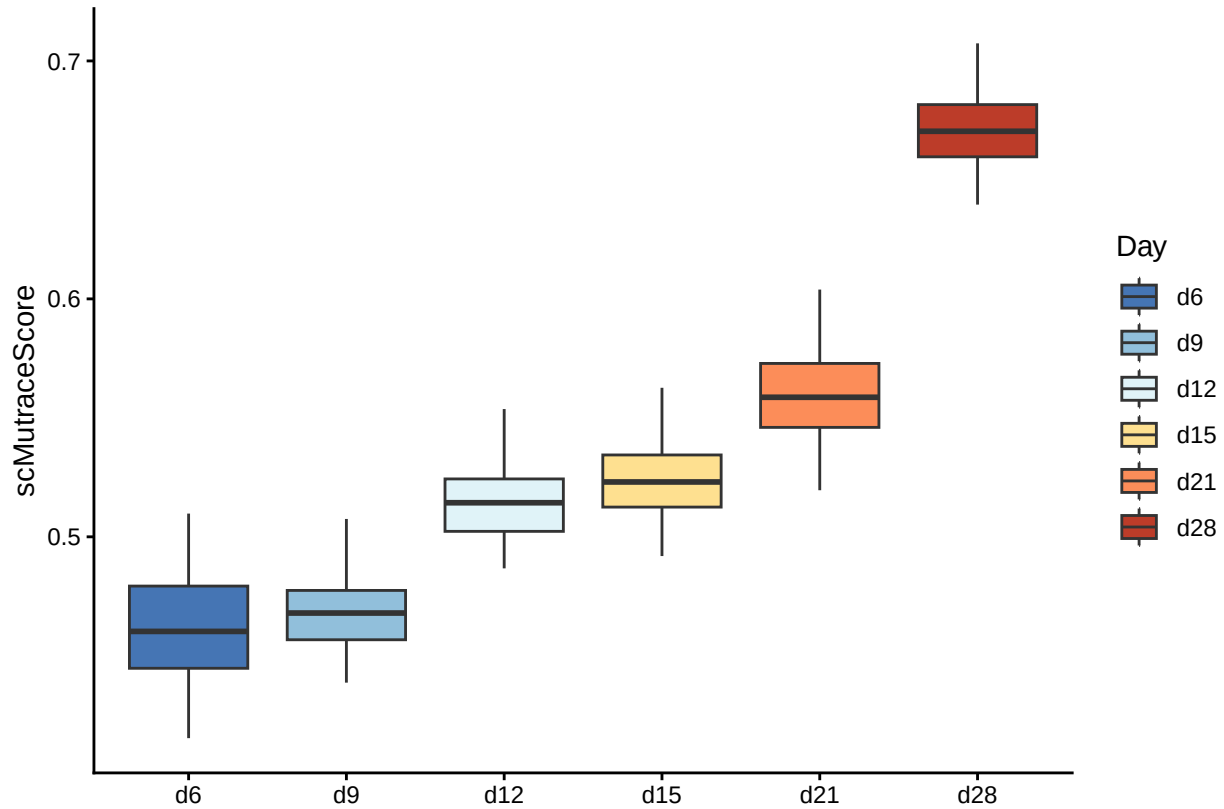
p_scMutModel_randomly
```



```
data_scMutModel_res_summary <- data_scMutModel_res %>%
  dplyr::group_by(iteration, Day) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScore, na.rm = TRUE)) %>%
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

data_scMutModel_res_summary_Qun <- data_scMutModel_res %>%
  dplyr::group_by(iteration, Day) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScoreQun, na.rm = TRUE)) %>%
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

ggplot(data = data_scMutModel_res_summary, mapping = aes(x = Day, y = mean_scMutScore, fill = Day)) +
  #geom_violin(trim = TRUE, alpha = 0.6) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
```



12 compareClonesScore

```
# generate data for comparision
MEF_meta_hf1_hf2 <- rbind(MEF_Meta_hf1, MEF_Meta_hf2)
res_index_scMut <- data.frame(Cellbarcode = rownames(res_index), scMutScore = res_index$scMutScore, Tag

res_index_scMut_meta <- merge(res_index_scMut, MEF_meta_hf1_hf2, by = "Cellbarcode")
res_index_scMut_meta$Day <- factor(res_index_scMut_meta$Day, levels = c("Day_0", "Day_3", "Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"))
res_index_scMut_meta <- res_index_scMut_meta[res_index_scMut_meta$Day %in% c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"),]

colNames_data_clone_score <- c("D0_tag", "D3_tag", "D13_tag", "Day", "Mean_Day_score", "Mean_Day_pseudo")
data_clone_score <- as.data.frame(matrix(ncol = length(colNames_data_clone_score)))
colnames(data_clone_score) <- colNames_data_clone_score
d0_tags <- unique(MEF_meta_hf1_hf2$CellTagD0_48k)

count_line = 0

for (i in 1:length(d0_tags)){
  d0_tag_tmp <- d0_tags[i]
  d3_tags <- unique(MEF_meta_hf1_hf2[MEF_meta_hf1_hf2$CellTagD0_48k == d0_tag_tmp,]$CellTagD3_48k)
  days_reprogramming <- unique(MEF_meta_hf1_hf2[MEF_meta_hf1_hf2$CellTagD0_48k == d0_tag_tmp,]$Day)
  for (day in 1:length(days_reprogramming)){
    if (length(d3_tags) >= 1){
```

```

for (j in 1:length(d3_tags)){
  d3_tag_tmp <- d3_tags[j]
  d13_tags <- unique(MEF_meta_hf1_hf2[MEF_meta_hf1_hf2$CellTagD0_48k == d0_tag_tmp & MEF_meta_hf1_hf2$CellTagD3_48k == d3_tag_tmp])
  if(length(d13_tags) >= 1){
    for (k in 1:length(d13_tags)){
      d13_tag_tmp <- d13_tags[k]
      count_line = count_line + 1
      data_clone_score[count_line,]$Day = days_reprogramming[day]
      data_clone_score[count_line,]$D0_tag = d0_tag_tmp
      data_clone_score[count_line,]$D3_tag = d3_tag_tmp
      data_clone_score[count_line,]$D13_tag = d13_tag_tmp
    }
  }else{
    count_line = count_line + 1
    data_clone_score[count_line,]$Day = days_reprogramming[day]
    data_clone_score[count_line,]$D0_tag = d0_tag_tmp
    data_clone_score[count_line,]$D3_tag = d3_tag_tmp
    data_clone_score[count_line,]$D13_tag = "NotAva"
  }
}
}
}

data_clone_score$ID <- paste(data_clone_score$D0_tag, data_clone_score$D3_tag, data_clone_score$D13_tag, sep = "_")
data_clone_score <- data_clone_score[data_clone_score$ID != "NotAva_NotAva_NotAva",]

for (i in 1:nrow(data_clone_score)){

  d0_tag_tmp <- data_clone_score[i,]$D0_tag
  d3_tag_tmp <- data_clone_score[i,]$D3_tag
  d13_tag_tmp <- data_clone_score[i,]$D13_tag
  day_tmp <- data_clone_score[i,]$Day

  data_clone_score[i,]$Mean_Day_score <- "NotAva"
  data_clone_score[i,]$Mean_Day_pseudotime <- "NotAva"

  id_check <- paste(d0_tag_tmp, d3_tag_tmp, d13_tag_tmp, sep = "_")
  data_clone_score[i,]$ID <- id_check

  scMutscore_tmp <- res_index_scMut_meta[res_index_scMut_meta$cloneTimeTag == id_check & res_index_scMut_meta$cloneTimeTag != "NotAva"]
  pseudotime_tmp <- res_index_scMut_meta[res_index_scMut_meta$cloneTimeTag == id_check & res_index_scMut_meta$cloneTimeTag != "NotAva"]

  scMutscore_tmp <- as.numeric(scMutscore_tmp[scMutscore_tmp != "NotAva"])
  pseudotime_tmp <- as.numeric(pseudotime_tmp[pseudotime_tmp != "NotAva"])
}

```

```

data_clone_score[i,]$Mean_Day_score <- mean(scMutscore_tmp)
data_clone_score[i,]$Mean_Day_pseudotime <- mean(pseudotime_tmp)
}

# filter
# clonetagDay
data_clone_score_clonetagDay <- data_clone_score[data_clone_score$Mean_Day_score != "NaN" & data_clone_score$Mean_Day_pseudotime != "NaN"]
ID_counts <- table(data_clone_score_clonetagDay$ID)
valid_IDs <- names(ID_counts[ID_counts >= 2])
data_clone_score_clonetagDay <- subset(data_clone_score_clonetagDay, ID %in% valid_IDs)

data_clone_score_clonetagDay$Mean_Day_score <- as.numeric(data_clone_score_clonetagDay$Mean_Day_score)
data_clone_score_clonetagDay$Mean_Day_pseudotime <- as.numeric(data_clone_score_clonetagDay$Mean_Day_pseudotime)

data_clone_score_clonetagDay$Day <- factor(data_clone_score_clonetagDay$Day, levels = c("Day_0", "Day_3"))

# reprogramming Day
p_clone_scMutraceScore <- ggplot(data_clone_score_clonetagDay, mapping = aes(x = as.factor(Day), y = Mean_Day_score)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = ID), color = "gray", alpha = 0.1) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

p_clone_pseudotime <- ggplot(data_clone_score_clonetagDay, aes(x = as.factor(Day), y = Mean_Day_pseudotime)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = ID), color = "gray", alpha = 0.1) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "Pseudotime") +
  theme_classic()

p_cell_scMutraceScore <- ggplot(data = res_index_scMut_meta, mapping = aes(x = Day, y = scMutScore, fill = Outcome)) +
  geom_boxplot(outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

p_cell_pseudotime <- ggplot(data = res_index_scMut_meta[res_index_scMut_meta$pseudotime != "NotAva"], mapping = aes(x = Day, y = pseudotime, fill = Outcome)) +
  geom_boxplot(outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "pseudotime") +
  theme_classic()

res_index_scMut_meta_success <- res_index_scMut_meta[res_index_scMut_meta$Outcome == "Successful",][,c("Day", "scMutScore")]
res_index_scMut_meta_deadend <- res_index_scMut_meta[res_index_scMut_meta$Outcome == "DeadEnd",][,c("Day", "scMutScore")]
res_index_scMut_meta_success_deadend <- rbind(res_index_scMut_meta_success, res_index_scMut_meta_deadend)

# Outcome
p_outcome_scMutraceScore <- ggplot(res_index_scMut_meta[res_index_scMut_meta$Outcome %in% c("DeadEnd", "Successful")], mapping = aes(x = Day, y = scMutScore, fill = Outcome)) +
  labs(x = "", y = "scMutScore") +
  theme_classic()

```



```

geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

p_outcome_pseudotime <- ggplot(res_index_scMut_meta[res_index_scMut_meta$Outcome %in% c("DeadEnd", "Successful")],
labs(x = "", y = "pseudotime") +
geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

p_reprogramming_scMutraceScore <- ggplot(res_index_scMut_meta[res_index_scMut_meta$reprogramming %in% c("F_reprogramming", "T_reprogramming")],
labs(x = "", y = "scMutScore") +
geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

p_reprogramming_pseudotime <- ggplot(res_index_scMut_meta[res_index_scMut_meta$reprogramming %in% c("F_reprogramming", "T_reprogramming")],
labs(x = "", y = "pseudotime") +
geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

timepoint_color <- c(Day_6 = '#7D5188', Day_9 = '#7D71A5', Day_12 = '#738DAF', Day_15 = '#6EA5B0', Day_18 = '#6A8C9E')
cluster_color <- c(c0 = '#D5393E', c1 = '#5A6B93', c2 = '#519376', c3 = '#6E8371', c4 = '#A55F7F', c5 = '#A55F7F', c6 = '#A55F7F')
outcom_color <- c(BAB = "lightgray", DeadEnd = "#7876B1FF", Successful = "#FFDC91FF")
reprogramming_colors <- c(F_reprogramming = '#7876B1FF', T_reprogramming = '#FFDC91FF', NotAva = "#d9d9d9")

p_umap_reprogramming <- ggplot(data = res_index_scMut_meta[res_index_scMut_meta$reprogramming %in% c("F_reprogramming", "T_reprogramming")],
scale_color_manual(values = reprogramming_colors) +
xlab("") + ylab("") +
theme_classic()

p_umap_outcome <- ggplot(data = res_index_scMut_meta[res_index_scMut_meta$Outcome %in% c("DeadEnd", "Successful")],
scale_color_manual(values = outcom_color) +
xlab("") + ylab("") +
theme_classic()

p_umap_cell_day <- ggplot(data = res_index_scMut_meta) + geom_point(mapping = aes(x = TSNE_1, y = TSNE_2)) +
scale_color_manual(values = color_settings_2) +
xlab("") + ylab("") +
theme_classic()

res_index_scMut_meta$cluster <- paste0("c", res_index_scMut_meta$cluster)
p_umap_cell_cluster <- ggplot(data = res_index_scMut_meta) + geom_point(mapping = aes(x = TSNE_1, y = TSNE_2)) +
scale_color_manual(values = cluster_color) +
xlab("") + ylab("") +
theme_classic()

p_umap_cell_Pseudotime <- ggplot(res_index_scMut_meta[res_index_scMut_meta$pseudotime != "NotAva",], aes(x = TSNE_1, y = TSNE_2)) +

```

```

geom_point(shape = 21, size = 2, alpha = 0.3) +
scale_fill_gradientn(colours = c("#313695", "#4575b4", "#74add1", "#abd9e9", "#e0f3f8", "#ffffbf", "#f4a460", "#d73027")) +
theme_classic() +
labs(fill = "Pseudotime")

p_umap_cell_scMutScore <- ggplot(res_index_scMut_meta[res_index_scMut_meta$scMutScore != "NotAva",], aes(x = pseudotime, y = scMutScore)) +
geom_point(shape = 21, size = 2, alpha = 0.3) +
scale_fill_gradientn(colours = c("#313695", "#4575b4", "#74add1", "#abd9e9", "#e0f3f8", "#ffffbf", "#f4a460", "#d73027")) +
theme_classic() +
labs(fill = "scMutScore")

data_scMutScore_pseudotime <- res_index_scMut_meta[,c("scMutScore", "pseudotime")]
data_scMutScore_pseudotime <- data_scMutScore_pseudotime[data_scMutScore_pseudotime$scMutScore != "NotAva",]
data_scMutScore_pseudotime$scMutScore <- as.numeric(data_scMutScore_pseudotime$scMutScore)
data_scMutScore_pseudotime$pseudotime <- as.numeric(data_scMutScore_pseudotime$pseudotime)
data_scMutScore_pseudotime$pseudotime_scale <- (data_scMutScore_pseudotime$pseudotime - min(data_scMutScore_pseudotime$pseudotime)) / (max(data_scMutScore_pseudotime$pseudotime) - min(data_scMutScore_pseudotime$pseudotime))

p_cor_scMutScore_pseudotime <- ggplot(data_scMutScore_pseudotime, aes(x = scMutScore, y = pseudotime_scale)) +
geom_point() +
geom_smooth(method = "lm", se = TRUE, color = "blue") +
stat_cor(method = "pearson", label.x = 0.5, label.y = 1) +
labs(x = "scMutScore", y = "Pseudotime") +
theme_classic()

p_cor_scMutScore_sumMut <- ggplot(res_index, aes(x = scMutScore, y = log2(sumMut))) +
geom_point() +
geom_smooth(method = "lm", se = TRUE, color = "blue") +
stat_cor(method = "pearson", label.x = 0.5, label.y = 1) +
labs(x = "scMutScore", y = "sum of matrix") +
theme_classic()

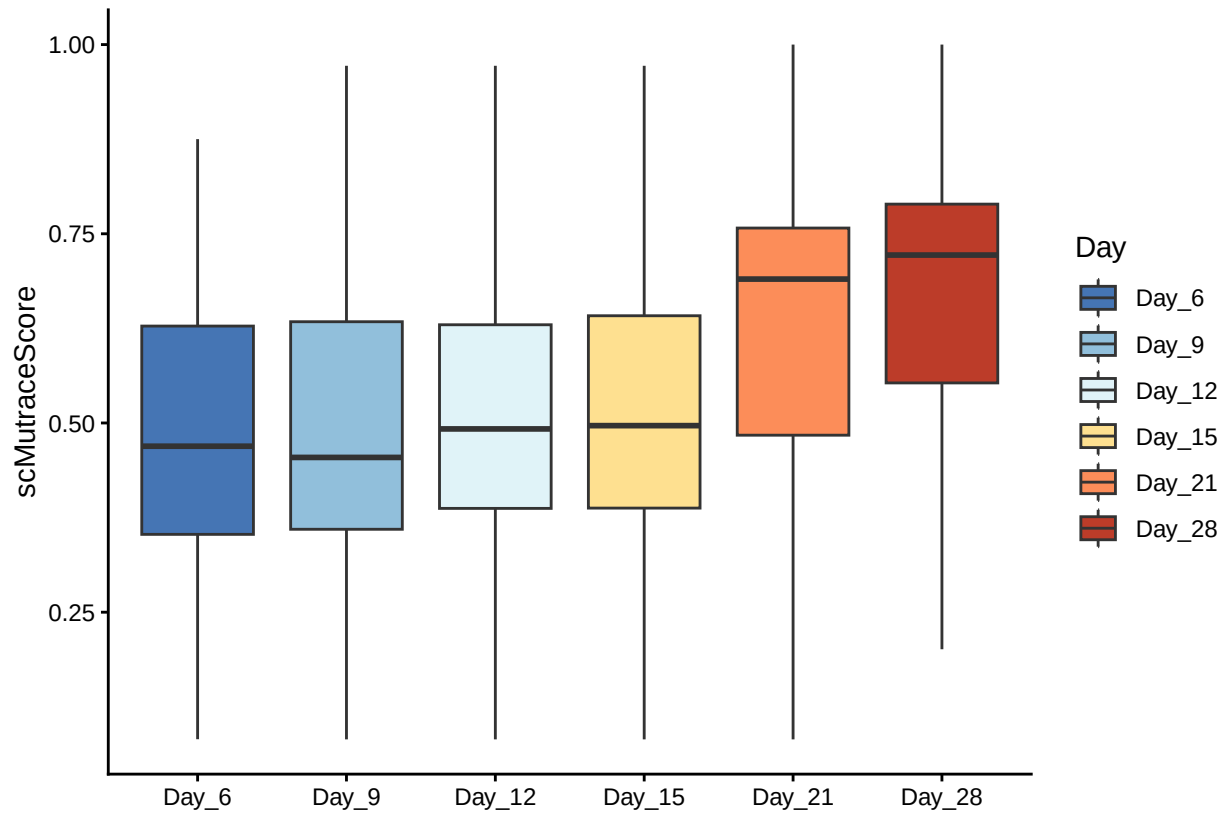
cor.test(data_scMutScore_pseudotime$scMutScore, data_scMutScore_pseudotime$pseudotime_scale)
##
## Pearson's product-moment correlation
##
## data: data_scMutScore_pseudotime$scMutScore and data_scMutScore_pseudotime$pseudotime_scale
## t = 24.639, df = 5007, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3039094 0.3533129
## sample estimates:
## cor
## 0.3288361
cor.test(res_index$scMutScore, res_index$sumMut)
##
## Pearson's product-moment correlation
##
## data: res_index$scMutScore and res_index$sumMut
## t = 24.382, df = 6916, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.2594998 0.3029015
## sample estimates:

```

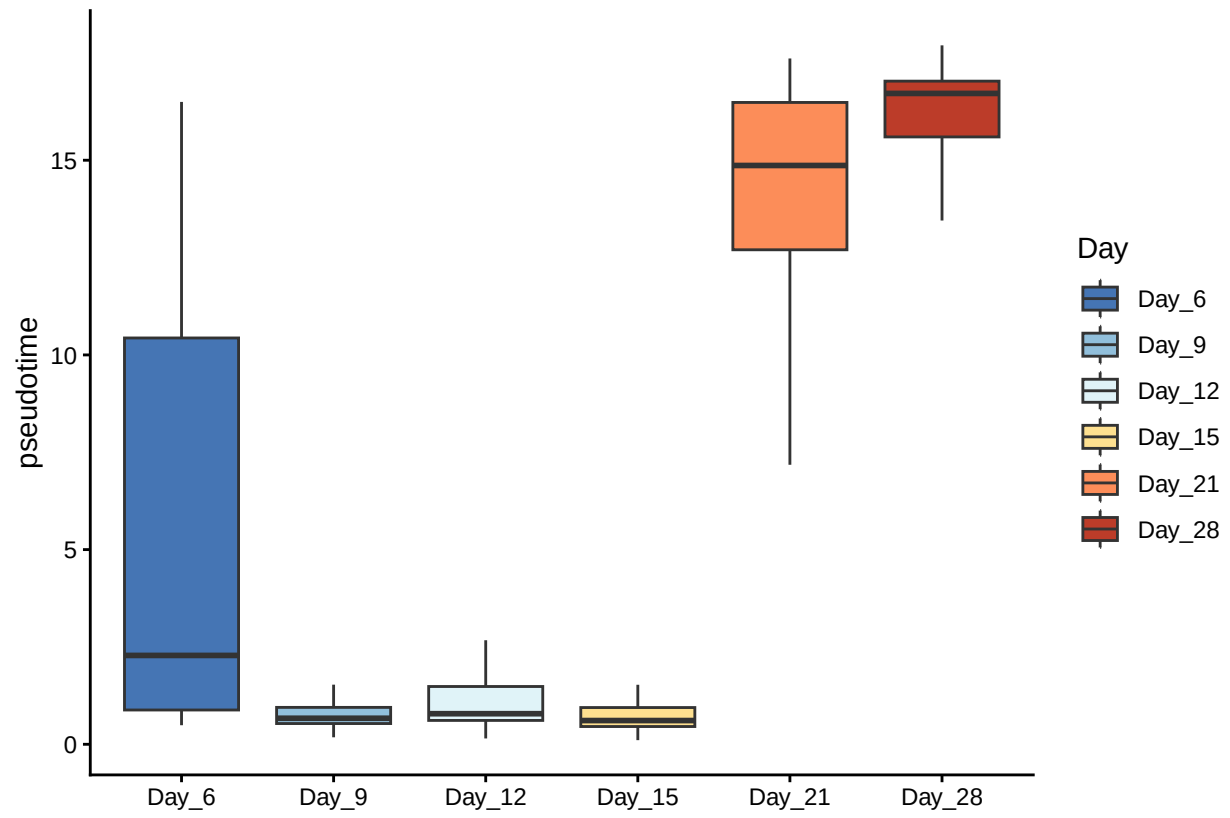
```
## cor
## 0.2813445
```

13 compareClonesScore figures

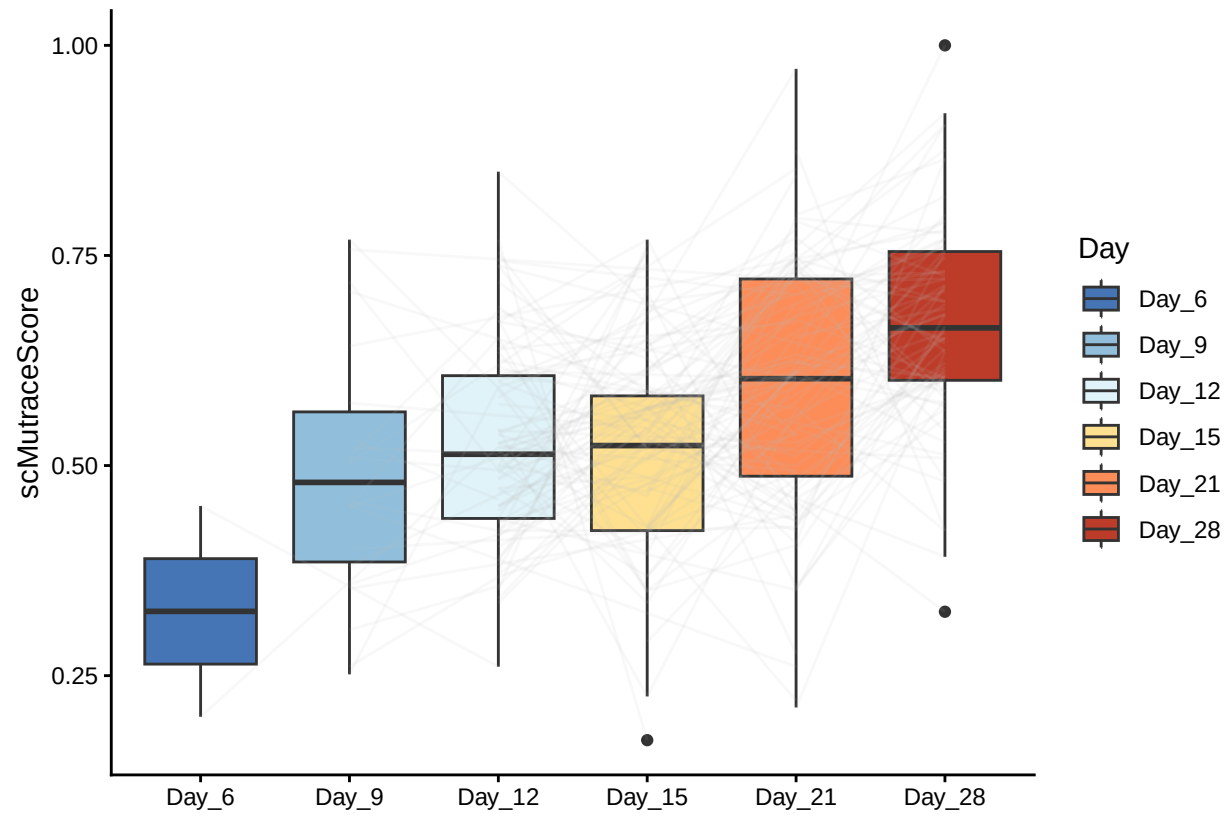
```
p_cell_scMutraceScore
```



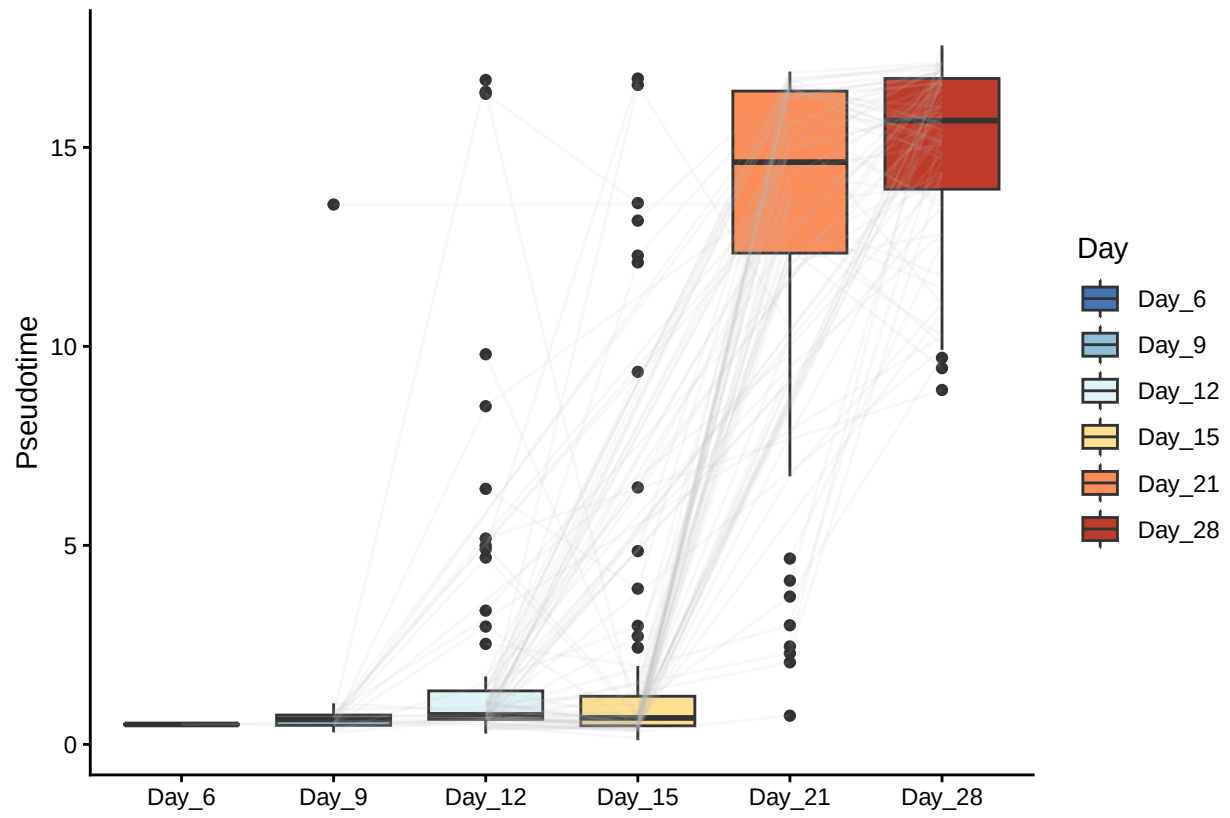
```
p_cell_pseudotime
```



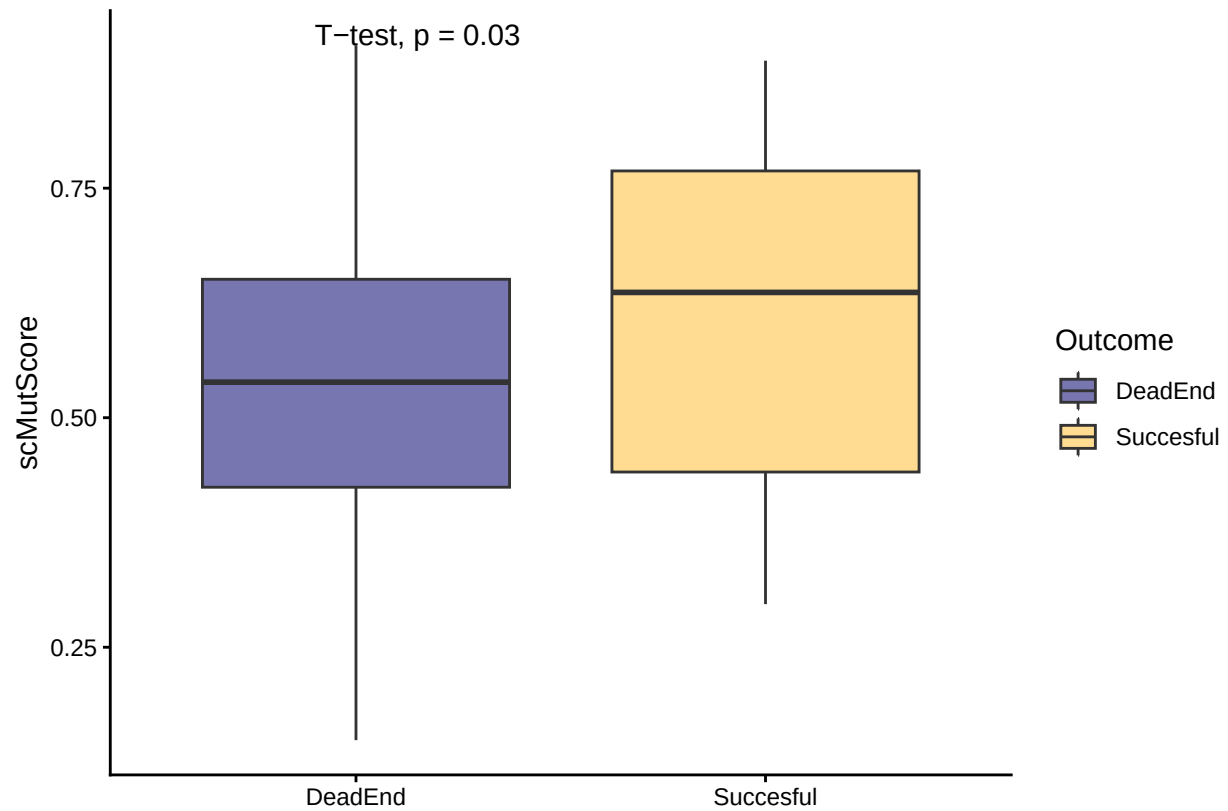
p_clone_scMutraceScore



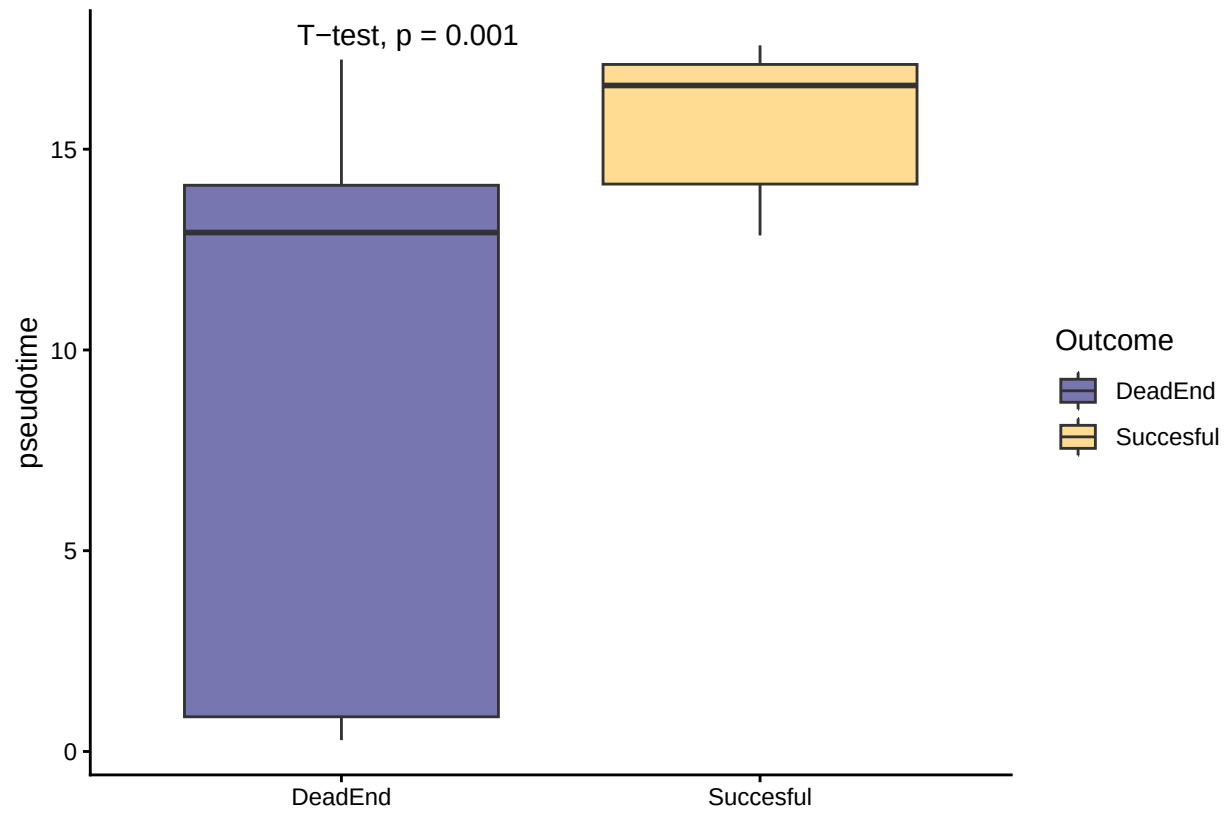
p_clone_pseudotime



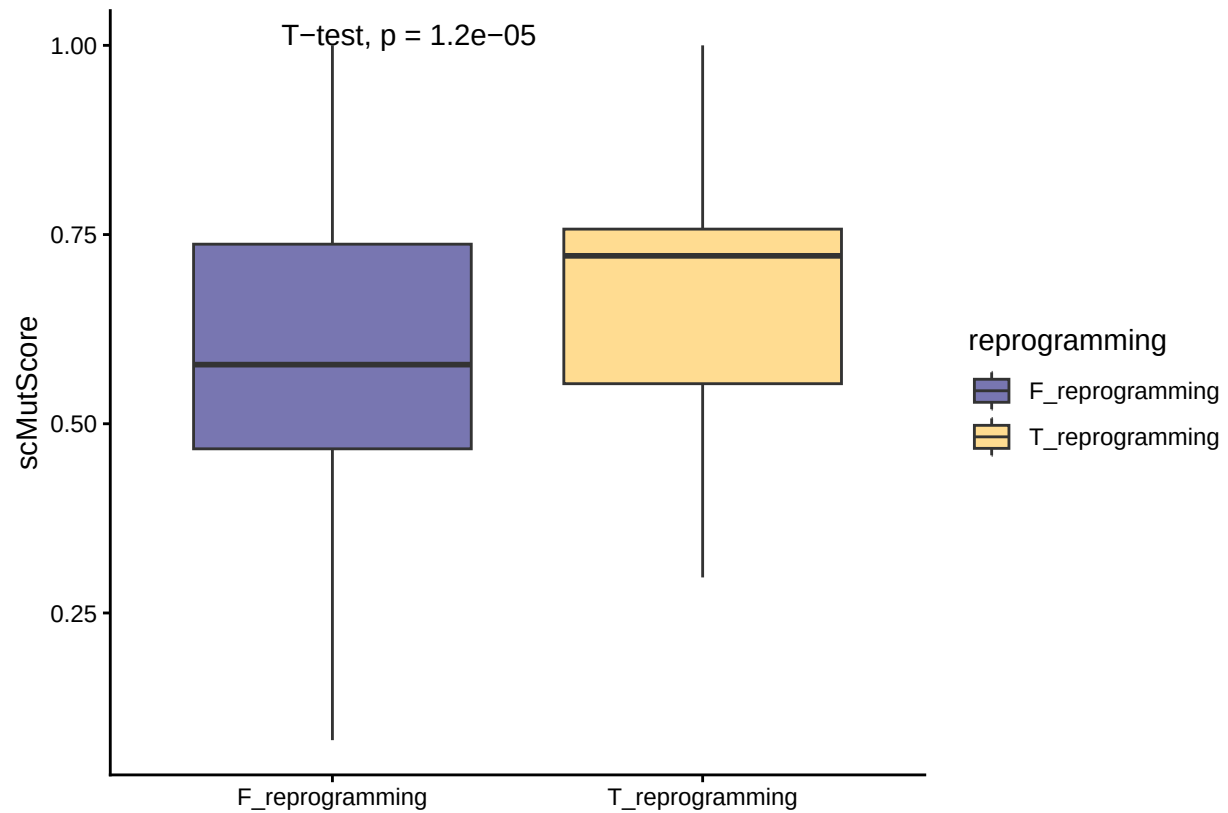
p_outcome_scMutraceScore



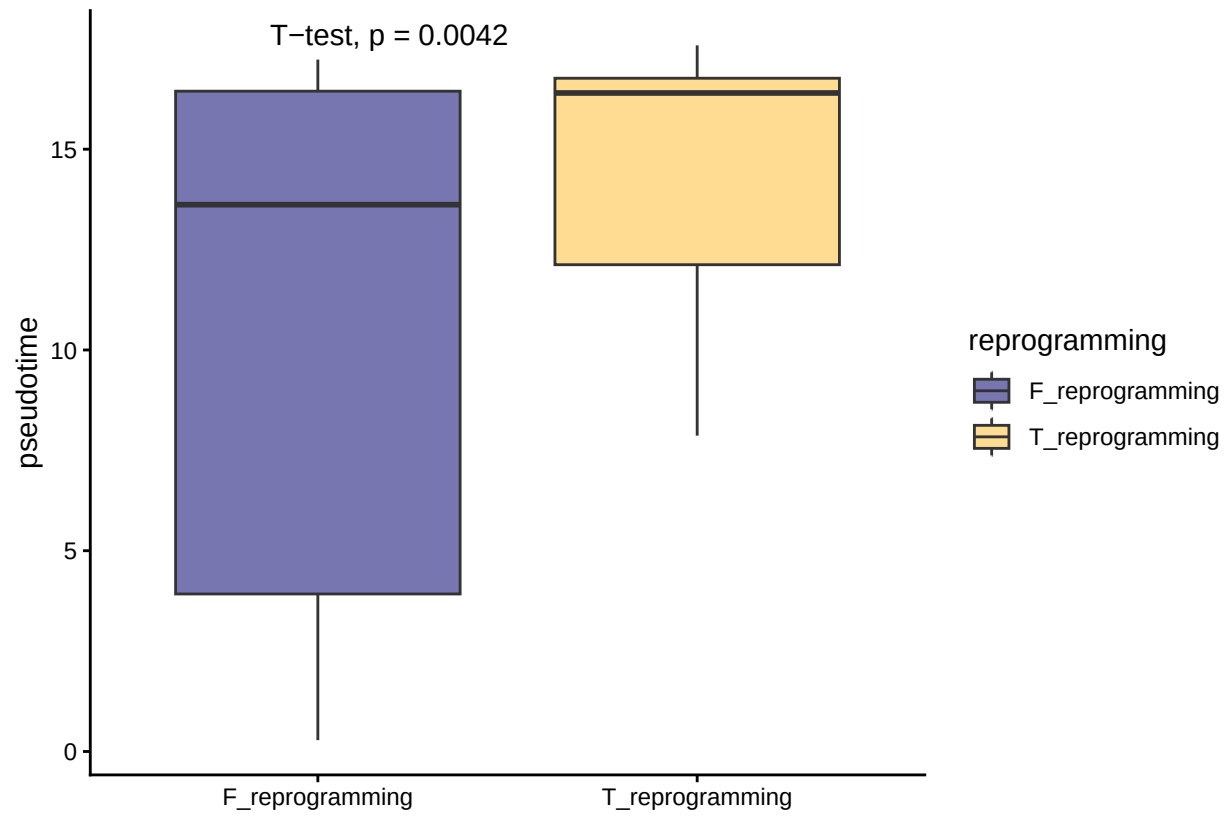
```
p_outcome_pseudotime
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning: Removed 102 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
## Warning: Removed 102 rows containing non-finite outside the scale range
## (`stat_compare_means()`).
```



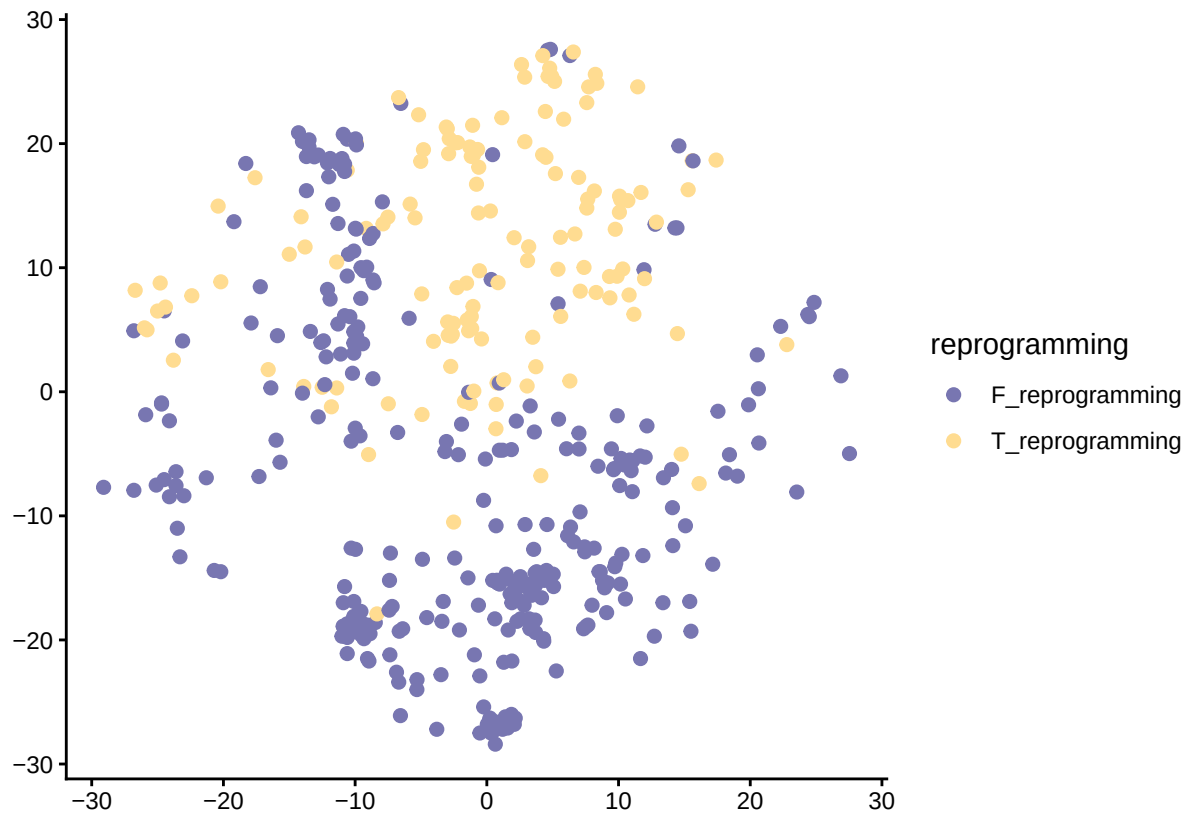
p_reprogramming_scMutraceScore



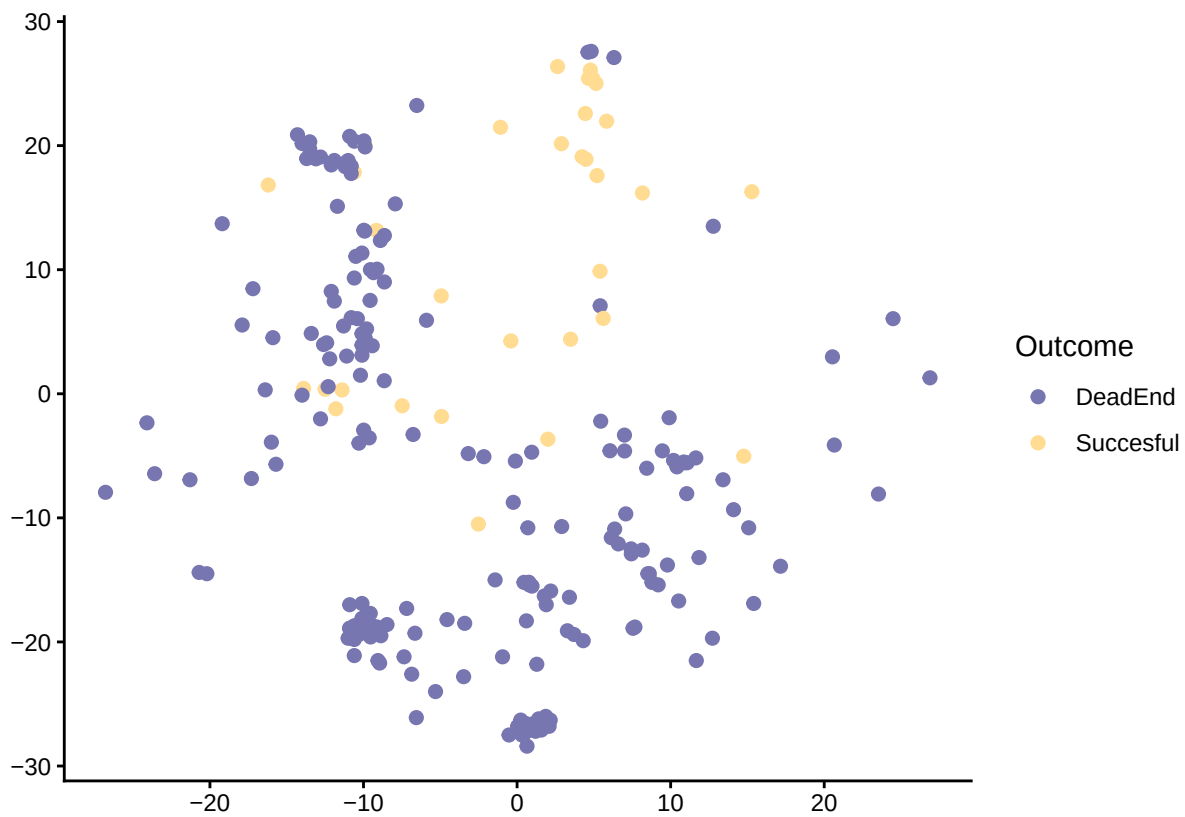
```
p_reprogramming_pseudotime
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning: Removed 135 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
## Warning: Removed 135 rows containing non-finite outside the scale range
## (`stat_compare_means()`).
```



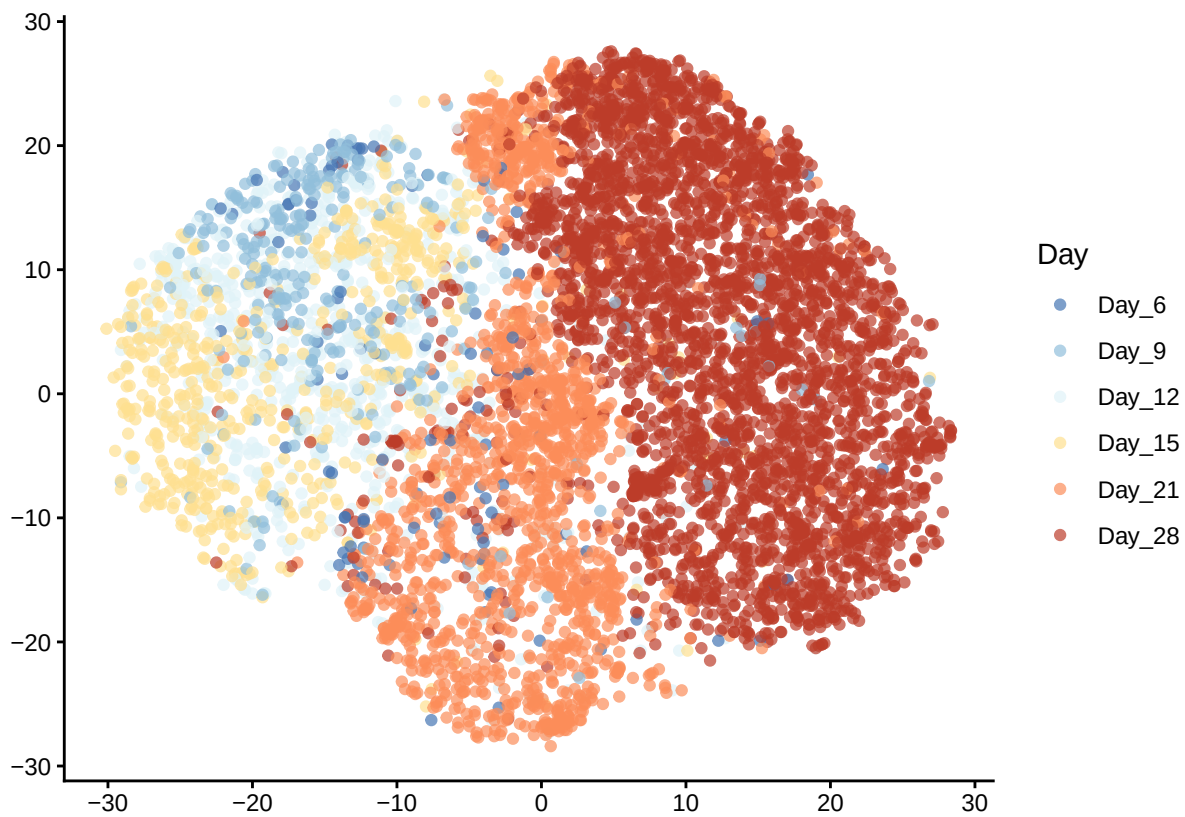
p_umap_reprogramming



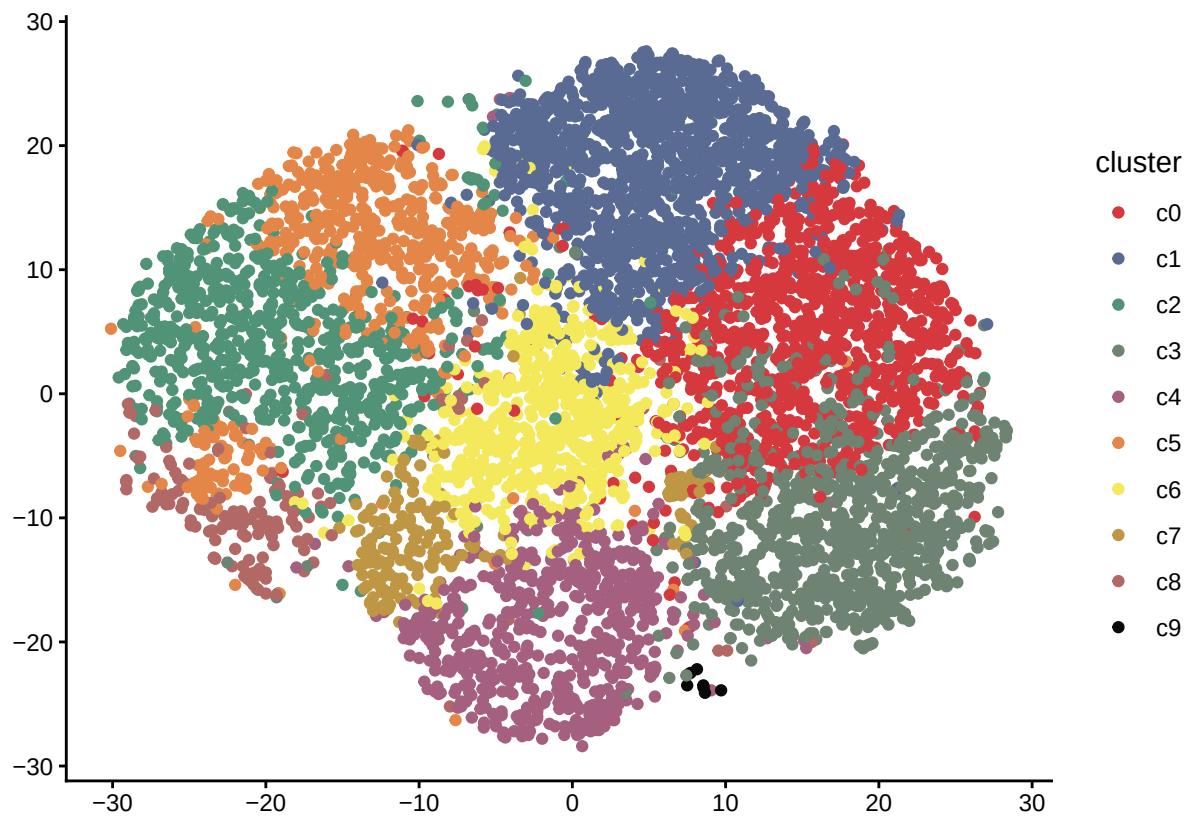
`p_umap_outcome`



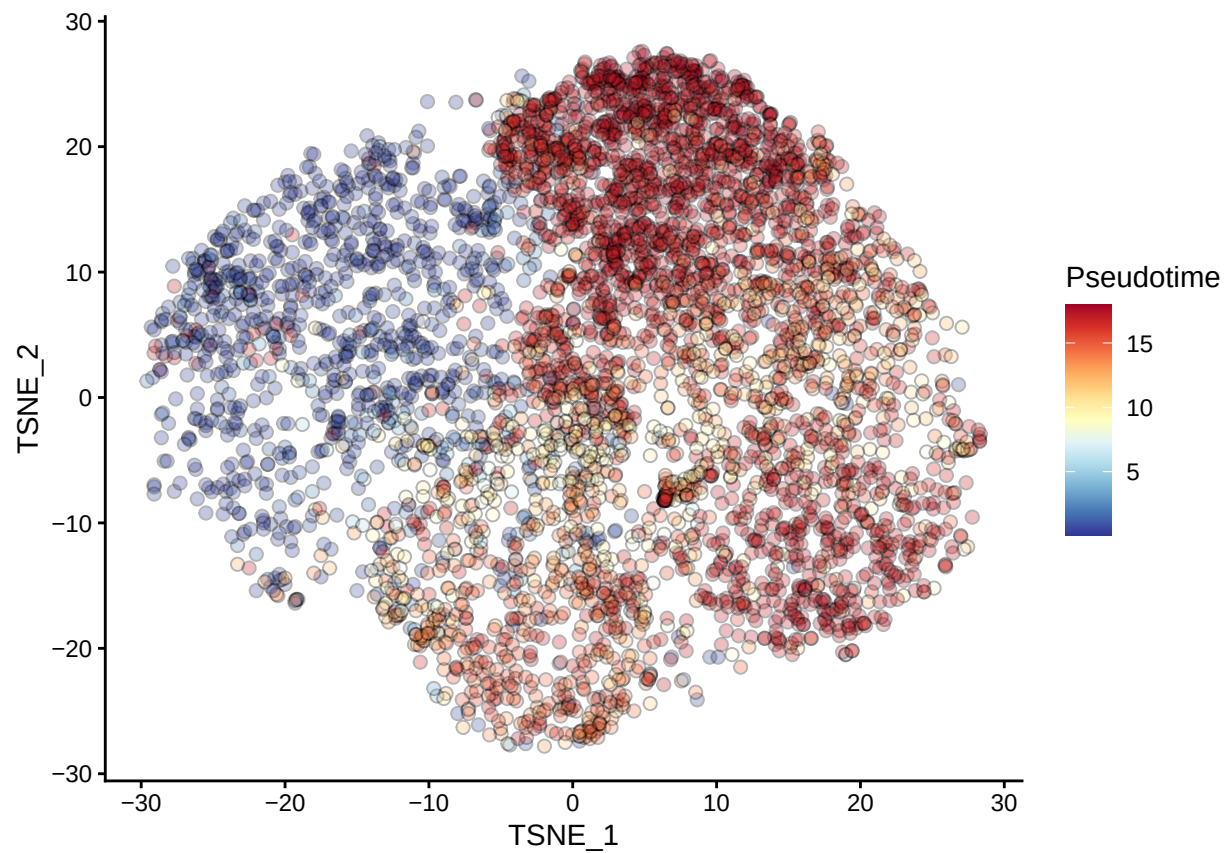
`p_umap_cell_day`



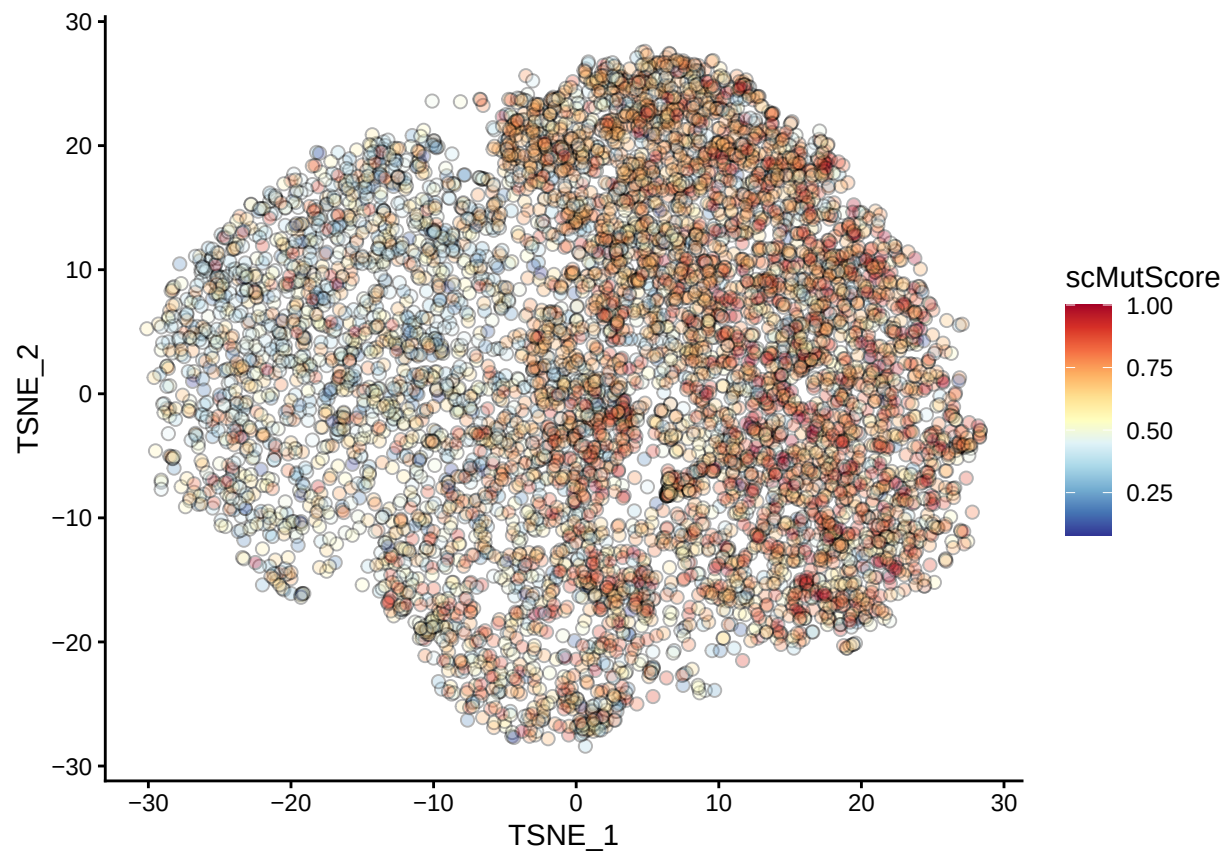
p_umap_cell_cluster



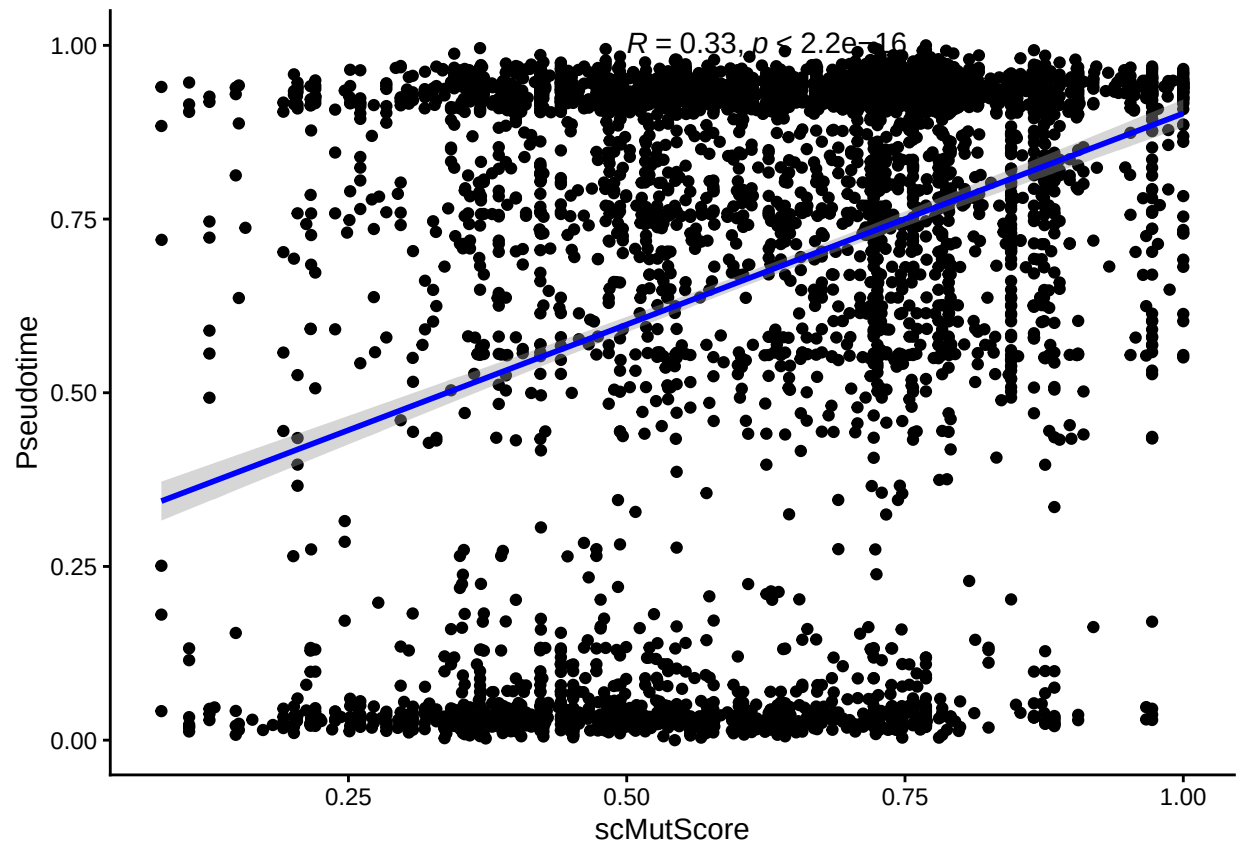
p_umap_cell_Pseudotime



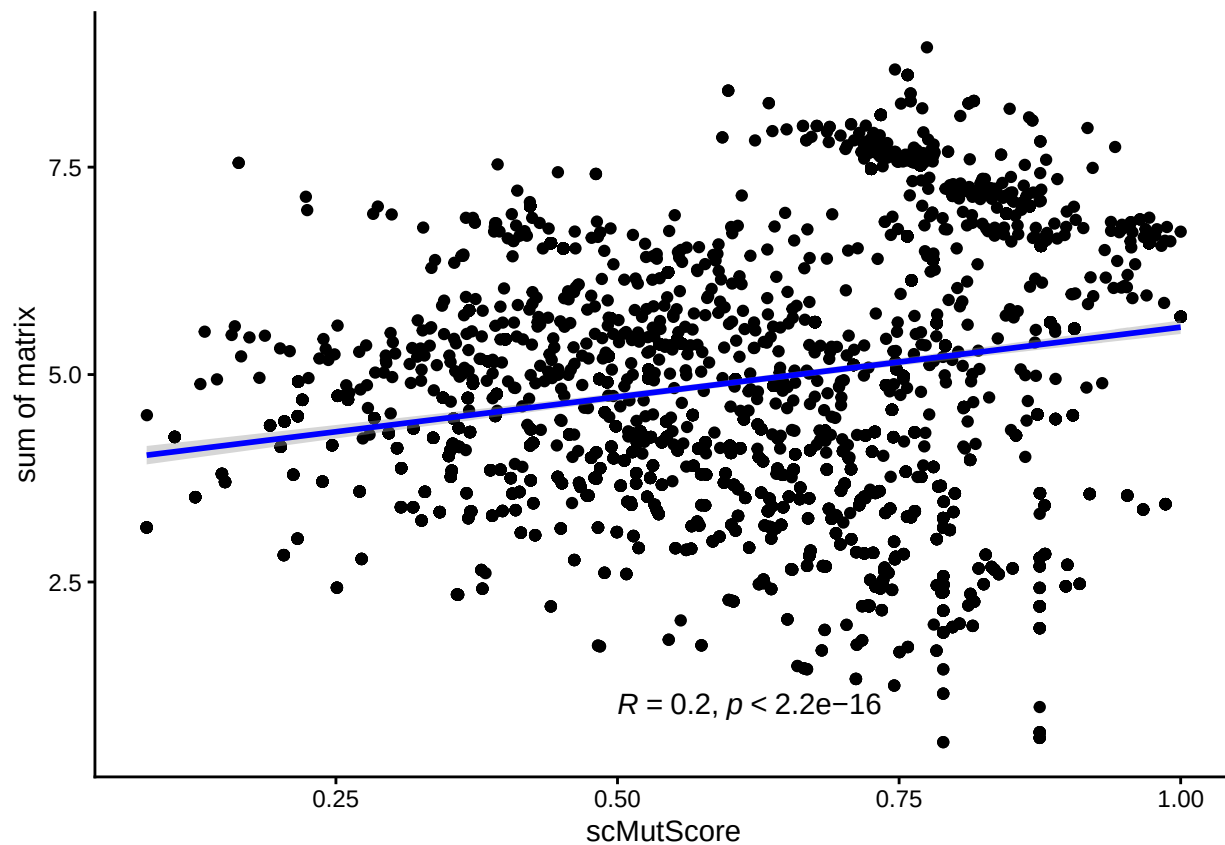
p_umap_cell_scMutScore



```
p_cor_scMutScore_pseudotime  
## `geom_smooth()` using formula = 'y ~ x'
```

```
p_cor_scMutScore_sumMut  
## `geom_smooth()` using formula = 'y ~ x'
```



14 save figures

```
# save data
pdf("../Figure/MEF/p_cell_scMutraceScore.pdf", width = 5.36, height = 3.42)
p_cell_scMutraceScore
dev.off()

pdf("../Figure/MEF/p_cell_pseudotime.pdf", width = 5.36, height = 3.42)
p_cell_pseudotime
dev.off()

pdf("../Figure/MEF/p_clone_scMutraceScore.pdf", width = 5.36, height = 3.42)
p_clone_scMutraceScore
dev.off()

pdf("../Figure/MEF/p_clone_pseudotime.pdf", width = 5.36, height = 3.42)
p_clone_pseudotime
dev.off()

pdf("../Figure/MEF/p_umap_cell_day.pdf", width = 5.36, height = 3.42)
p_umap_cell_day
dev.off()
```

```
pdf("../Figure/MEF/p_umap_cell_scMutScore.pdf", width = 5.36, height = 3.42)
p_umap_cell_scMutScore
dev.off()

pdf("../Figure/MEF/p_umap_cell_Pseudotime.pdf", width = 5.36, height = 3.42)
p_umap_cell_Pseudotime
dev.off()

pdf("../Figure/MEF/p_success_faile_scMutraceScore.pdf", width = 5.36, height = 3.42)
p_reprogramming_scMutraceScore
dev.off()
```

15 check randomly clone scMutScore

```
selected_Days <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")
set.seed(2025)
data_random_cells <- res_index_scMut_meta[sample(nrow(res_index_scMut_meta), 1000, replace = TRUE), c("scMutScore", "Day")]
data_random_cells$Day <- "Day_6"

for (i in 2:length(selected_Days)){
  random_rows_1 <- res_index_scMut_meta[sample(nrow(res_index_scMut_meta), 50), c("scMutScore", "Day")]
  random_rows_1$Day <- selected_Days[i]
  data_random_cells <- rbind(data_random_cells, random_rows_1)
}

data_random_cells$Day <- factor(data_random_cells$Day, levels = c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"))

p_random_cell_days <- ggplot(data = data_random_cells, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = TRUE) +
  #stat_summary(fun.data="mean_sdl", fun.args = list(mult=1), geom="pointrange", color = "black") +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutScore") +
  theme_classic()

data_random_clones <- data_clone_score_clonetagDay[sample(nrow(data_clone_score_clonetagDay), 400, replace = TRUE), c("Mean_Day_score", "Day")]
data_random_clones$Day <- "Day_6"

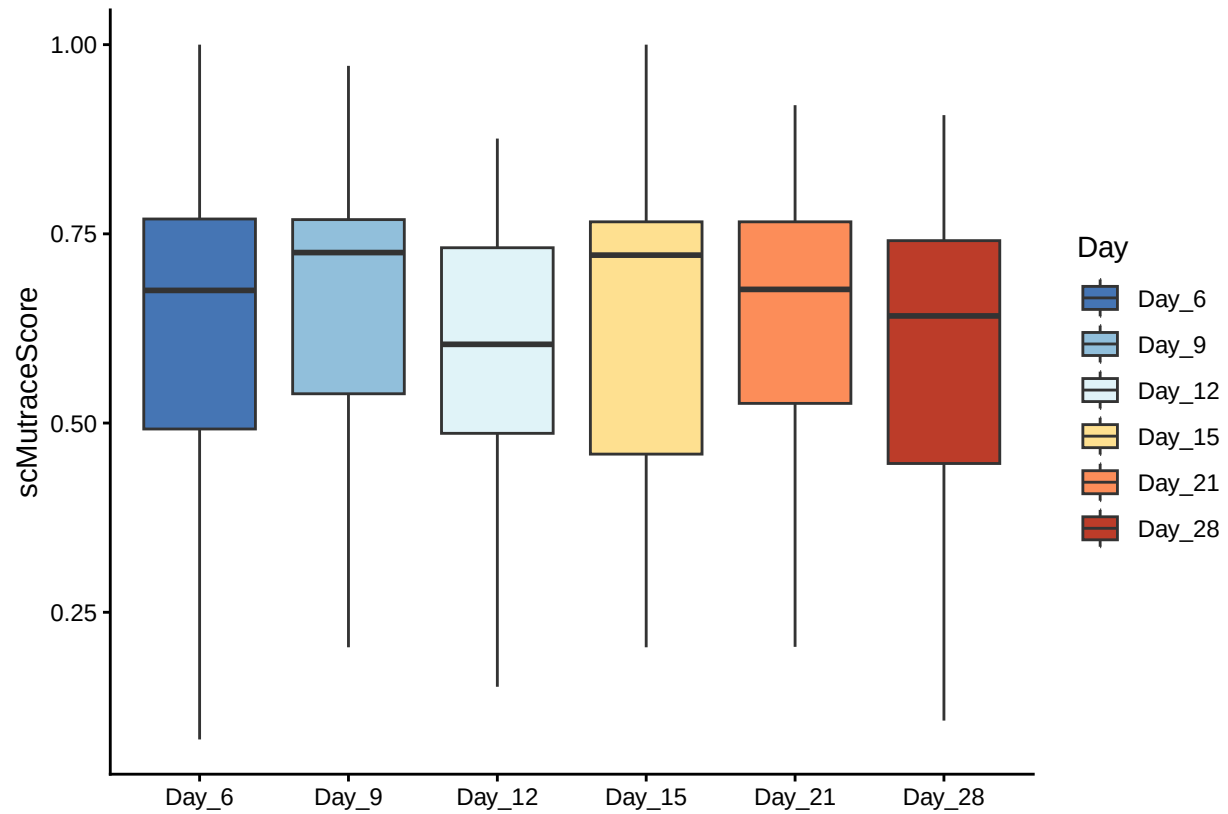
for (i in 2:length(selected_Days)){
  random_rows_1 <- data_clone_score_clonetagDay[sample(nrow(data_clone_score_clonetagDay), 50), c("Mean_Day_score", "Day")]
  random_rows_1$Day <- selected_Days[i]
  data_random_clones <- rbind(data_random_clones, random_rows_1)
}

data_random_clones$Day <- factor(data_random_clones$Day, levels = c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"))

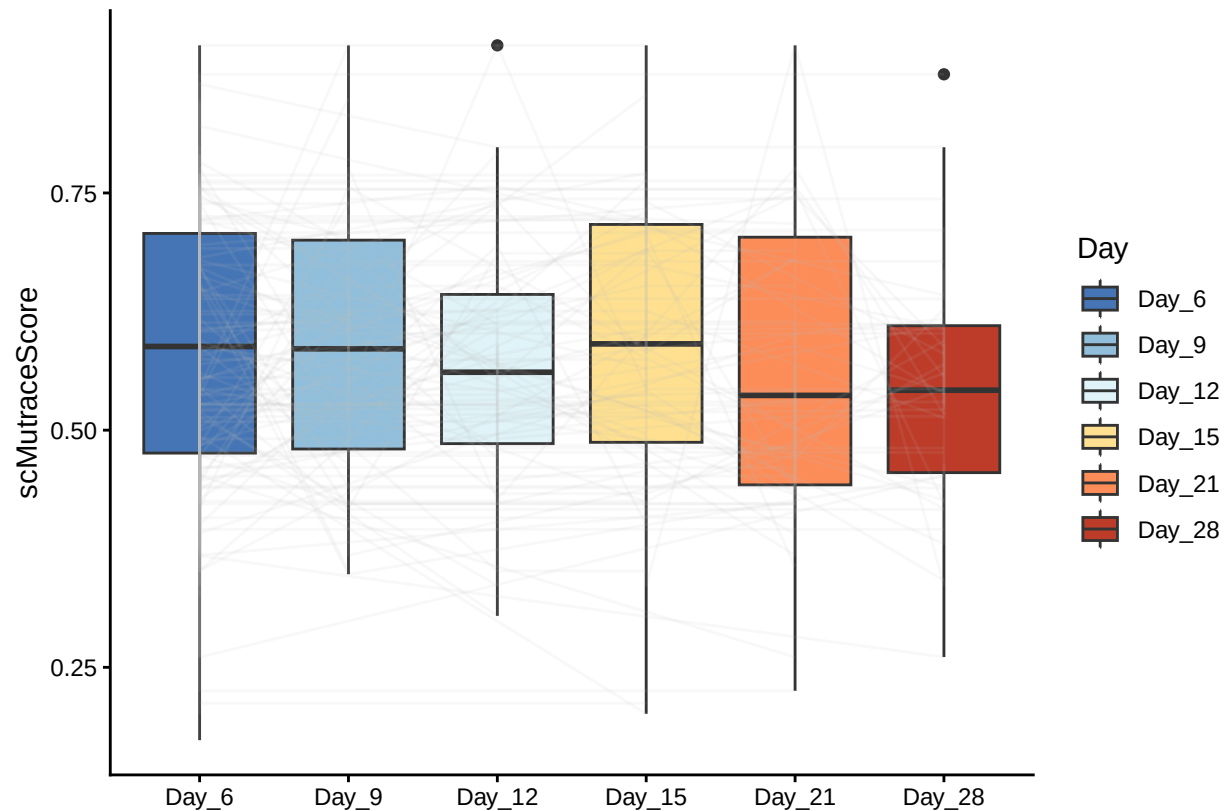
p_random_clone_days <- ggplot(data_random_clones, mapping = aes(x = as.factor(Day), y = Mean_Day_score, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = ID), color = "gray", alpha = 0.1) +
  #geom_point(size = 1, shape = 21, color = "black") +
```

```
scale_fill_manual(values = color_settings_2) +
labs(x = "", y = "scMutraceScore") +
theme_classic()
```

p_random_cell_days



p_random_clone_days



```
library(clinfun)
# cells scMutScore
res_index_selected <- res_index_scMut_meta[res_index_scMut_meta$Day %in% c('Day_6', 'Day_9', 'Day_12',
res_index_selected$Day <- factor(res_index_selected$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15', 'Day_21', 'Day_28'))
jonckheere.test(x = res_index_selected$scMutScore, g = res_index_selected$Day, alternative = "increasing")
## Warning in jonckheere.test(x = res_index_selected$scMutScore, g = res_index_selected$Day, : Sample size > 100 or data are not independent
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 10520356, p-value < 2.2e-16
## alternative hypothesis: increasing
## JT = 10520356, p-value < 2.2e-16
## alternative hypothesis: increasing

res_index_selected <- res_index_scMut_meta[res_index_scMut_meta$Day %in% c('Day_6', 'Day_9', 'Day_12',
res_index_selected <- res_index_scMut_meta[res_index_scMut_meta$pseudotime != "NotAva",]
res_index_selected$Day <- factor(res_index_selected$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15', 'Day_21', 'Day_28'))
jonckheere.test(x = as.numeric(res_index_selected$pseudotime), g = res_index_selected$Day, alternative = "increasing")
## Warning in jonckheere.test(x = as.numeric(res_index_selected$pseudotime), : Sample size > 100 or data are not independent
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
```

```

##
## data:
## JT = 6857494, p-value < 2.2e-16
## alternative hypothesis: increasing
## JT = 6857494, p-value < 2.2e-16
## alternative hypothesis: increasing

# clone scMutscore
data_clone_score_clonetagDay_selected <- data_clone_score_clonetagDay[data_clone_score_clonetagDay$Day %> 0]
data_clone_score_clonetagDay_selected$Day <- factor(data_clone_score_clonetagDay_selected$Day , levels = c("Day_6", "Day_9", "Day_12", "Day_15"))
jonckheere.test(x = data_clone_score_clonetagDay_selected$Mean_Day_score, g = data_clone_score_clonetagDay_selected$Day, alternative = "increasing")
## Warning in jonckheere.test(x = data_clone_score_clonetagDay_selected$Mean_Day_score, : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 31093, p-value = 8.216e-15
## alternative hypothesis: increasing
## JT = 31093, p-value = 8.216e-15
## alternative hypothesis: increasing
jonckheere.test(x = as.numeric(data_clone_score_clonetagDay_selected$Mean_Day_pseudotime), g = data_clone_score_clonetagDay_selected$Day, alternative = "increasing")
## Warning in jonckheere.test(x = as.numeric(data_clone_score_clonetagDay_selected$Mean_Day_pseudotime), : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 37857, p-value < 2.2e-16
## alternative hypothesis: increasing
## JT = 37857, p-value < 2.2e-16
## alternative hypothesis: increasing

# data_random_cells
data_random_cells$Day <- factor(data_random_cells$Day , levels = c("Day_6", "Day_9", "Day_12", "Day_15"))
jonckheere.test(x = data_random_cells$scMutScore, g = data_random_cells$Day, alternative = "increasing")
## Warning in jonckheere.test(x = data_random_cells$scMutScore, g = data_random_cells$Day, : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 135420, p-value = 0.657
## alternative hypothesis: increasing
## JT = 135420, p-value = 0.657
## alternative hypothesis: increasing

#data_random_clones
data_random_clones$Day <- factor(data_random_clones$Day , levels = c("Day_6", "Day_9", "Day_12", "Day_15"))
jonckheere.test(x = data_random_clones$Mean_Day_score, g = data_random_clones$Day, alternative = "increasing")
## Warning in jonckheere.test(x = data_random_clones$Mean_Day_score, g = data_random_clones$Day, : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##

```

```
## Jonckheere-Terpstra test
##
## data:
## JT = 58798, p-value = 0.9372
## alternative hypothesis: increasing
## JT = 58798, p-value = 0.9372
## alternative hypothesis: increasing
```

16 tracing allClones

```
res_index_scMut_meta_MEF <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k != "NotAva" & res_index_scMut_meta$CellTagD3_48k != "NotAva" & res_index_scMut_meta$CellTagD13_48k != "NotAva",]

res_index_scMut_meta_D3 <- res_index_scMut_meta[res_index_scMut_meta$CellTagD3_48k != "NotAva" & res_index_scMut_meta$CellTagD13_48k != "NotAva",]

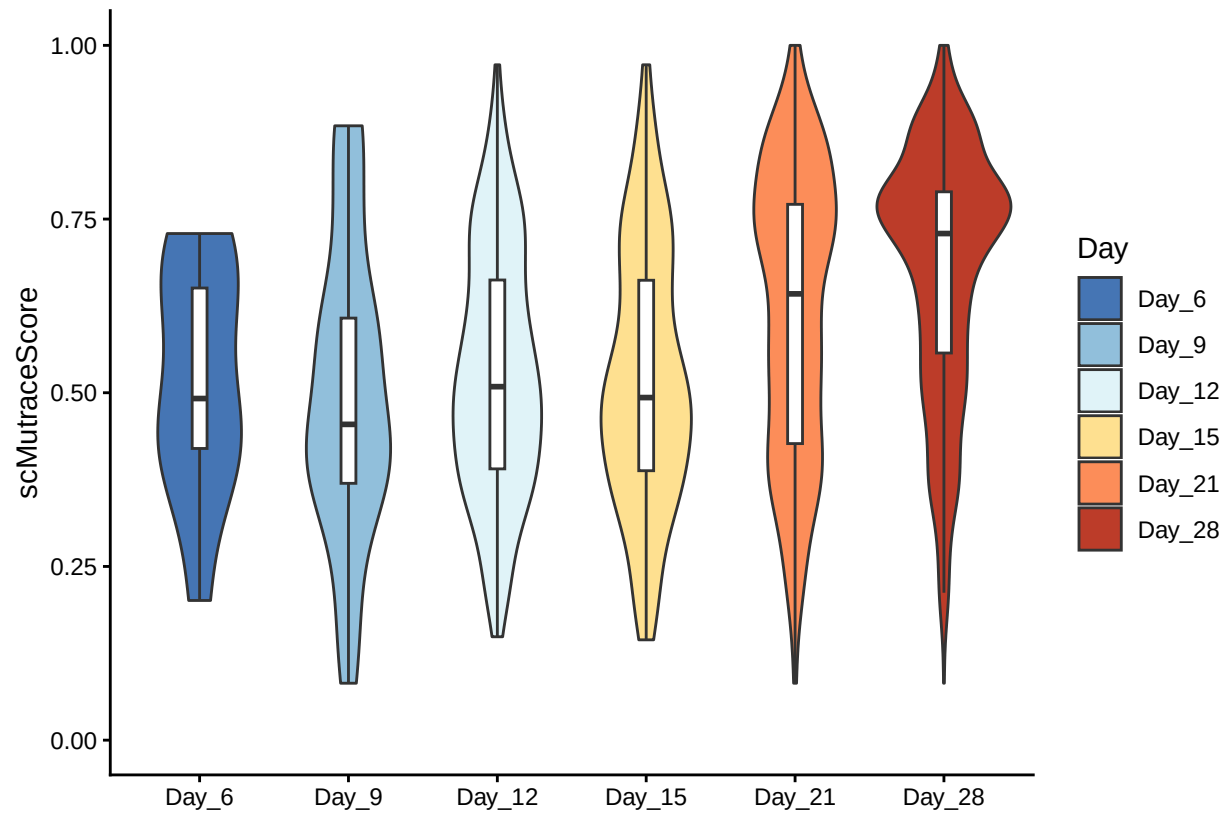
res_index_scMut_meta_D13 <- res_index_scMut_meta[res_index_scMut_meta$CellTagD13_48k != "NotAva",]

p_cell_MEF <- ggplot(data = res_index_scMut_meta_MEF, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMuttraceScore") +
  theme_classic()

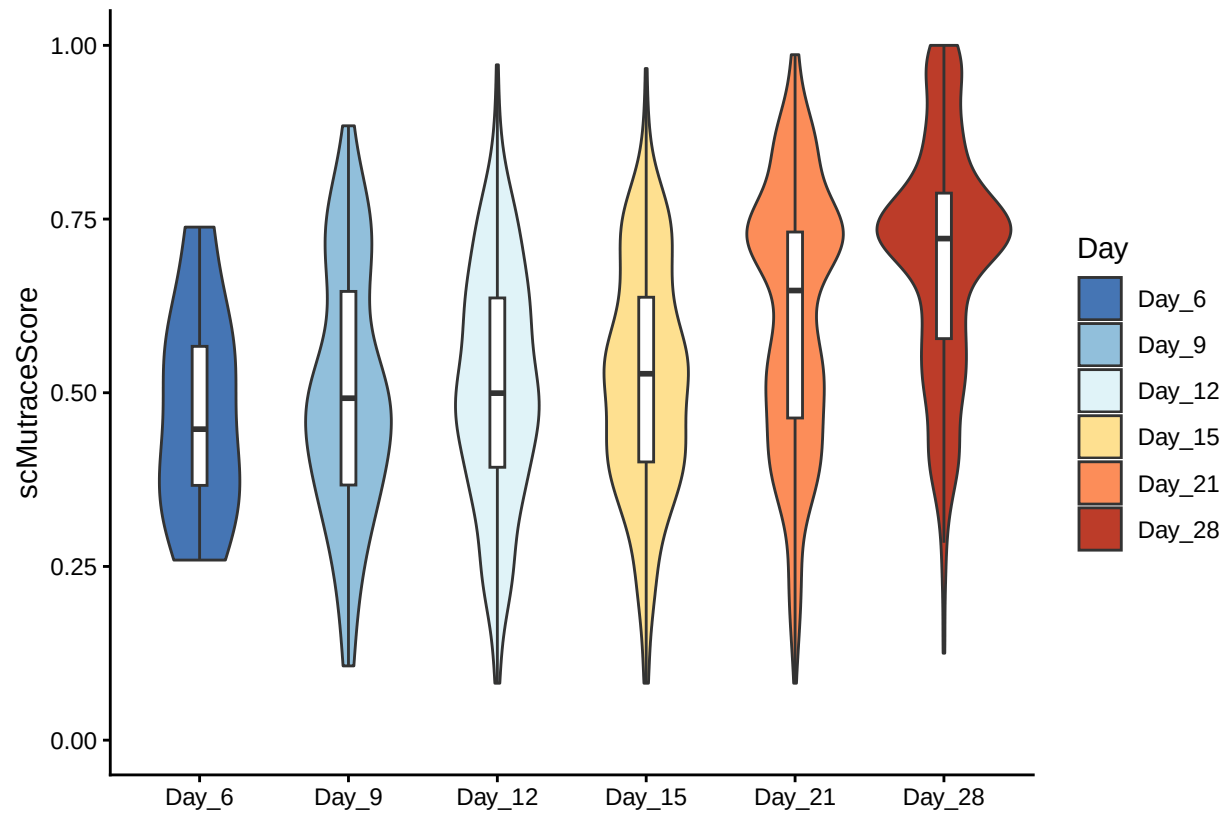
p_cell_D3 <- ggplot(data = res_index_scMut_meta_D3, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMuttraceScore") +
  theme_classic()

p_cell_D13 <- ggplot(data = res_index_scMut_meta_D13, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  #stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMuttraceScore") +
  theme_classic()

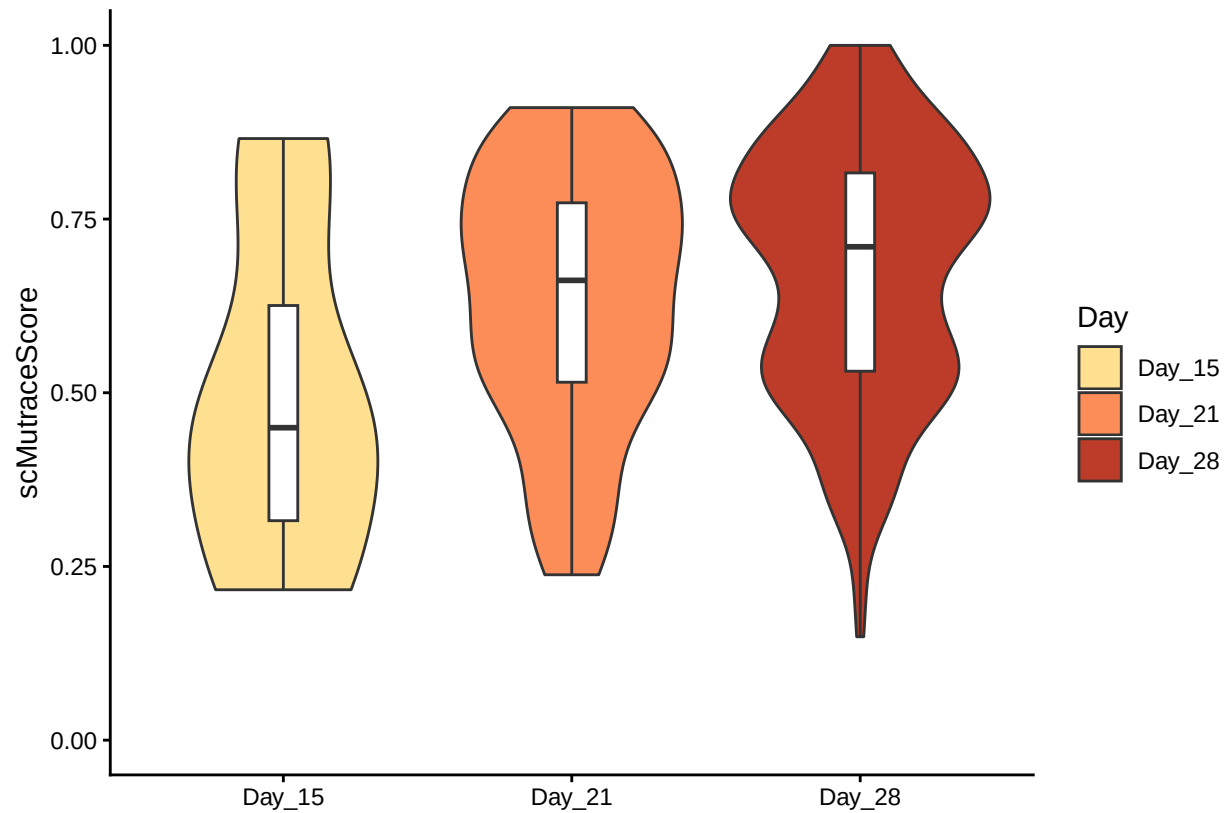
p_cell_MEF
```



p_cell_D3



p_cell_D13



17 randomlySelectMut

```
color_settings <- c(d0 = "#FFDC91FF", d3 = "#7876B1FF", d6 = "#4575b4",
  d9 = "#91b9db", d12 = "#e0f3f8", d15 = "#fee090",
  d21 = "#fc8d59", d28 = "#BC3C29FF")

color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
  Day_9 = "#91b9db", Day_12 = "#e0f3f8", Day_15 = "#fee090",
  Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

col_names_days <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28", "sampleSize")
reps_random <- 100
row_names_reps <- paste("iter", c(1:reps_random), sep = "_")

sample_num <- c(10, 30, 50, 100, 150)

data_random_muts2cell_hf1 <- as.data.frame(matrix(nrow = reps_random * length(sample_num), ncol = length(sample_num)))
data_random_muts2cell_hf2 <- as.data.frame(matrix(nrow = reps_random * length(sample_num), ncol = length(sample_num)))

colnames(data_random_muts2cell_hf1) <- col_names_days
colnames(data_random_muts2cell_hf2) <- col_names_days

#rownames(data_random_muts2cell_hf1) <- row_names_reps
```

```

#rownames(data_random_muts2cell_hf2) <- row_names_reps

hf1_muts <- data_CellPerMut_hf1$Location
hf2_muts <- data_CellPerMut_hf2$Location

set.seed(2025)
line_count <- 0
for (i in 1:reps_random){

  for (j in sample_num){

    line_count <- line_count + 1
    hf1_muts_tmp <- sample(hf1_muts, j, replace = TRUE)
    hf2_muts_tmp <- sample(hf2_muts, j, replace = TRUE)

    hf1_cells <- retrieveCellsByLocation(MEF_CSM_hf1, hf1_muts_tmp)$Mut
    hf2_cells <- retrieveCellsByLocation(MEF_CSM_hf2, hf2_muts_tmp)$Mut

    data_random_muts2cell_hf1[line_count, "Day_6"] <- sum(str_count(hf1_cells, "hf1.d6"))
    data_random_muts2cell_hf1[line_count, "Day_9"] <- sum(str_count(hf1_cells, "hf1.d9"))
    data_random_muts2cell_hf1[line_count, "Day_12"] <- sum(str_count(hf1_cells, "hf1.d12"))
    data_random_muts2cell_hf1[line_count, "Day_15"] <- sum(str_count(hf1_cells, "hf1.d15"))
    data_random_muts2cell_hf1[line_count, "Day_21"] <- sum(str_count(hf1_cells, "hf1.d21"))
    data_random_muts2cell_hf1[line_count, "Day_28"] <- sum(str_count(hf1_cells, "hf1.d28"))
    data_random_muts2cell_hf1[line_count, "sampleSize"] <- j

    data_random_muts2cell_hf2[line_count, "Day_6"] <- sum(str_count(hf2_cells, "hf2.d6"))
    data_random_muts2cell_hf2[line_count, "Day_9"] <- sum(str_count(hf2_cells, "hf2.d9"))
    data_random_muts2cell_hf2[line_count, "Day_12"] <- sum(str_count(hf2_cells, "hf2.d12"))
    data_random_muts2cell_hf2[line_count, "Day_15"] <- sum(str_count(hf2_cells, "hf2.d15"))
    data_random_muts2cell_hf2[line_count, "Day_21"] <- sum(str_count(hf2_cells, "hf2.d21"))
    data_random_muts2cell_hf2[line_count, "Day_28"] <- sum(str_count(hf2_cells, "hf2.d28"))
    data_random_muts2cell_hf2[line_count, "sampleSize"] <- j
  }
}

data_random_muts2cell_hf1$iteration <- rownames(data_random_muts2cell_hf1)
data_random_muts2cell_hf2$iteration <- rownames(data_random_muts2cell_hf2)

data_random_muts2cell_hf1_long <- data_random_muts2cell_hf1 %>%
  pivot_longer(
    cols = starts_with("Day"),
    names_to = "Day",
    values_to = "CellNum"
  )

data_random_muts2cell_hf2_long <- data_random_muts2cell_hf2 %>%
  pivot_longer(
    cols = starts_with("Day"),
    names_to = "Day",
    values_to = "CellNum"
  )

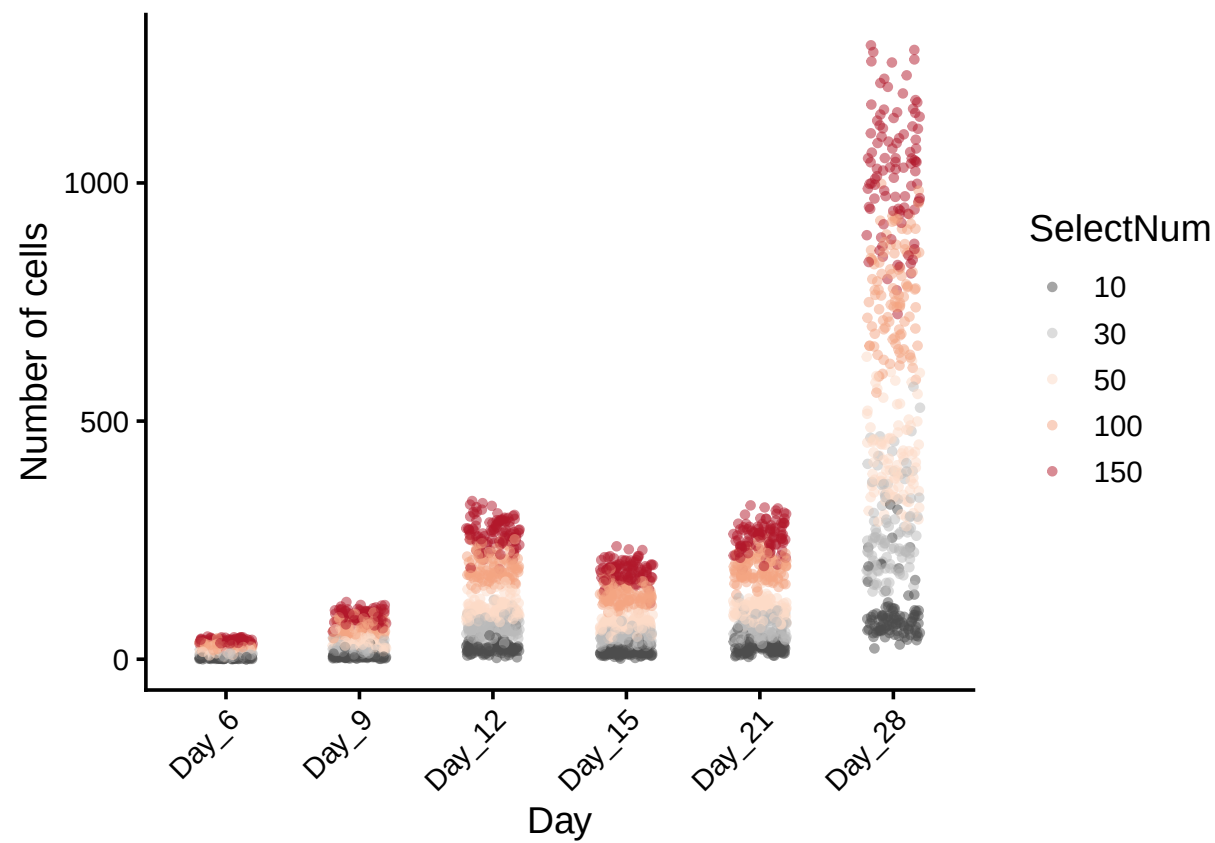
```

```
data_random_muts2cell_hf1_long$Day <- factor(data_random_muts2cell_hf1_long$Day, levels = c("Day_6", "D
data_random_muts2cell_hf2_long$Day <- factor(data_random_muts2cell_hf2_long$Day, levels = c("Day_6", "D
```

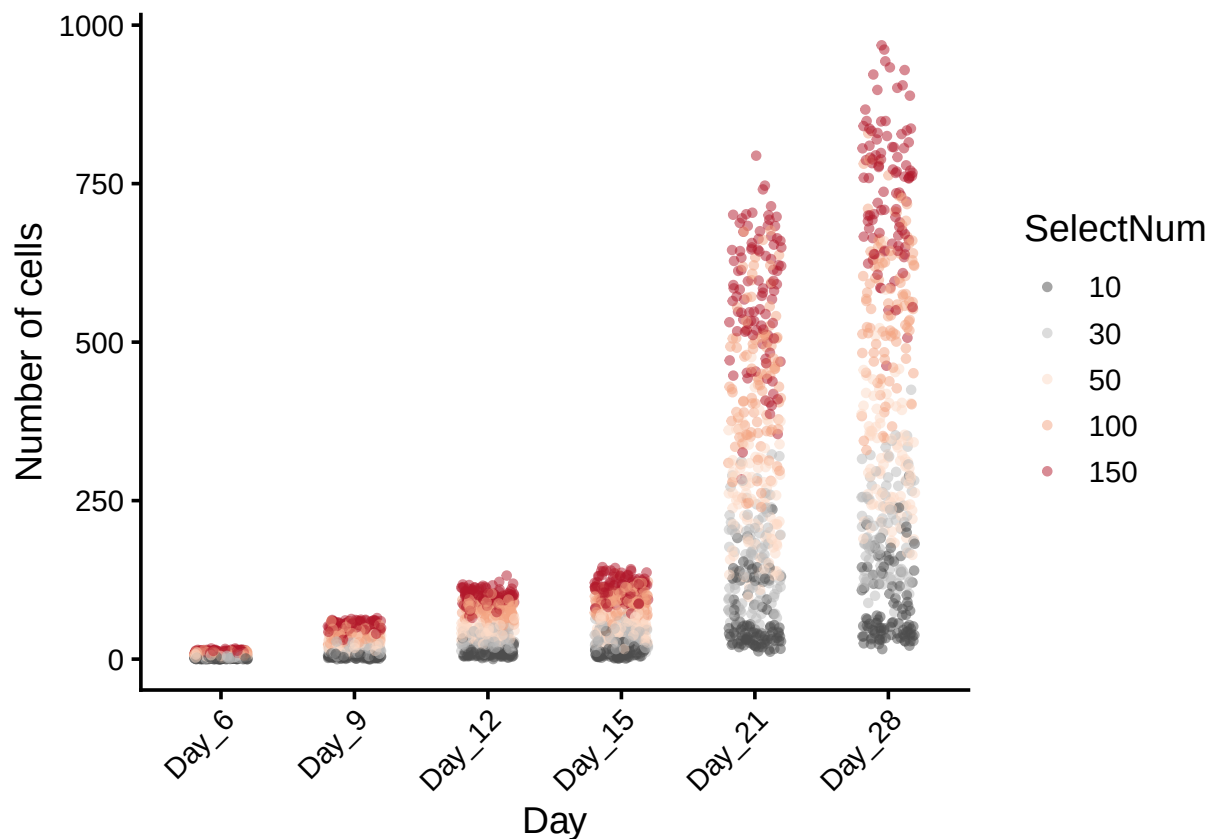
```
p_Mut2Cell_hf1 <- ggplot(data_random_muts2cell_hf1_long,
  aes(x = Day, y = CellNum, color = factor(sampleSize))) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  ylab("Number of cells") +
  xlab("Day") +
  labs(color = "SelectNum") +
  scale_color_manual(values = c(
    "10" = "#4d4d4d",
    "30" = "#bababa",
    "50" = "#fddbc7",
    "100" = "#f4a582",
    "150" = "#b2182b"
  )) +
  theme_classic(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
p_Mut2Cell_hf2 <- ggplot(data_random_muts2cell_hf2_long,
  aes(x = Day, y = CellNum, color = factor(sampleSize))) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  ylab("Number of cells") +
  xlab("Day") +
  labs(color = "SelectNum") +
  scale_color_manual(values = c(
    "10" = "#4d4d4d",
    "30" = "#bababa",
    "50" = "#fddbc7",
    "100" = "#f4a582",
    "150" = "#b2182b"
  )) +
  theme_classic(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
p_Mut2Cell_hf1
```



p_Mut2Cell1_hf2



18 lineageTracingExample

```
selected_start_cloneID <- 2352
data_clone_score_selected <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k == selected_start_
colnames(data_clone_score_selected) <- c("Cellbarcode", "scMutScore", "D0_tag", "D3_tag", "D13_tag")

# Categorization of all situations
data_clone_score_selected_long <- bind_rows(
  # C1: D3 == NotAva & D13 != NotAva → D0 → D13
  data_clone_score_selected %>%
    dplyr::filter(D3_tag == "NotAva" & D13_tag != "NotAva") %>%
    dplyr::select(source = D0_tag, target = D13_tag),
  # C2: D3 == NotAva & D13 == NotAva → D0 → D0_self
  data_clone_score_selected %>%
    dplyr::filter(D3_tag == "NotAva" & D13_tag == "NotAva") %>%
    dplyr::mutate(target = paste0(D0_tag, "_self")) %>%
    dplyr::select(source = D0_tag, target = target),
  # C3: D3 != NotAva & D13 != NotAva → D0→D3
  data_clone_score_selected %>%
    dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
```

```

dplyr::select(source = D0_tag, target = D3_tag),

# C4: D3 != NotAva & D13 != NotAva → D3→D13
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D3_tag, target = D13_tag),

# C5: D3 != NotAva & D13 == NotAva → D0→D3
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::select(source = D0_tag, target = D3_tag),

# C6: D3 != NotAva & D13 == NotAva → D3→D3_self
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::mutate(target = paste0(D3_tag, "_self")) %>%
  dplyr::select(source = D3_tag, target)
)

# Counting the number of pairs
clone_links <- data_clone_score_selected_long %>%
  dplyr::group_by(source, target) %>%
  dplyr::summarise(value = n()) %>%
  dplyr::arrange(desc(value))
## `summarise()` has grouped output by 'source'. You can override using the
## `.groups` argument.

clone_links <- as.data.frame(clone_links)

clone_nodes <- data.frame(
  name=c(as.character(clone_links$source),
  as.character(clone_links$target)) %>% unique()
)

clone_nodes$group <- ""

for (i in 1:nrow(clone_nodes)){
  checkNode <- as.character(clone_nodes[i,"name"])
  if (checkNode %in% as.character(data_clone_score_selected$D0_tag)){
    clone_nodes[i,"group"] <- "MEF"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D3_tag)){
    clone_nodes[i,"group"] <- "D3"
  }

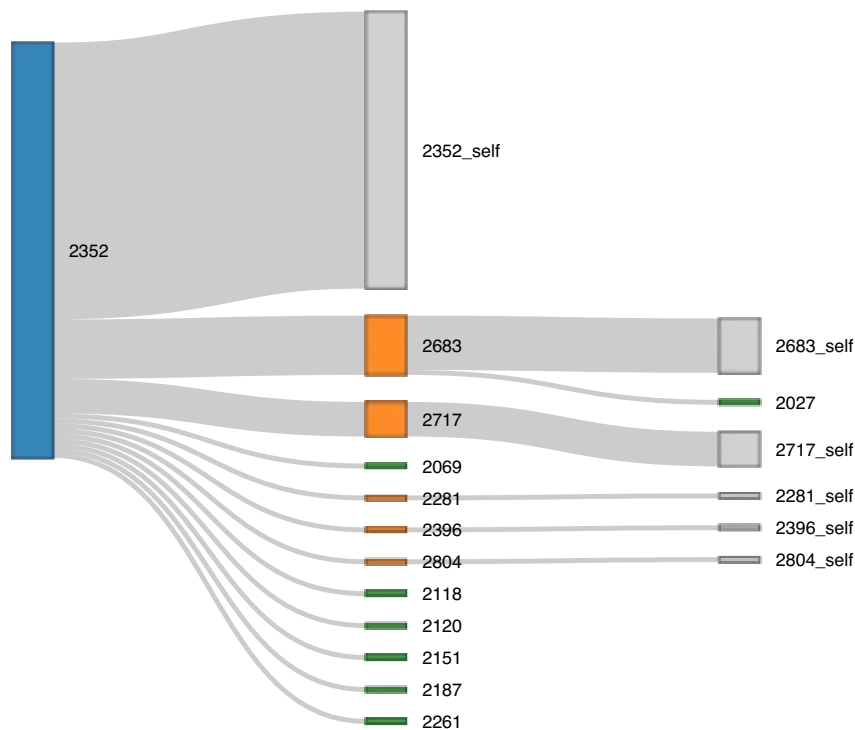
  if (checkNode %in% as.character(data_clone_score_selected$D13_tag)){
    clone_nodes[i,"group"] <- "D13"
  }
}

clone_links$IDsource <- match(clone_links$source, clone_nodes$name)-1
clone_links$IDtarget <- match(clone_links$target, clone_nodes$name)-1

```

```
p_sankeyNetwork <- sankeyNetwork(Links = clone_links, Nodes = clone_nodes,
  Source = "IDsource", Target = "IDtarget",
  Value = "value", NodeID = "name",
  sinksRight=FALSE,
  colourScale = JS("d3.scaleOrdinal()
    .domain(['MEF', 'D3', 'D13'])
    .range(['#1f77b4', '#ff7f0e', '#2ca02c'])"),
  NodeGroup = "group")

p_sankeyNetwork
```



19 lineageTracingExample with scMutScore

```
selected_start_cloneID <- c(124, 487, 2352)

data_clone_score_selected <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID]
colnames(data_clone_score_selected) <- c("Cellbarcode", "scMutScore", "D0_tag", "D3_tag", "D13_tag")

# Categorization of all situations
data_clone_score_selected_long <- bind_rows(
  # C1: D3 == NotAva & D13 != NotAva → D0 → D13
```



```

data_clone_score_selected %>%
  dplyr::filter(D3_tag == "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D0_tag, target = D13_tag),

# C2: D3 == NotAva & D13 == NotAva → D0 → D0_self
data_clone_score_selected %>%
  dplyr::filter(D3_tag == "NotAva" & D13_tag == "NotAva") %>%
  dplyr::mutate(target = paste0(D0_tag, "_self")) %>%
  dplyr::select(source = D0_tag, target),

# C3: D3 != NotAva & D13 != NotAva → D0→D3
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D0_tag, target = D3_tag),

# C4: D3 != NotAva & D13 != NotAva → D3→D13
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D3_tag, target = D13_tag),

# C5: D3 != NotAva & D13 == NotAva → D0→D3
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::select(source = D0_tag, target = D3_tag),

# C6: D3 != NotAva & D13 == NotAva → D3→D3_self
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::mutate(target = paste0(D3_tag, "_self")) %>%
  dplyr::select(source = D3_tag, target)
)

# Counting the number of pairs
clone_links <- data_clone_score_selected_long %>%
  dplyr::group_by(source, target) %>%
  dplyr::summarise(value = n()) %>%
  dplyr::arrange(desc(value))
## `summarise()` has grouped output by 'source'. You can override using the
## `.groups` argument.

clone_links <- as.data.frame(clone_links)

clone_links <- clone_links[!grepl("_self", clone_links$target), ]

#clone_links <- clone_links[clone_links$value > 1,]

clone_nodes <- data.frame(
  name=c(as.character(clone_links$source),
  as.character(clone_links$target)) %>% unique()
)

clone_nodes$group <- ""

```

```

for (i in 1:nrow(clone_nodes)){
  checkNode <- as.character(clone_nodes[i,"name"])
  if (checkNode %in% as.character(data_clone_score_selected$D0_tag)){
    clone_nodes[i,"group"] <- "MEF"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D3_tag)){
    clone_nodes[i,"group"] <- "D3"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D13_tag)){
    clone_nodes[i,"group"] <- "D13"
  }
}

# cal node scMutScore
clone_nodes$scMutScore <- 0

data_MEF_D3_D13 <- data.frame()

for (i in 1:nrow(clone_nodes)){
  group_tmp <- clone_nodes[i, "group"]
  node_tmp <- clone_nodes[i, "name"]
  print(node_tmp)
  print(group_tmp)
  if (group_tmp == "MEF"){
    data_mef_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID &
    if(nrow(data_mef_tmp) == 0){
      data_mef_tmp_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k == selected_start_cloneID &
      data_mef_tmp_tmp$Tag <- "MEF"
      clone_nodes[i, "scMutScore"] <- min(data_mef_tmp_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_mef_tmp_tmp)
    }else{
      data_mef_tmp$Tag <- "MEF"
      clone_nodes[i, "scMutScore"] <- min(data_mef_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_mef_tmp)
    }
  }
  if (group_tmp == "D3"){
    data_d3_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID &
    if(nrow(data_d3_tmp) == 0){
      data_d3_tmp_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k == selected_start_cloneID &
      data_d3_tmp_tmp$Tag <- "D3"
      clone_nodes[i, "scMutScore"] <- min(data_d3_tmp_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_d3_tmp_tmp)
    }else{
      data_d3_tmp$Tag <- "D3"
      clone_nodes[i, "scMutScore"] <- min(data_d3_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_d3_tmp)
    }
  }
  if (group_tmp == "D13"){
    data_d13_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID &

```

```

data_d13_tmp$Tag <- "D13"
clone_nodes[i, "scMutScore"] <- min(data_d13_tmp$scMutScore)
data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_d13_tmp)
}
}
## [1] "487"
## [1] "MEF"
## [1] "124"
## [1] "MEF"
## [1] "2352"
## [1] "MEF"
## [1] "204"
## [1] "D3"
## [1] "240"
## [1] "D3"
## [1] "212"
## [1] "D3"
## [1] "2683"
## [1] "D3"
## [1] "767"
## [1] "D3"
## [1] "488"
## [1] "D3"
## [1] "2717"
## [1] "D3"
## [1] "543"
## [1] "D3"
## [1] "779"
## [1] "D3"
## [1] "83"
## [1] "D13"
## [1] "93"
## [1] "D13"
## [1] "770"
## [1] "D3"
## [1] "784"
## [1] "D3"
## [1] "104"
## [1] "D13"
## [1] "43"
## [1] "D13"
## [1] "47"
## [1] "D13"
## [1] "99"
## [1] "D13"
## [1] "2069"
## [1] "D13"
## [1] "2118"
## [1] "D13"
## [1] "2120"
## [1] "D13"
## [1] "2151"
## [1] "D13"

```

```

## [1] "2187"
## [1] "D13"
## [1] "2261"
## [1] "D13"
## [1] "2281"
## [1] "D3"
## [1] "2396"
## [1] "D3"
## [1] "2804"
## [1] "D3"
## [1] "178"
## [1] "D13"
## [1] "2027"
## [1] "D13"
## [1] "110"
## [1] "D13"
## [1] "522"
## [1] "D3"
## [1] "128"
## [1] "D13"

data_MEF_D3_D13$Tag <- factor(data_MEF_D3_D13$Tag, levels = c("MEF", "D3", "D13"))
clone_nodes$group <- factor(clone_nodes$group, levels = c("MEF", "D3", "D13"))
data_MEF_D3_D13$Day <- factor(data_MEF_D3_D13$Day, levels = c("Day_0", "Day_3", "Day_6", "Day_9", "Day_12"))

network <- graph_from_data_frame(d=clone_links, vertices=clone_nodes, directed=T)

color_settings_network_1 <- c( D13 = "#e41a1c", D3 = "#377eb8", MEF = "#4daf4a")

color_settings_network_2 <- colorRampPalette(c("#fee8c8", "#fdbb84", "#e34a33"))

colors <- color_settings_network_2(5)

breaks <- unique(quantile(V(network)$scMutScore, probs = seq(0, 1, length.out = 6)))

value_group <- cut(V(network)$scMutScore, breaks = breaks, include.lowest = TRUE, labels = FALSE)

my_color_2 <- colors[value_group]

edge_width <- log2(E(network)$value + 0.1)

vertex_frame_color <- color_settings_network_1[V(network)$group]

p_selected_clone_MEF <- ggplot(data = data_MEF_D3_D13[data_MEF_D3_D13$Tag == "MEF",], mapping = aes(x = clone_ID, y = scMutScore)) +
  geom_violin(trim = FALSE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.05, fill = "white", outliers = FALSE) +
  #stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = paste0("scMutraceScore_", selected_start_cloneID)) +
  theme_classic()

p_selected_clone_D3 <- ggplot(data = data_MEF_D3_D13[data_MEF_D3_D13$Tag == "D3",], mapping = aes(x = clone_ID, y = scMutScore)) +
  geom_violin(trim = FALSE) + coord_cartesian(ylim = c(0, 1)) +

```

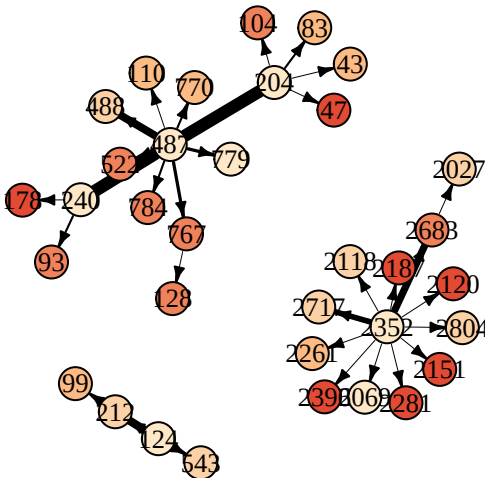
```

geom_boxplot(width = 0.05, fill = "white", outliers = FALSE) +
#stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
scale_fill_manual(values = color_settings_2) +
labs(x = "", y = paste0("scMutraceScore_", selected_start_cloneID)) +
theme_classic()

p_selected_clone_D13 <- ggplot(data = data_MEF_D3_D13[data_MEF_D3_D13$Tag == "D13",], mapping = aes(x =
geom_violin(trim = FALSE) + coord_cartesian(ylim = c(0, 1)) +
geom_boxplot(width = 0.05, fill = "white", outliers = FALSE) +
#stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
scale_fill_manual(values = color_settings_2) +
labs(x = "", y = paste0("scMutraceScore_", selected_start_cloneID)) +
theme_classic()

plot(network,
      vertex.color=my_color_2,
      vertex.label.color = "black",
      vertex.label.cex = 0.8,
      vertex.size = 15,
      #vertex.frame.width = 3,
      #vertex.frame.color = vertex_frame_color,
      edge.color = "black",
      edge.arrow.size = 0.4,
      edge.width = edge_width)

```



```
col_fun <- colorRampPalette(c("#fee8c8", "#fdbb84", "#e34a33"))
col_seq <- col_fun(length(my_color_2))
image(z = matrix(1:length(my_color_2), ncol = 1), col = col_seq,
      axes = FALSE, main = "Gradient Legend")
axis(4, at = seq(0, 1, length.out = 5), labels = round(seq(0, 1, length.out = 5), 2))
```

Gradient Legend



20 ModelFigure

```
m1 <- c(1, 1, 1, 1, 1, 1, 1, 1)
m2 <- c(0, 1, 1, 1, 1, 1, 1, 1)
m3 <- c(0, 0, 1, 0, 0, 0, 0, 0)
m4 <- c(0, 0, 0, 1, 1, 1, 1, 1)
m5 <- c(0, 0, 0, 0, 1, 1, 1, 1)
m6 <- c(0, 0, 0, 0, 0, 1, 0, 0)
m7 <- c(0, 0, 0, 0, 0, 0, 1, 0)
m8 <- c(0, 0, 0, 0, 0, 0, 0, 1)

t1 <- c(1, 0, 0, 0, 0, 0, 0, 0)
t2 <- c(1, 1, 0, 0, 0, 0, 0, 0)
t3 <- c(1, 1, 1, 1, 0, 0, 0, 0)
t4 <- c(1, 1, 1, 1, 0, 0, 0, 0)
```

```

t5 <- c(1, 1, 0, 1, 1, 0, 0, 0)
t6 <- c(1, 1, 0, 1, 1, 1, 1, 1)

test_Matrix_cm <- data.frame(m1 = m1, m2 = m2, m3 = m3, m4 = m4, m5 = m5, m6 = m6, m7 = m7, m8 = m8)
test_Matrix_mt <- data.frame(t1 = t1, t2 = t2, t3 = t3, t4 = t4, t5 = t5, t6 = t6)
rownames(test_Matrix_cm) <- c("c1", "c2", "c3", "c4", "c5", "c6", "c7", "c8")

test_Matrix_ct <- as.matrix(test_Matrix_cm) %*% as.matrix(test_Matrix_mt)
rownames(test_Matrix_ct) <- c("c1", "c2", "c3", "c4", "c5", "c6", "c7", "c8")

res <- cal_MultiSpe(test_Matrix_ct)

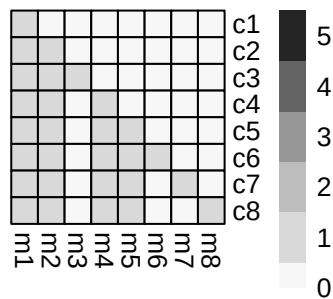
cal_scMutScore(test_Matrix_cm, test_Matrix_mt, NA)
## [1] 8 8
## [1] 8 6
## $diversity
##      t1      t2      t3      t4      t5      t6 Diversity
## c1  0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## c2  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c3  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c4  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c5  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c6  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c7  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c8  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
##
## $score
##      Shannon  Simpson      Gini sumMut      Tau      GiniTau scMutScore
## c1  1.791759 0.8333333 0.0000000      6 0.0000000 0.000000000 0.00000000
## c2  1.767761 0.8264463 0.07575716     11 0.1000000 0.007575716 0.06691216
## c3  1.737983 0.8165680 0.16666590     13 0.3333333 0.055555299 0.48481273
## c4  1.736741 0.8177778 0.14444387     15 0.2000000 0.028888773 0.25320566
## c5  1.711547 0.8096886 0.20588163     17 0.3500000 0.072058569 0.62733159
## c6  1.692020 0.8024691 0.24073994     18 0.4800000 0.115555170 1.00000000
## c7  1.692020 0.8024691 0.24073994     18 0.4800000 0.115555170 1.00000000
## c8  1.692020 0.8024691 0.24073994     18 0.4800000 0.115555170 1.00000000
##
##      Diversity sumMut_norm GiniTau_norm scMutScoreQun
## c1  0.0000000 0.0000000 0.0000000 0.0000000
## c2  0.8333333 0.4166667 0.0655593 0.4166667
## c3  0.8333333 0.5833333 0.4807686 0.5833333
## c4  0.8333333 0.7500000 0.2499998 0.7500000
## c5  0.8333333 0.9166667 0.6235859 0.9166667
## c6  0.8333333 1.0000000 1.0000000 1.0000000
## c7  0.8333333 1.0000000 1.0000000 1.0000000
## c8  0.8333333 1.0000000 1.0000000 1.0000000
##
## $matrixD
##      t1 t2 t3 t4 t5 t6
## c1  1  1  1  1  1  1
## c2  1  2  2  2  2  2
## c3  1  2  3  3  2  2
## c4  1  2  3  3  3  3
## c5  1  2  3  3  4  4

```

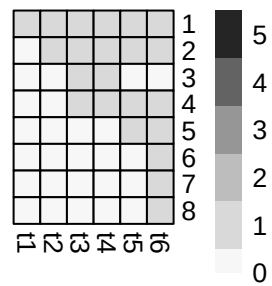
```
## c6 1 2 3 3 4 5
## c7 1 2 3 3 4 5
## c8 1 2 3 3 4 5
```

```
my_colors <- c("#f7f7f7", "#d9d9d9", "#bdbdbd", "#969696", "#636363", "#252525")
breaks <- c(0, 0.5, 1.5, 2.5, 3.5, 4.5, 5.5)
```

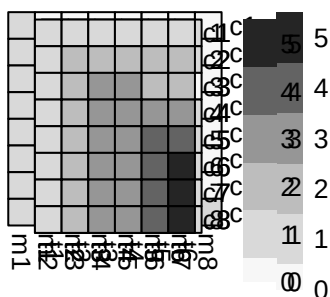
```
p_Matrix_cm <- pheatmap(test_Matrix_cm, cluster_rows = F, cluster_cols = F, color = my_colors, breaks =
```



```
p_Matrix_mt <- pheatmap(test_Matrix_mt, cluster_rows = F, cluster_cols = F, color = my_colors, breaks =
```

```
p_Matrix_ct <- pheatmap(test_Matrix_ct, cluster_rows = F, cluster_cols = F, color = my_colors, breaks =
p_Matrix_cm
p_Matrix_mt
p_Matrix_ct
```



21 sessionInfo

```
sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] clinfun_1.1.5      patchwork_1.3.2    lubridate_1.9.4    forcats_1.0.0
## [5] purrr_1.1.0        readr_2.1.5        tibble_3.3.0       tidyverse_2.0.0
```

```
## [9] networkD3_0.4.1      tidygraph_1.3.1      reshape2_1.4.4      pheatmap_1.0.13
## [13] corrplot_0.95        stringr_1.5.2        ggpmisc_0.6.2        ggpp_0.5.9
## [17] webshot_0.5.5        ggraph_2.2.2         igraph_2.1.4         Seurat_5.3.0
## [21] SeuratObject_5.2.0   sp_2.2-0             ggpubr_0.6.1         ggplot2_4.0.0
## [25] tidyr_1.3.1          dplyr_1.1.4          vegan_2.7-1          permute_0.9-8
## [29] ineq_0.2-13          FSA_0.10.0
##
## loaded via a namespace (and not attached):
## [1] RcppAnnoy_0.0.22      splines_4.5.1         later_1.4.4
## [4] polyclip_1.10-7       fastDummies_1.7.5     lifecycle_1.0.4
## [7] rstatix_0.7.2         processx_3.8.6        globals_0.18.0
## [10] lattice_0.22-7        MASS_7.3-65           backports_1.5.0
## [13] magrittr_2.0.4        plotly_4.11.0         rmarkdown_2.30
## [16] yaml_2.3.10           httpuv_1.6.16         sctransform_0.4.2
## [19] spam_2.11-1           spatstat.sparse_3.1-0 reticulate_1.43.0
## [22] cowplot_1.2.0         pbapply_1.7-4         RColorBrewer_1.1-3
## [25] abind_1.4-8           Rtsne_0.17            tweenr_2.0.3
## [28] data.tree_1.2.0       ggrepel_0.9.6         irlba_2.3.5.1
## [31] listenv_0.9.1         spatstat.utils_3.2-0  MatrixModels_0.5-4
## [34] goftest_1.2-3         RSpectra_0.16-2       spatstat.random_3.4-2
## [37] fitdistrplus_1.2-4    parallelly_1.45.1     codetools_0.2-20
## [40] ggforce_0.5.0         tidymodels_1.2.1      farver_2.1.2
## [43] viridis_0.6.5         matrixStats_1.5.0     spatstat.explore_3.5-3
## [46] jsonlite_2.0.0        progressr_0.16.0      Formula_1.2-5
## [49] ggribes_0.5.7         survival_3.8-3        tools_4.5.1
## [52] ica_1.0-3             Rcpp_1.1.0            glue_1.8.0
## [55] gridExtra_2.3         xfun_0.53             mgcv_1.9-3
## [58] withr_3.0.2           fastmap_1.2.0         SparseM_1.84-2
## [61] callr_3.7.6           digest_0.6.37         timechange_0.3.0
## [64] R6_2.6.1             mime_0.13             colorspace_2.1-2
## [67] scattermore_1.2       tensor_1.5.1          dichromat_2.0-0.1
## [70] spatstat.data_3.1-8   generics_0.1.4        data.table_1.17.8
## [73] graphlayouts_1.2.2    httr_1.4.7            htmlwidgets_1.6.4
## [76] uwot_0.2.3           pkgconfig_2.0.3       gtable_0.3.6
## [79] lmtest_0.9-40         S7_0.2.0              htmltools_0.5.8.1
## [82] carData_3.0-5         dotCall64_1.2          scales_1.4.0
## [85] png_0.1-8            spatstat.univar_3.1-4 knitr_1.50
## [88] rstudioapi_0.17.1     tzdb_0.5.0            nlme_3.1-168
## [91] zoo_1.8-14           cachem_1.1.0          KernSmooth_2.23-26
## [94] parallel_4.5.1        miniUI_0.1.2          pillar_1.11.1
## [97] grid_4.5.1           vctrs_0.6.5           RANN_2.6.2
## [100] promises_1.3.3        car_3.1-3             xtable_1.8-4
## [103] cluster_2.1.8.1       evaluate_1.0.5        mvtnorm_1.3-3
## [106] cli_3.6.5            compiler_4.5.1        rlang_1.1.6
## [109] future.apply_1.20.0    ggsignif_0.6.4        labeling_0.4.3
## [112] ps_1.9.1             plyr_1.8.9            stringi_1.8.7
## [115] viridisLite_0.4.2     deldir_2.0-4          lazyeval_0.2.2
## [118] spatstat.geom_3.6-0    quantreg_6.1          Matrix_1.7-4
## [121] RcppHNSW_0.6.0        hms_1.1.3            future_1.67.0
## [124] shiny_1.11.1          ROCR_1.0-11          broom_1.0.10
## [127] memoise_2.0.1         polynom_1.4-1
```